PSYCHIATRIC NOTES

FOR MEDICAL STUDENTS

2ND EDITION

Kurdistan Region – Iraq
Ministry of Higher Education and Sientific Research
University of Sulaimani
Faculty of Medical Sciences
School of Medicine
Department of Medicine

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BY

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Psychiatric Notes for Medical Students

Second Edition

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FOREWORD

Medical education is an important tool to create doctors well equipped with knowledge, medical ethics and clinical skills. Over the years there have been a lot of guides for medical students in various fields of General Medicine and various specialties related to medicine. Most of the guides were written to suit the requirements of students throughout the world.

This book is the first attempt to help medical students during their training in the medical schools. Most of the teachers and students in the medical schools are challenged by lack of time and lack of simplified resources to cover the syllabus of mental health. The large size of textbooks and shortage of access to them in our locality created the need for a valuable book like this one.

The author tried to gather the most important and updated material required for medical students in a very concise way avoiding unnecessary details. This book fills a very important gap in the field of medical education. It helps students in various allied sciences like Nursing, Dentistry and Pharmacy in their years of training too.

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PREFACE to the first edition

Throughout my academic life, both as a medical student and a faculty member, I perceived students' penchants to rely more on lectures notes, materials, and modules rather than textbooks. Such a partiality may be explicable since notes and modules offer more condensed, up-to-minute, examinations-oriented, as well as lecturers' notes relevant information. Furthermore, we have to acknowledge that the fatty textbooks may sound pretty boring at least for undergraduate medical students.

With the new advanced technologies, and the availability of innovative teaching materials from computerized tablets, notes, as well as smart phones and boards, writers and publishers alike, are in charge to consider a 'fitness round' for their heavy writings in order to keep survival of their materials.

This book may act as a connecting bridge between lectures and books isles. In that sense, it adopted lecture-notes format, yet in a book-frame. Therefore, it may serve as'reconciliation' between medical students and book-reading. Such 'lecture-format' is evident since the first chapter due to the end of the work.

The current book is, basically, addressing undergraduate medical students. For that reason, great deal of efforts made by the author to avoid comprehensive tackling of clinical disorders. Instead, concise and succinct summarizing of clinical situations was considered. Moreover, not all mental disorders psychiatrists faced, heard, or read about were addressed here. Instead, only well-known psychiatric morbidities I felt relevant to medical students as they may come across, after graduation, in their internship placements, were illuminated.

The book takes in three parts: part I, treats general aspects of psychiatry over ten chapters, starting from a brief historical overview, then presenting general psychiatric disorders, and closing with a hand out of psychiatric emergencies. Part II, addresses different specialties in psychiatry like substance-related disorders. Part III, however, is dedicated to different modes of therapy in psychiatry. The book, then, seal by a couple of appendices; the first is a patient assessment template, whereas the second is a table of prescriptions' abbreviations.

In order to keep medical students up-to-date, the newly released version of American Psychiatric Association's (APA) Diagnostic and Statistical Manual of Mental Disorders (DSM) fifth edition (May 2014) was adopted for the definitions and descriptions of clinical features of entire clinical disorders presented in this book. Here, I would like to acknowledge that almost all descriptions and diagnostic criteria were replicated literally from DSM-5.

August 2014

PREFACE to the second edition

The foremost motive encouraged me to consider the second edition of this textbook was the growing demands from different universities for additional copies. As the University of Sulaimani printed only 300 copies of the first edition, which was below these aggregating requests, I found the delivery of the second edition to be indispensable as soon as possible.

Since the first edition of this book was published, I have had several useful remarks and suggestions for modifications and improvements. I have done my best to accommodate these in this revised edition.

The first two chapters of the previous edition were merged together, several graphs and diagrams were more clearly edited, and on the top of every thing, an outstanding new addition is the presence of a comprehensive Index.

It was my pleasure for the book to perceive the increasing attentions of medical institutes and students alike. I hope this book will continue to play a part in facilitating medical education.

September 2015

ACKNOWLEDGEMENTS

It is a must to thank my friend and colleague Dr. Afram Muhammad Hassan for his sponsorship of this new edition. He is renowned for his continuous support for academic and educational projects in Sulaimani.

My special thanks are also forwarded to my postgraduate students and colleagues who generously offered their review, corrections, and suggestions upon the first edition.

Hereby, I would like to echo my gratitudes to the contributors of the first edition; particularly, Professor Nazar Muhammad Muhammad-Amin, Assistant Professor Sirwan Kamil Ali, as well as Assistant Lecturer Ranji Shorsh Rauf Muhammad for their exceptional scientific and linguistic reviews.

PART I

GENERAL ASPECTS OF PSYCHIATRY

01 A BRIEF INTRODUCTION TO AND CLINICAL ASSESSMENT IN PSYCHIATRY

Historical Overview

Although the current phenomenological as well as medical scenes of psychiatry can, merely, be traced back to a couple of centuries ago, history of psychiatry dated back to, much earlier, several centuries BC. There are evidences of referring to mental disturbances in Egyptian papyri and Old Testament. Early mental disorders' manuals were created by Greeks. Hippocrates postulated very early insight to the physiological bases of mental disorders. Furthermore, several psychiatric terms which, more or less, retained their usage in modern psychiatry like Melancholia, Mania, Phobia, Hypochondriasis, Hysteria and so forth were germinated in Ancient Greece.

Avicenna, the Persian physician, developed many Galen's ideas about mental disorders in his *Canon of Medicine*. Baghdad was the home of first specialty hospitals for mental disorders in 705 AD. Several physicians, apart from Avicenna, wrote on mental disorders during the medieval Islamic period including Rhazes.

In Europe, first asylums were built during 13th century AD which lacked any kind of therapy. It was not until early 1800s when the French physician Philippe Pinel introduced the *humane treatment* approach for those suffering from mental disorders; for which, many consider him to be the *Father* of modern psychiatry.

Psychiatry vs. Mental Disorder

The term 'Psychiatry' derived from psych- (from Ancient Greek *psykhē*: soul) and –iatry (from Ancient Greek *iātrikos*: medical, *iāsthai*: to heal). Therefore; psychiatry, in verbatim, stands for 'Medical Treatment of Soul'.

The term was coined by the German physician Johann Christian Reil, in 1808, devoting to a medical specialty addressing diagnosis, treatment, as well as prevention of mental disorders.

Mental Disorder can be defined as a clinically significant behavioral or psychological syndrome associated with distress or disability, not just an expected response to a particular event or limited to relations between a person and society. The phrase is preferred to 'mental disease' or 'mental illness,' since the terms disease and illness are basically associated with organic conditions and many mental disorders, however, can purely be psychogenic or physiological in origin.

Etymologically, the term *mental* derived from the Latin *mēns* which means *mind*. Therefore, *Mental Disorders* might be translated as *Mind Disorders*. The expression 'mental disorder' may not be a genuine description since it implies a distinction between *mental/mind* and *physical* disorders that is a reductionistic relic of mind/body dualism. Nevertheless, in our waiting for a new descriptive phrase, the *Mental Disorder* idiom will be adopted throughout this book.

Classification of Mental Disorders

Classification is the key system in organizing any discipline. In medicine, including psychiatry, classification serves in distinguishing different diagnoses so that clinicians and scientists build a common language, organize their treatment plan, and hypothesize possible underlying etiological factors for the clinical conditions.

Hippocrates was the first who tried to classify mental disorders. Since then, several systems of classification have been proposed. However, the two most important psychiatric classifications, nowadays, are the *Diagnostic and Statistical Manual of Mental Disorders* (DSM) developed by American Psychiatric Association (APA) and the *International Classification of Disease* (ICD) developed by World Health Organization (WHO).

Since this book is more dedicated to undergraduate medical students, and to avoid possible confusion between both systems, this book will rely solely on one system of classification, DSM.

In 1952, APA published the first edition of DSM (DSM-I). Since then, six further editions have been published: DSM-II (1968), DSM-III (1980), DSM-III revised (1987), DSM-IV (1994), DSM-IV-text revision (TR) (2000), and most recently DSM-5 (2014). In order to keep students most up-to-date, this book will bond with the latest version of DSM, DSM-5.

Prevalence of Mental Disorders

Epidemiologic Catchment Area (ECA), which is the largest study of prevalence of psychiatric disorders, surveyed more than 20,000 individuals across five states of US during 1980-1985. One year prevalence of any mental disorder was 28.1%. Anxiety disorders were the most common mental disorder followed by substance-abuse and mood disorders.

Most recent national survey of mental disorders' prevalence, conducted by Iraqi Ministry of Health in collaboration with WHO and World Mental Health (WMH) survey initiative during 2007-8, surveyed 4,332 Iraqis. The study revealed back 18.8% lifetime prevalence of any mental disorder in Iraq. Likewise, anxiety disorders were the most common class of disorders. However, major depression was the most common disorder.

Clinical Assessment in Psychiatry

Psychiatric assessment, like medicine, requires systematic gathering of signs and symptoms in order to conclude a provisional diagnosis or a set of possible differential diagnoses. However, unlike medicine, pretty arts and crafts are necessary to apply such an organized evaluation.

Instructors may efficiently teach interview techniques; however, arts and crafts are more difficult to be taught. Rather, they acquired by practice and experience.

The assessment is not limited to symptoms-checklists. But, it's a dexterity of conducting an interpersonal communication. No technology is devised yet to replace the individual interviewer in interviewing the interviewee.

Nonetheless, interview is not a social tittle-tattle. To a certain extent, it's a professional interaction equipped towards employing the patient in a therapeutic relationship.

Characteristics of Good Interviewer

There are numerous characteristics that amount psychiatrists to be qualified as 'good' interviewers. Below are handfuls of them:

A. **Be empathetic and sympathetic:**

   Empathy and sympathy are, often, understood and used interchangeably. Empathy refers to the capability to place oneself in the emotional viewpoint of another while maintaining one's own viewpoint. If the interviewer cannot maintain one's emotional viewpoint, then the condition is best referred as '*identification*', which indeed harms, more than helps, the patient.

Sympathy, alternatively, refers to how far the interviewer cares and reacts to what the patient is suffering from.

B. Be professional:

Empathy and sympathy may be offered to a patient, more efficiently, by a family member or a friend. What upgrades the interviewer is his/er professionalism. The assessor should be aware of the entire process, budget an interview time appropriately so as covering the whole aspects of the assessment, and keep formality. Therefore, it's not only 'what' to ask, but, also, 'how' to ask. No question, gesture, or comment should be slipped out without rationale.

C. Be confident:

Interviewers should own abilities to convince the patient that nobody else would help him/er better. However, psychiatrists asked to avoid 'dogmatism'.

D. Be neutral:

Be 'Objective' not 'Subjective'. Sticking one to the back row and keeping a non-judgmental attitude toward the patient is the best strategy. Many patients feel they have been judged, especially those with psychotic symptoms. Further, never install your believes and principles in patient's mind.

Above characteristics are necessary to build up trust in patients which is crucial for any therapeutic adherence. (Fig 1-1)

Assessment Setting

Psychiatrists might interview patients in different settings, ranging from their own private clinic, out-patient unit and in-patient ward of a psychiatric hospital, non-psychiatric medico surgical ward, a medico legal committee, patient's own home, and even in prison.

Wherever, before starting assessment and questioning the patient, it's worth describing the adequate interview setting:

A. The room:

1. The room should be fairly calm and not located in the center of a busy unit. Also, it's NOT advisable to provide background music, even the soft and relaxing one, unlike believes of some psychotherapists, to avoid distraction of both the patient and the therapist from the content of interview.

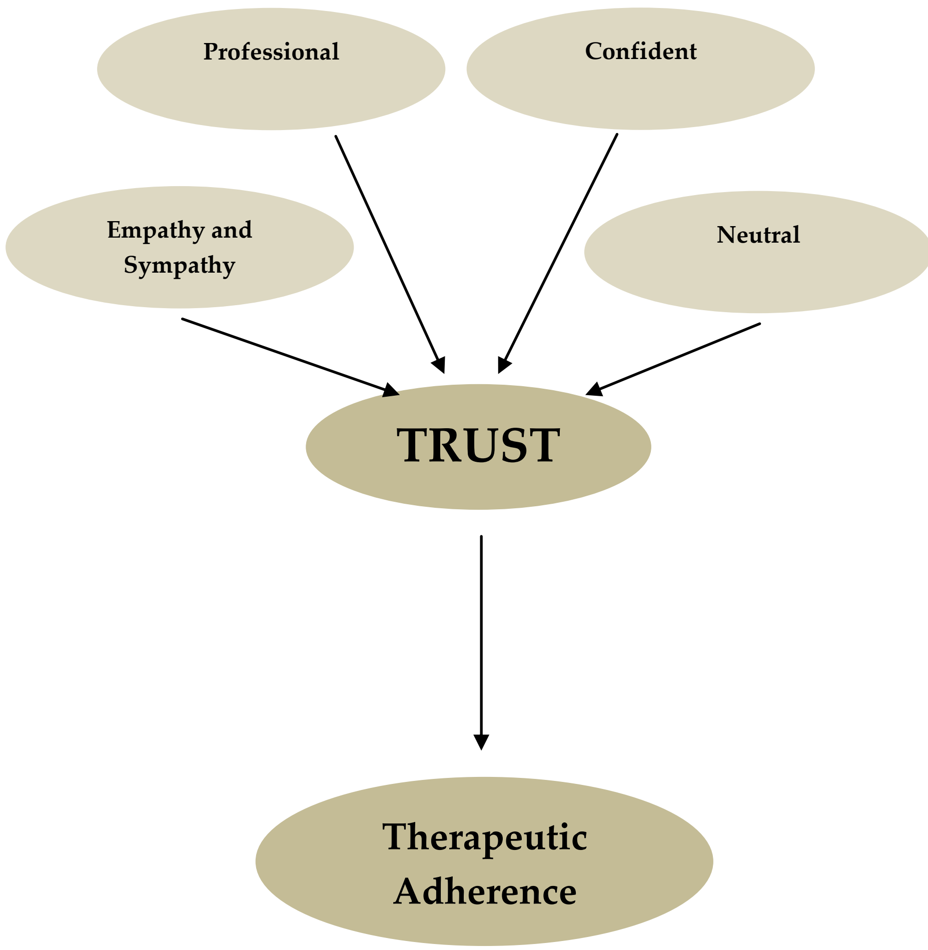

Fig 1-1: Characteristics of good interviewer

2. Privacy is one of the basic patients' rights. Therefore, the room supposed to be relatively isolated, and the interviewer has to ensure no interruption while assessing the patient. However, psychiatrists must increasingly consider the risks of aggression and violence from patients' side. Hence, securing a second-hand help is not a trivial idea.
3. Keep the interview room as simple as possible by avoiding attractive decoration which might turn out to missiles! Also, ornaments might give clues about the educational and cultural background of the therapist, the condition which throw out the psychiatrist's neutrality (see above for the characteristics of good interviewer).

4. Interviewer's security is not guaranteed by the 'Spartan' room only. In addition, several other measures are necessary; for instance, keep telephone cables as short and fixed to ground as possible, handle a portable alarm, and keep yourself between the patient and the door.

B. The seat:

1. Physicians, including psychiatrists, frequently advocate the presence of desk in their clinic. However, desk does not help the patient anyhow. Rather, it's more reassuring for the therapist since it offers an authority figure which is attractive for many psychiatrists! Though, it's not helpful to interview the patient across outsized, long, and obstructing desk given that it creates both physical and psychological barriers between both partners.
2. Adopting a bed for the patient to lie down and relax on while interviewing might deem gorgeous. However, it's no longer more than a myth. Letting the patient lying in front of the examiner may be, rather, embarrassing especially with opposite-gender patients. Instead, nothing is better than simply inviting the patient to a chair seat. The chairs of therapist and patient should be of the same quality and color. It's recommended for the seats to be high and comfortable. The low and soft chairs (Sofa) are better to be avoided because they impose an uncomfortable posture and make standing exit more difficult, particularly for elderly, obese, and patients with Parkinsonism.

C. The attire of the interviewer:

The dress of a psychiatrist, whether formal (white coat) or not, faced more than expected debates. It depends on the individual health institute regulations.

Many psychiatrists argue for the 'non-formality' in their dress as long it offers more similarity, familiarity, and friendly-looking perceptions by psychiatric patients. These, making patients at ease, are undoubtedly important in commencing and maintaining proper doctor-patient relationship, especially when targeting particular populations like children or adolescents.

Nonetheless, several health institutes still stress on formal dressing of their staffs, including psychiatrists. The notion behind the uniform medical wears is to keep the perception of patients to their mental suffering within the medical context. The doctors' uniform might role in insight-orientation, and suggests a medical origin of their distresses similar to a gastric upset or chest discomfort for instance.

D. Duration of the interview:

How long the interview may last is abound with the purpose of the assessment. Roughly estimation of a 30 minutes minimum may be mandatory, particularly for the first-time evaluation.

E. Introductions:

Finally, before starting the assessment, both the interviewer and interviewee have to introduce themselves to each other.

1. It's not uncommon for patients to be referred by other specialties, or forced by family members, for instance, to you without being made clear that you are a psychiatrist! Patient's unexpected perception of the interviewer as a psychiatrist might raise unanticipated reactions of fear, anxiety, or anger which, possibly, complicated by poor doctor-patient relationship. Therefore, it's necessary not only to introduce '*who*' but '*what*' you are too. Transparency is, probably, the best shortcut to ensure proper welcome.
2. Then, invite the patient to take a seat. Do not leave the patient to make an awkward social assumption!
3. How to address the interviewee? It depends on the cultural and educational background of the patient. Adopting some formal addresses like 'Mr', 'Mrs' or 'Miss' might be appropriate, principally for those who seem to be elder than you. But, some culturally-oriented addresses, rather, may be preferred. For instance, 'Haji', 'Kaka', or 'Khatoon' as they are appropriate. Some patients do prefer to call them, respectfully, by their profession; take for example, 'Mamosta', 'Dr', 'Professor', 'General' and so forth. Once you felt confused about how to address the patient, then nothing is better than addressing him/er by a given first name. Nevertheless, it does not harm if you ask the patient how s/he would like to be called.
4. Keeping the first couples of minutes 'silent' while you are indulged in reading the referral letter or past case notes is rude and frustrating. Hence, such kind of preparation should be done, in advance, before patient's entry to the room.
5. It's not unusual for psychiatric patients to be accompanied by another person(s) who may be a family member, friend, spouse, or even a policeman. It's worthwhile to interview both the patient and the companion. However, the question is when to interview the second person? Despite the preference of some psychiatrists to interview the companion prior to the patient,

it's advisable to see the patient before any accompanied person, so as to avoid possible bias, and to grant permission from the patient for obtaining information by proxy. Other psychiatrists do prefer interviewing both together. Although such kind of practice might be reassuring, but you may miss crucial, otherwise, sensitive information for which the patient would be reluctant to disclose in front of a third party.

6. Questions: if the patient doesn't attempt to unveil his/er experiences, then nothing is better than questioning them. However, interviewer ought not to turn out a dialogue to an interrogation.

   Roughly speaking, there are three kinds of questions: Open-ended, Closed-ended, and Leading-questions.

   Open-ended question is a kind of question for which no particular answer is expected. It's the best chance for the patient to narrates his/er answer. Best example of such kind of query is "what problem brought you to hospital?", or "how is your sleep?" It's prudent to employ open-ended questions during the first half of an interview.

   This kind of question might divulge the most reliable information of the interview; therefore keep considerable time of the interview for this style of free talk.

   Closed-ended question is a kind of question for which particular answer is expected, like 'yes', 'no', or 'name of person, an object, or an address'.

   This sort of query is useful when time is short, when an open-ended question does not work, or to clarify more details after a prior open-ended question has been answered. It's desirable to keep it to the second half of interview.

   Leading question is a sort of question for which the questioner includes the answer within it. For instance, "then you hear voices talking to you, do you?" It's wise to avoid or sparingly adopt this mode of question. However, it may be reasonable when you feel a patient does conceal information from you (instance for the above question is when the patient denies 'auditory hallucinations' while he looks talking to him/er self and there are clear evidences from his/er behaviors that there are voices arguing with him/er).

History Taking

Demographic data

Parallel to elsewhere in other medical specialties, history taking in psychiatry has to set up by obtaining detailed demographic data (Box 1-1).

The purpose of getting through whole meticulous profile data is not barely to let the patient introduce him/er self. Rather, demographic data help psychiatrist to attain preliminary sketches about diagnosis, therapy, as well as prognosis of the condition. For instance, memory impairment in an 18 year-old patient might draw the first-attention of the interviewer toward ‘anxiety’ or ‘depression’ symptom; however, the same problem in an 80 year-old client would suggest an organic brain disorder sign like ‘dementia’.

Box 1-1: Demographic Data
Name
Age
Gender
Education
Occupation
Address
Marital status
Number and gender of Children

Chief/Present compliant(s)

Chief complaint is the principal reason which forced the patient to visit you. It’s not unusual for the complaint to be more than one. However, it is not exceptional for the patient to reply your question by “I have no complaint”!

A. Employ **Open-ended questions** here. Best starting question may be: “what problem brought you to hospital?” Don’t forget to ask for the duration of each complaint.
B. Write what the patient tells you in verbatim.

History of present illness (HOPI)

HOPI is where the patient has to narrate his/er most recent experiences/sufferings. Hence, it’s wise to dedicate majority of the interview time to HOPI. Moreover, HOPI is the most reliable part of the interview to conclude a diagnosis. However, to avoid long irrelevant tale by the patient, it’s suitable for the psychiatrist to guide and control the story by a couple of, relatively, open-ended questions.

In ‘Chief Complaint’ and ‘HOPI’ you are asked to avoid psychiatric technical terms (see below) like ‘Hallucination’ or ‘Delusion’. As a substitute, convey the experiences in patient’s terms.

Explore each chief complaint, in detail, by asking about **(3WA2CSH)** in sequel: (Fig. 1-2)

A. **When** it came into view (onset).
B. **Why** it occurred (precipitating event or factor). Principally, precipitating factors are stressful events, like death, illness, marriage, divorce, failure in exam, loss of job, and so forth. It is convenient to remind the patient for the possibility of any preceding psychosocial stressful events.
   However, precipitating factor might be, only, patient’s justification of his/er current experience and not the factual scientific one. For instance, a patient may contribute his excessive night sleep to lack of noon nap. Whether patient’s explanations sound scientific or not, it’s worthy to consider them because, at their minimum, they give clues about the ways of patient’s logic thinking and rationalization.
C. **What** is the complaint (describe the symptom in detail). For example, if the patient’s presenting complaint is ‘headache’, then you have to ask for the quality, quantity, and location of the pain, as well as whether it’s radiating to elsewhere or not. If another patient’s chief compliant is ‘poor sleep’, then ask for whether s/he was suffering from difficulty in falling to sleep, early morning wakening, frequent night wakening, non-refreshing sleep and so forth.
D. **Associated symptoms:** what other symptoms occurred parallel to the main complaint?
E. **Course** of the present illness (evolution): how the condition went forward from the onset up to the time of interview? What factors aggravate/mitigate the condition? And during which time of the 24 hours the complaints were worse / better?
F. **Consequences**/Impacts of the present illness. How the current condition affected the socio-occupational status. For instance a patient may state “my poor sleep turned me down, inattentive, in the class”.
G. **Scanning questions:** these questions are, fairly, proportional to ‘Systematic Review’ in medicine. Whatever the experience of the patient, the interviewer has to test out presence/absence of the following: suicidality, anxiety, depression, obsession, psychotic symptoms, unusual experiences, as well as sexual and sleep problems.

H. **Help:** lastly, check out for any pre-visit help seeking behaviors; like visiting a general practitioner (GP), another psychiatrist, faith healers, herbalists, and so forth. Did their advices/therapies lessen/worsen the problem?

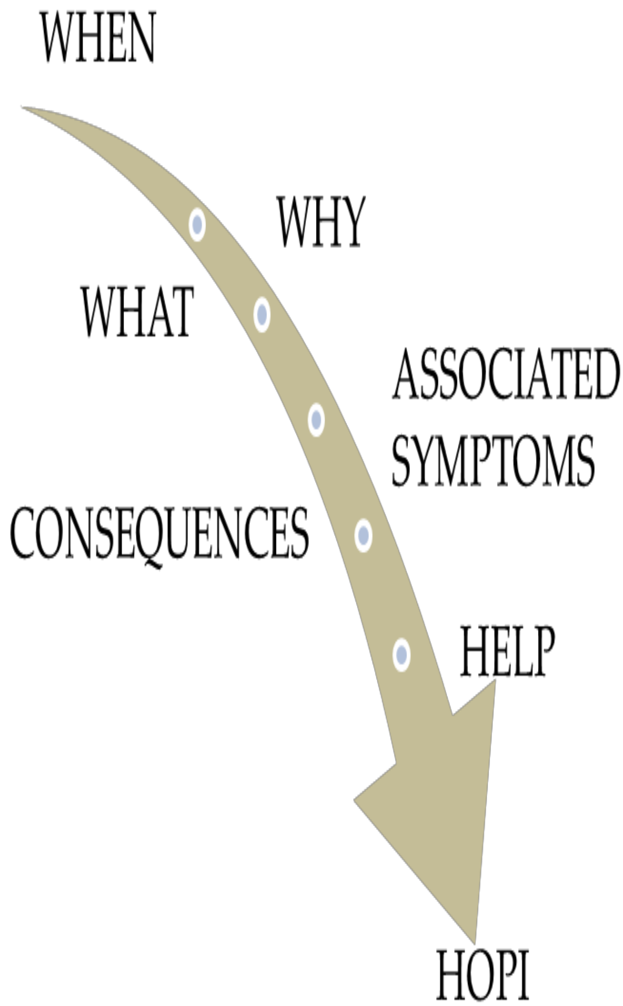

Fig 1-2: Sequel of HOPI

Past psychiatric history: (tables 1-1 and 1-2)

The best place to record past psychiatric problems, if any, is immediately after completing HOPI. Because past history, frequently, influences the current presentation. Further, it's not infrequent for the recent presentation to be the latest episode of the past experiences.

Most junior psychiatrists and trainees find the differentiating period between past-history and HOPI, more or less, confusing! Especially when they interview patients with chronic psychiatric disorders like 'schizophrenia' or 'mood disorders'. In other occasions, the interviewer might be misled by the patient, when s/he attempts to narrate the entire experience in a single chronicle.

Nevertheless, for detecting the time-zone border between past and present experiences, it's worthwhile to ask the patient "when did you feel well/healthy last time? And for how long?"

Past psychiatric history guides you in: tuning the diagnosis of the current presentation, thinking about adjusting which kind of therapy does fit the patient, as well as helping you in forecasting the outcome (prognosis) of the most recent episode.

What kind of past information to ask over? It is indispensable to record the quality and quantity of experiences/disorders, sorts of therapies received by the patient like pharmaco-, psycho-, or electroconvulsive-therapies, extends of effects as well as side-effects of therapies, and history of suicidality.

Additionally, the interviewer should not forget recording non-recognized instances. This is factual, relatively, in milder episodes of 'depression' when patients never attempted to seek help. Also, constantly remind yourself to assess the in-between remission periods, stressing more on the level of socio-occupational functionality.

Medico-surgical history

List all major physical illnesses and treatments chronologically. Stress more on history of: epilepsy, migraine, diabetes, thyroid dysfunctions, cardiovascular disorders, jaundice, nervous system infections, and head injury. Don't forget to record past and current medicines the patient might receive; especially, hormonal- and steroid-like substances.

Family history: (table 1-3)

For the first degree relatives (parents, children, siblings, and spouse), you have to document: Demographic data (see above), social circumstances, quality of relation with the patient, major health problems, history of psychiatric disorders, and history of alcohol or illicit drugs misuses. Don't forget to look at history of psychiatric disorders among more distant relatives.

Table 1-1: Past psychiatric history*

		Occasions							
		1	2	3	4	5	6	7	8
Approximate date									
Diagnosis (experience)									
Received Treatment?	No								
	Yes								
If yes:	Where								
	What								
	Benefits								
	Side-effects								
Suicide									

*note: you can add/delete 'occasions' column as its pertinent.

Table 1-2: Past psychiatric history/ an example

		Occasions			
		1	2	3	4
Approximate Date		Jul 2005	Sep 2007	Feb 2008	Apr 2010
Diagnosis (experience)		Depression	Major Depression	Major Depression	Resistant Depression
Received Treatment?	No	**X**			
	Yes		**X**	**X**	**X**
If yes:	Where		Clinic	Home	Hospital
	What		Psychotherapy	Drugs	Psychotherapy Drugs ECT
	Benefits		Moderate	Moderate	Optimal
	Side-effect		Costly	Sedation	Amnesia Weight gain
Suicide		Ideas	None	Ideas	attempt

Table 1-3: Family History*								
	Father	Mother	Children		Siblings		Spouse	Others
			#1	#2	#1	#2		
Name								
Age								
Education								
Occupation								
Address								
Marital status								
Social circumstances								
Relation with the patient								
Major health problems								
Psychiatric history								
Alcohol/drug misuse								
Notes								

*note: you can add/delete 'children' and 'siblings' columns as it's pertinent.

Personal history

This history aims to understand the patient in more detail as an individual phenomenon. It, also, helps the assessor to understand why (etiology) the patient suffers from the current problem. The interviewer has to focus on certain aspects of personal profile of the interviewee by utilizing close-, yet non-leading, questions; otherwise, the patient might make a scene of a lengthy life story.

We can divide the personal sketch of the patient to:

A. Childhood: record the following:
   1. Pregnancy and mode of delivery.
   2. Any problems around birth.
   3. Delays in milestones.
   4. Any early mal conduct or unusual temper.
   5. Early relations to the rest of family members or caregivers.

B. School age: report the details of primary and secondary school era:
   1. Age of entrance.
   2. Achievements and qualifications.
   3. Relations with peers and teachers.
   4. Any history of: school phobia, truancy, bullying, etc…

C. Educations: report details of:
   1. All qualifications.
   2. Achievements.
   3. Did s/he enjoy his/er study.
   4. Major difficulties s/he faced.

D. Occupations: list the following:
   1. Each job s/he employed chronologically.
   2. Periods of unemployment.
   3. Job satisfaction.
   4. Reasons of leaving jobs.
   5. Other difficulties at work.
E. Relationships: stress on quality and quantity of the listed social relationships:
   1. Family.
   2. Friends.
   3. Colleagues.
   4. Spouse.
   5. Children.
F. Sexual history: search for the following aspects of sexual profile:
   1. Age of menarche/ puberty.
   2. First sexual experience, including masturbation.
   3. Sexual orientation and preferences.
   4. Any sexual difficulties.
   5. Any occasion of past abuse.
   6. Sexual offences which should be spared to the 'Forensic History'.

      However, be careful in knocking such a sensitive and private aspect of your patient. Should you judge the client not to be ready, or would not accept talking about this confidential side, for instance with over religious Muslim elderly women, then never hesitate to bridge over to the next station of evaluation.
G. Leisure and habits:

   These two terms, frequently, misheard interchangeably. Leisure activities are the free-time, non-professional, pleased activities like reading, sporting, playing music, etc… But, Habits are systematic repeated semi-automatic behaviors that help people cope with different life situations inattentively. Habit may be seen as a 'social' synonym to 'misuse and dependence'. Any kind of dependence (addiction) starts with a pleasurable act (Leisure), which by time turn down to a desperate addiction (Habit). Any leisure activity has the chance to cross over to the evil island of 'habit'.

   Best instances of 'habitual' leisure are: nicotine, illicit drugs and substances, and alcohol uses, which by time may turn down to severe dependence.

However there are several leisure activities which have less likelihood to be misused. Should still do, they have less negative health impacts, like sport, playing music, reading, etc…

When assessing leisure, don't be confused by wishes and fantasies. For instance, a client may tell you "I like reading books in my off-times", but s/he might never did read a book. Do not forget to list all the sorts and extents of leisure and habits.

H. Forensic history: record all offences and crimes whether convicted or not.

I. Current social circumstance: ask about the triad: finance, housing, and social support from present family scene.

Pre-morbid personality

Personality is the distinctive and characteristic pattern of thoughts, reactions, attitudes, and behaviors which contours a unique outline of an individual.

Here, you have to record the usual, pre-morbid, personality traits rather than the illness-modifying characters. It's not strange if you come across kinds of confusion from patient's understanding when s/he may describe the disorder-flagged prospect. For instance, an otherwise 'cheerful' 'social' and 'optimistic', yet, currently depressed patient may point out his/er personality round to the opposite side!

Pre-morbid personality helps the psychiatrist in thinking about predisposing etiological factors, for example those who are paranoid may be more prone to psychotic disorders, while those who are perfectionists more towards obsession or depressive disorders and so forth. Moreover, knowing underlying personality guides the therapist toward better management and offers clues about prognoses of the present clinical disorder.

Best way to assess personality is to ask "how would people who know you, describe you in your best day" or "would you describe your characters in your best day before the current problem?"

It is astute, however, to inquire patient's personality from one of the companions as well.

It's not necessary to label the personality of the patient like 'Paranoid', 'Schizotypal', 'Histrionic', etc…. Instead, adopting descriptive traits like 'Sensitive/Mistrustful', 'Odd/Eccentric', 'Dramatizer/Emotional', etc…respectively may be fairly sufficient.

Examination

Prior to attempting physical examination which is analogous to the rest of medicine, psychiatrists ought to look at the current 'mental state' of their patients. This inspection called **Mental State Examination**.

Mental state examination (MSE)

MSE is the most difficult part of assessment for students; though the most straightforward, undemanding, and already done piece. Majority of MSE already passed over while you questioned the patient in 'history taking'. What distinguishes MSE from 'history taking' is the fact that now, in MSE, you have to record what you perceive in front of you. Whereas in 'history taking', you got through patient's anecdote from the 'past' to 'present'.

In other words, you should implement 'present tenses' while recording MSE, but both 'past and present tenses' for 'history taking'.

Unlike 'history taking', in MSE records your observation using 'Technical' terms like 'Hallucination' or 'Delusion'. Nonetheless, try to avoid the term 'Normal' until it relays your approach. Likewise, avoid negative expressions akin to 'NO Hallucination', 'NO Delusion', etc…except for exceedingly noteworthy 'negatives' like 'NO current suicidal thoughts' for example.

Overall, in MSE, barely communicate what you 'detected' beside remarkable 'absents' to your colleagues and tutor.

Appearance and behavior

This very early step of MSE is, comparatively, analogous to 'Inspection' in medicine. However; now, you are in charge to 'inspect' the mental state of the client from his/er outer shell.

The difference between appearance and behavior is, relatively, equivalent to that between a photograph and a movie.

A. Appearance: observe the following aspects:
   1. Consciousness: it's not mandatory to record Glasgow Coma Scale (GCS) here. Rather, an overall evaluation of the level of consciousness is enough which may range from drowsy to hyper aroused. GCS can be assessed later, if necessary, under 'neurological examination'.
   2. Body physical appearance: this may include body built, cleanliness, tattoos, evidences of weight loss or self-harm, and any apparent medical illness.

3. Eye-to-eye contact: This may vary from appropriate to poor, odd, inappropriate contacts.
4. Dress: this may be outlandish, eccentric clothing, dirty, neglected, unsuitable to weather, or extraordinary. Sometimes, patients may cover their self-injured arms by wearing long shirts or gloves. Also, it's not unusual for patients with 'anorexia nervosa' to mask their weight loss by dressing extra-layers of clothes.

B. Behavior: observe whether the patient is agitated, retarded or own some obsessive-compulsive, stereotyped, or any other odd movements. Also trace his/er attitude toward examination which may be friendly, apprehensive, hostile and so forth. Rarely, there may be what is called '**Echopraxia**' in which the patient repeats what the examiner does in front of him/er. Echopraxia is one of the rare symptoms of 'Catatonia'.

Speech

Speech is one of the means of assessing the 'form of thought' (see below). However, here you are in charge to assess the abnormalities of the 'mechanic' of speech per se. Should any oddity in speech reflects disorder of thought, the interviewer has to record it elsewhere under the section of (thought).

Likewise, don't record neurologically-determined abnormality of speech production here like 'dysphasia', 'stammering', 'dysarthria' and so forth. Adjourn these observations to the section of (physical examination).

Instead, record the overall psychologically-determined properties of speech right here. For instance, the following aspects might guide you about what to look after while assessing the speech of the patient:

A. Spontaneity: in depression for example, the patient may only speaks in response to questions. In mania, however, the patient possibly spontaneously talks about different topics.
B. Speed: could be 'pressured' in mania or 'retarded' in depression.
C. Loudness: relatively 'low-volume' in depression and 'loud/high-volume' in mania.
D. Quantity: saying too little in 'depression' or too much in 'mania' or 'alogia' which is the poverty of speeches content. Alogia, indeed, is limitation of the quality and quantity of the speech of the patient. The patient replies to questions via very brief and only handful words. For example, when you ask the patient an

open-end question, the reply may be only by one or few words (brief).

You may observe an additional, yet relatively rare, speech abnormality called '**Echolalia**' in which the patient repeats what the examiner asks him/er. E.g. "what is your name?" the patient would reply "what is your name?" this may happen in 'catatonia'.

Mood and affect

Despite the continuous debate about the difference between 'mood' and 'affect', medical students may find it easier to consider 'mood' as 'the subjective, more sustainable, longer-lasting emotion that is less prone to external and internal stimuli; whereas 'affect' is 'the objective emotion that observed by the assessor'. Affect, unlike mood, may rapidly vary over shorter periods of time and it's easily influenced by both external and internal stimuli.

It's not necessary for the 'affect' to reflect 'mood'. For instance, you might appear happy while you are, relatively, feeling low.

When you are recording 'mood', you have to question the patient about his/er subjective feelings. However, you have not to forget your own assessment of the patient's mood.

A. Affect can be assessed over the following headings:
   1. Type: includes; anxious, sad, happy, disgusted, angry, detached, etc...
   2. Stability: e.g. labile, indifference, and blunted affects:
      a. Labile is when the affect shifts rapidly from one pole to another in a very short time. Differential diagnoses of labile affect include organic brain conditions like dementia, during an extreme stress, hypomania, drunkenness, or mixed affective state.
      b. Indifference: it's named, sometime, 'belle indifference' when there is masked emotion. The patient appears to be careless about what happens to him/er. Main differential diagnoses are Conversion Disorder (Hysteria), intoxication, or frontal lobe damage.
      c. Blunted (flat): Flat affect means limited expression of emotion. For instance, the patient may reach a state that you cannot conclude any emotions from his appearance. You cannot judge whether he is happy, gloomy, irritable, etc…Further, the patient may reach the state of blunted affect (no emotions at all) (dead pan face). [Technically, **flattening** refers to a loss of ability to feel- or at least to express emotions and, **blunting**, to a reduced sensitivity

to others. However, this distinction is not clinically relevant]. Differential diagnoses include schizophrenia, severe depression, Parkinson's disease, or sometime hypothyroidism.

B. Mood can be assessed over the following headings:
1. Type: includes; depression, hypomania, and mania (chapter 3), or anxiety and irritability which may occur in several psychiatric morbidities as well as in normal people. 'Alexithymia' is another mood condition describes the inability to feel or explain any kind of mood. This can occur in stroke, post-traumatic stress disorder (PTSD), or in schizophrenia. 'Euthymic mood' is used when mood seems to be within normal range.
2. Intensity/severity: this ranges from mild, moderate, severe, to severe with psychotic symptoms.
3. Congruity: In average (not mentally-ill) people, mood goes parallel to thought. For instance, if someone's thought and opinion are of optimistic nature, the mood would be high too, and if the thought is pessimistic, the mood would be down as well. For delusional thoughts (see below), the above congruency may occur also, for which it called **Mood-congruent delusion**. E.g. someone with grandiose delusion and elevated mood; another with delusion of guilt and depressed mood; another with delusion of persecution and anxious and irritable mood.
   However, it's possible for the delusional idea to be **Mood Non-Congruent**. Example may be a patient with grandiose delusion and depressed mood. Mood Non-congruent delusion amounts more toward the diagnosis of schizophrenia and schizophreniform disorders. While Mood-Congruent delusion more toward the diagnoses of Mood Disorders, Delusional Disorders, and Schizo-affective Disorder.

Thought

Thought is usually assessed through patient's speech and to a lesser extend via patient's writing or behavior. There are two components of thought: the content and the form. Each one has to be assessed by the interviewer.

A. Thought's content: the contents of any thought are group of ideas. These ideas might deviate from normal average thinking of mankind. The deviation should be assessed against patient's socio-cultural and educational backgrounds, as well as whether such a deviation complicated by socio-occupational impairment, or/and distresses to the patient or to his/her surroundings.

Abnormalities of thought content may be quantitative or qualitative. Quantitative abnormalities range from 'overvalued idea' to 'phobia' or 'obsessional idea'. Whereas qualitative abnormality of thought content includes, exclusively, 'delusional ideas'.

Quantitative abnormalities amount toward 'anxiety- and mood-related disorders', whereas qualitative deviation amounts, rather, toward 'psychotic disorders'. However, it's not exceptional for psychotic disorders to present with an associated quantitative changes as well.

What is Delusion? Delusion is, relatively, **fixed**, and **false** beliefs. **Fixed** means the idea cannot be changed by reasoning (unshakable). **False** means the idea is inconsistent with the cultural and educational backgrounds of the patient. Yet, delusions may have a culturally based content. For instance, Kurdish people who believe that they are the target of influence by jinni! However, Jinni possession might be a culturally acceptable idea due to illiteracy. Nevertheless, such an idea, which is **false**, is subjected to modification through debates, arguments, and education in healthy minded people, i.e. it's shakeable and **not fixed**. However, such believes remains consistent and unshakable among psychotic people. Therefore, both poles, **fixed and false**, are necessary for the definition of **Delusion**.

Types of delusion: Delusion can be classified according to:

1. **Time of occurrence** to (Primary vs. Secondary delusions):
   a. **Primary delusion** is a delusion that exists without any preceding psychopathology. Examples of primary delusion are:
      i. **Delusional perception** is giving delusional meaning to normally perceived stimuli. E.g., 25 year-old man thought his friend accuse him to be homosexual, because he offered him a salty biscuit. Here, the biscuit is a normally perceived stimulus; though, his interpretation is delusional.
      ii. **Sudden delusional idea** is any delusional theme (see below) which develops completely and suddenly in the patient's mind, out of blue.
      iii. **Delusional mood:** the subject feels his familiar surroundings have changed in a puzzling way which he

may be unable to describe, though seems to be especially significant for him.

b. **Secondary delusion** is a delusion which is preceded by another psychopathology, as a depressed patient who developed belief that he sinned greatly. Here the depressed mood is PRIMARY, and the delusion of sin is SECONDARY. The delusion might be secondary to another psychotic symptom, like hallucination.

2. **Nature of the delusion** to (Bizarre vs. Non-Bizarre (Systematic) delusions):
   Delusions said to be **bizarre** if they are clearly implausible, not understandable, and not derived from ordinary life experiences. For instance, an individual's belief that a stranger has removed his internal organs and replaced them with someone else's organs. In contrast, **non-bizarre** delusions involve situations that can conceivably occur in real life as being followed, poisoned, or deceived by spouse. Bizarre delusions more account for schizophrenia, while Non bizarre delusions may occur in many disorders, like schizophrenia, delusional disorders, and mood disorders.
3. **Mood congruency** to (Mood-congruent vs. Mood-non-congruent delusions):
   (See above).
4. **Content (theme) of the delusional idea** to:

a. Problem with surrounding (others) **(I am good, others not!)**: examples are

i. **Delusion of persecution:** others conspired him/er.

ii. **Delusion of reference:** others refer to, or talk about him/er.

iii. **Passivity phenomena (Passivity delusions) (delusion of Control):** others control his/er mind and/or body. Examples include:

iv. **Thought withdrawal:** others withdraw his/er thoughts.

v. **Thought insertion;** others insert thoughts not one's own.

vi. **Thought broadcasting:** whatever s/he thinks about is broadcasted via radio, TV, internet, etc…

vii. **Made affect:** one's feelings and emotions are made by others.

viii. **Made impulse:** the same.

ix. **Made will (volition):** the same

x. **Somatic passivity:** others, jinni, aliens, Mafias, and so forth control his/er body organ, replace, move parts, etc…

xi. **Grandiose delusions:** I am superior to others. This includes:

- **Delusion of grandiose ability:** the subject thinks s/he is chosen by some power or by destiny because of his/er unusual talents. Hence, the patient may think that s/he is a talented artist, scientist, etc…
- **Delusion of grandiose identity:** the patient believes he is famous, rich, titled, from royal decent, etc…
- **Religious delusion:** patient feels he is saint, prophet, or even God. **NOTE:** Grandiose ability amounts more towards manic episode, while Religious one towards schizophrenia.

b. Problem with self **(I am bad/poor, others not!)**

i. **Delusion of sin and guilt:** patient thinks s/he is guilty of any minor or probable wrong or of some misfortune happens elsewhere.

ii. **Nihilistic delusion:** the person believes that part(s) of his/er body, the family, or even the entire world has been disappeared or destroyed.

iii. **Delusion of poverty:** patient feels that s/he is poor, knows nothing, or incompetent despite evidences of contrary.

NOTE: These delusions are more in Depressive disorder with psychotic feature.

c. Other themes: like Delusion of love, delusion of illness, etc...

What are the Non-delusional, quantitative, abnormalities of thought content? Non-delusional abnormalities of thought content differ from delusion in that the person still hold insight for what s/he is suffering from. These include:

a. **Overvalued ideas** are strongly held, govern the thoughts but are not necessarily illogical or culturally inappropriate. Because of the associated feeling tone, overvalued idea takes priority over other ideas consistently for long period of time. It may occur in themes similar to that of delusion (see above).

b. **Phobia** is an exaggerated, irrational fear from particular object or situation which occurs only on exposure to or anticipation of the fearful stimulus. The fear is clearly recognized by the person to be unreasonable (see relevant chapter).

c. **Obsessions** are ideas, images, or memories persistently and recurrently intrude to the patient's mind. The patient recognizes the senselessness of these obsessional themes and continuously tries to resist them. However, both obsessions and patient's resistances are sources of distress to the patient.

B. Thought's form: The form of any thought refers to the coherence and logic structure of that thought - the 'how' or 'mechanics' of thinking- which makes the conversation easy to follow. The form of thought is manifested through its expression: by observing the speech, writing, or sometime the odd behaviors of the patient. Examples of formal thought disorder (FTD) include:

1. **Circumstantiality** is an over-inclusive, extremely-detailed answer to the question. When occurred alone, it doesn't amount to a specific pathology. Instead, it may be kind of personality trait or an elevated mood.
2. **Tangentiality** is replying to a question in an oblique manner. The answer may be related to the question in some distant way. For example: "where are you from? Well, you know, Erbil is a big city; now the city became more multicultural than before, so far, I don't feel much uncomforted…"
3. **Loosening of thoughts' association:** Loosening means the associations between subsequent ideas become more fragile. It may range from mild **Derailment**, to more **Knight Move Speech**, to an extreme **Incoherence** of speech. **Derailment** means gradual deviation of patient's talk from the track so as, at the end of his talk, the examiner may get lost!! **Knight Move speech** means the answer of the patient suddenly shifts to a new track, in 90 degree angle, merely like chess's Knight Move. Sometimes, the patient's talk may be incomprehensible due to an extreme loosening **(incoherence)**, incoherence of speech may reach to the stage of what is named **'Word salad'.**
4. **Flight of ideas** is when the patient's speech jumps rapidly from one topic to another. Its almost always associated with pressured talk. Flight of ideas is strongly suggestive of manic episode of mood disorder.
5. **Perseveration** means when the patient's thought or behavior jammed and would not shift on to different topics. It occurs mainly in organic brain disorder, but sometime may occur in schizophrenia or depression too.
6. **Thought blocking** is when the train of thoughts stops, suddenly, for seconds and up to minutes. The patient, then after, can't remember what s/he was trying to say.

Perception

Human being has various sensory systems. Those well known to common people are: Auditory, Visual, Olfactory, Tactile, and Gustatory systems. Yet, many other sensory modalities do exist, like pain, balance, temperature, etc…

These sensory systems work like Camera's lens, receiving and capturing scene from environment. Therefore, normally, there should be an external stimulus for our body Camera lens to work out.

Perception is the mind's interpretations of our body's cameras inputs; i.e., interpretation of external stimuli. There are two main abnormal perceptual conditions; illusion and hallucination. **Illusion** is the misinterpretation of actual external stimuli. Its when our mind misguide us; for instance, a tree might looks like a human shadow in a dark night. Illusion may occur in an exhausted and anxious but otherwise normal people. Common abnormal presentations of illusion are in delirious patients or during alcohol or substances intoxications.

Hallucination, on the other hand, is perception without actual external stimuli. Its when our mind delivers out scenes without its reliance on body's cameras. Hallucination may be experienced by normal people as in 'dreaming', at the onset of sleep 'hypnogogic hallucination', or when someone wakes up from sleep 'hypnopompic hallucination'. However, 'abnormal' hallucination does exist in several organic as well as functional morbidities. There are parallel hallucinations to each single sensory modality; for instance, auditory, visual, tactile, gustatory, olfactory hallucinations and so forth. Apart from auditory hallucinations which are prevalent in psychotic disorders like schizophrenia, other modalities' hallucinations are more amount to non-functional organic neuropsychiatric conditions like delirium, dementia, epilepsy, migraine, brain tumor, and so forth. Nevertheless, their existence does not rule out functional psychiatric disorders.

Examples of common auditory hallucinations in psychotic disorders are:

A. **Second person auditory hallucination:** the heard voice directly addresses and talks to the patient.

1. **Running commentary:** A voice commenting on patient's thoughts, attitude, behavior, etc…For example a voice says "you are failure".

2. **Commanding (ordering) voices:** Voices command particular behavior from the patient which might be dangerous like ordering the patient to kill oneself or others.

B. **Third person auditory hallucination:** two or more voices talking with each other. This kind of hallucination might convey 'commentary' or 'commanding' messages to the patient, moreover.

C. **Thought echo:** is hearing one's thoughts spoken aloud.

Cognitive functions assessment (CF)

Cognition in mental state examination refers to brain's information-processing like awareness, attention, concentration, and memory. Cognitive functions assessment deems the most difficult aspect of patients' assessment to undergraduate students. However, should you could not detect obvious signs of CF during your history taking, then you would be in charge to, only, record few basic tests described below.

It would be much better if you try to introduce the CF tests to the patient before starting so as to avoid unfamiliarity or embarrassment by patient's side. For instance; "There are few questions we have to ask everyone to test concentration and memory. They should not take long time if you do not mind. Are you happy with this please?"

A. **Level of consciousness:** It's the person's ability of awareness to oneself and surrounding. Consciousness is essential for other brain's higher and lower functioning. When it is impaired, then all other brain functions might be impaired as a consequence.

   If there are no clear evidences of impairment, then you would not ask for formal assessment. Only you need to describe the level of consciousness been observed by you.

   Should there be an evidence for even mild impairment, it's advisable to step forward toward more formal assessment of consciousness, for example applying 'Glasgow Coma Scale (GCS)'.

B. **Orientation:** This is one's awareness to time, place, and person in sequence. Orientation to time includes awareness to the passage of time as well, and it's the most sensitive sort of orientation. Hence, if a person became disoriented, then 'time' is the first one to be impaired and 'person' is the last and hardest one to be lost. In other words, if a person was disoriented to persons, ultimately s/he would be disoriented to time and place as well. Notwithstanding, it's possible for a patient to be disoriented to time, but not place and persons, or disoriented to both time and place but preserving his/er orientation to persons.

Disorientation commonly accompanies impairment of consciousness and it is more amount toward organic syndrome like 'Delirium'.

1. Time's disorientation: "would you tell me what time it is without looking at your watch?", "Do you know what date/month/season/year we are in?"
2. Place's disorientation: "can you tell me where we are now?", "could you tell me the name of this clinic/hospital/place?", "what city/region/country are we in?"
3. Person's disorientation: this can be assessed since the start of the interview by asking about patient's own name or to identity an accompanied relative or friend.

C. Attention and concentration: Attention is the ability to focus on a stimulus or duty. Best example is when a teacher calls out your name in the class and asks you a question, you would continue to keep attention to answer the question until the teacher picks up another student's name.

Concentration is the capability to maintain attention for a period of time. Both attention and concentration are assessed together by the same question during the interview.

Best way to assess is to ask a 'reverse' sequence of a particular theme like: "can you tell me in a reverse order the days of the week/the months of the year?", "would you count down from 20 backwards to 1?", "could you spell out the letters of 'Kurds' both for- and backwards?"

More difficult question would be what is known as 'Serial Subtraction'. For example: "can you subtract 7 from 100 downwards to 2?", or "can you subtract 3 from 20 downwards to 2?" Should the patient do it correctly, then stop him/er after 4-5 subtractions.

For all questions, it would be better if you attempt to explain to the patient the question by solving an example prior to the patient's one.

D. Memory: Classification of memory witnessed some divergent views by different authors. Many authors divided memory to 3 types; immediate, short-term, and long-term memories. Others, however, replaced the term 'immediate' with 'sensory' memory.

From clinical point of view, it is more convenient to classify memory to 2 kinds; Short-term memory (immediate recall), and Long-term memory (delayed recall).

Short-term memory is the kind of memory which preserves information for about 30 seconds. Best mean for assessing this sort of memory is to mention 3-5 'random' items, like numbers, name of place or person or food, etc… clearly in front of the patient taking in account that you informed the patient in advance that you will ask him/er to repeat these names immediately after you both for- and backwards.
Short term memory has 6-7 digit spans; i.e., it has the ability to maintain 6-7 items at once.
Long-term memory, on the other side, is any kind of information which kept after few minutes from shifting to a new task. Best way to assess this class of memory is to ask about previous items have been asked in short-term memory assessment after about 5 minutes. However, you have to inform the patient that you are going to ask him/er after five minutes in advance. Thus, its possible to asses both varieties of memory by the same test.
Another kind of long-term memory assessment is by asking about some sort of general knowledge like "what is the name of current president?", "what was the highlight of newsletters for the last 2 weeks?" and so forth. You can modify these questions according to the cultural and educational background of the patient.
E. Insight: It is patient's awareness to what s/he is suffering from. It remains intact in most psychiatric disorders. However, insight might be, comparatively, impaired in psychotic disorders, few kinds of personality disorders, dementia, and some other organic brain conditions.
When the insight is preserved, the cooperation between the therapist and the patient would be more likely. While assessing insight, you have to keep in mind that it is not matter of whether insight is intact or lost. Rather, the student and therapist alike are in charge to objectively 'describe' the insight through asking the patient the following questions:

1. Is the patient aware that there is something wrong?
2. Is the problem within the patient or external?
3. If there is something wrong, does the patient think it is a result of a disorder?
4. If it is a disorder, is it physical or psychological?
5. If it is psychological, can it be helped?
6. Is the patient willing to accept help? If so, what kind of help does s/he accept?

Final Steps

Before your decision(s) on the client's therapeutic fate, it is worthy to double check the following:

A. **Your reactions or feelings toward the patient:** Your feelings might guide you toward more accurate diagnosis. For instance a manic patient make you laugh and happy, depressed patients may turn you down too, psychotic patients may confuse you, and sometimes, patients with personality disorders might make you feel uncomfortable or even angry. Nevertheless, never depend solely on your rapport.

B. **Summarize your main findings.**

C. **Ask a final question:** "do you have something else to tell me I forgot to ask for?"

D. **Physical examination** including neurological examination as it is appropriate.

02 PSYCHOTIC DISORDERS

What is Psychosis (Psychoses-Pl)?

In order to make sense of what does meant by psychosis, it's necessary to recognise the meaning of REALITY from psychiatric point of view.

Reality: in everyday usage means 'the state of things as they actually exist'. The term *reality*, in its widest sense, includes everything that is whether or not observable or comprehensible.

In psychiatry, however, reality referred to how an average human being does perceive the subsistence through:

A. His/er sensory modalities (Auditory, Visual, Gustatory, Olfactory, and Tactile sensory modalities). *Perception*
B. How does s/he think, explain, or rationalize what happens in his/er surroundings. *Thinking Processes*
C. And how s/he behaves and reacts in response to his/er surroundings. *Behaviors and emotions*

Psychosis

The word *psychosis* was first used by Ernst von Feuchtersleben in 1845 as a substitute to 'insanity' and 'mania'. It was used to distinguish disorders which were thought to be disorders of the mind opposite to 'neurosis', which was thought to stem from a disorder of the nerves (table 2-1).

Table 2-1: Differentiation between psychosis and neurosis:

#	Psychosis	Neurosis
Definition	Disorder of mind	Disorder of nerve
Changes occur	Are qualitative	Are quantitative
Psychopathology	Thought processes, and perceptions mainly	Emotion mainly
Symptoms	Delusions, FTDs, Hallucinations, and lack of insight	Anxiety and/or depressive symptoms
Examples	Schizophrenia and other Psychotic Disorders	Anxiety Disorders, Obsessive-Compulsive Disorder, Mood Disorders.

Psychosis (from the Greek ψυχή 'psyche', for mind or soul, and -ωσις '-osis', for abnormal condition), literally stands for abnormal condition of the mind. In psychiatry, however, psychosis can be defined as a mental state characterized by the 'loss/Impairment of contact with reality'.
Acknowledging that we can stay in contact with reality through our perceptions, thinking processes, and our behaviors and emotions; loss of such contact (with reality), clinically manifests through abnormalities of one or more of our perceptions, thinking processes, and behaviors and emotions. Therefore, psychotic symptoms comprise: Hallucination, Delusion, Formal Thought Disorder, Abnormal or Odd emotions, and/or Abnormal or Catatonic Behaviors (see chapter 1). An individual might judge to be 'Psychotic' should s/he suffers from one or more of the above abnormalities.
Additional important feature of Psychosis is the 'lack of Insight'. What is unreal for us, is reality for the patient; i.e. the hallucinations and delusions are reality for the patient, and the patient would not regard them to be products of his/er mind.

Differential diagnoses of psychosis
Whenever one or more psychotic features were evident, conditions displayed in (Box 2-1) might be the underlying possibilities.

Schizophrenia Spectrum and Other Psychotic Disorders include five main clinical conditions; Schizophrenia, Schizophreniform Disorder, Brief Psychotic Disorder, Schizoaffective Disorder, and Delusional Disorders. Nevertheless, two, rather shared, additional conditions are further classified with this category; Catatonia, which may couple several psychiatric disorders, and Schizotypal Personality Disorder, which more recently recognized to be a common disorder between psychotic as well as personality disorders.
Moreover, similar to any psychiatric disorder, DSM recognized additional classes of psychotic disorders secondary to substances/drugs and/or medical conditions.

Box 2-1: Differential diagnoses of psychosis

A. Normal events in healthy subjects:

(NOTE: at least10-20% of general population may experience individual psychotic symptoms at some point in their lives, especially as adolescents, but do not develop a psychotic illness).

1. Hypnagogic Hallucination; hallucinations while falling into sleep.
2. Hypnopompic Hallucination; hallucinations while awakening from sleep.
3. Dreaming while sleepy.
4. Cloudiness in perception; like hallucinations or illusions in dark night.
5. Extensive fears or emotions might lead to hallucination in an otherwise healthy subject.
6. Expectations of the occurrence of some events may lead to relevant hallucinations or illusions.
7. Bereaved subjects.

B. Medical conditions:

1. Neurological disorders:
   a. Head Injuries
   b. Brain tumors
   c. Infections *(encephalitis, meningitis)*
   d. Epilepsy, *specially Complex Partial (Temporal Lobe) Epilepsy*
   e. Systemic Lupus Erythematosus
   f. Multiple Sclerosis
   g. Parkinson Disease
2. Medical disorders:
   a. Endocrinal dysfunctions *(Cushing Syndrome, Thyrotoxicosis, Hypoglycemia)*
   b. Electrolyte disturbances (*Hyper/Hypo Ca, Na, K, Mg)*
   c. Vitamins deficiencies
   d. Infections (like *HIV)*
3. Organic Psychiatric Conditions
(Neurocognitive disorders and Delirium)

C. Substances/Drugs induced:

1. Alcohol
2. Illicit substances/drugs *(Amphetamine, Cocaine, LSD, Cannabis, etc...)*
3. Prescribing pills *(Anti-Parkinson drugs (L-dopa), Bromocriptine), (Anticholinergics (Benzhexol, Atropine)), (Benzodiazepine withdrawal), etc…*

D. **Functional psychiatric disorders:**
1. Schizophrenia
2. Schizophreniform Disorder
3. Brief Psychotic Disorder
4. Schizoaffective Disorder
5. Delusional Disorders
6. Mood Disorders (Depression or Mania) with Psychotic Features
7. Some Personality Disorders may present with temporary psychotic features, like: Paranoid, Schizoid, Schizotypal, and Borderline Personality Disorders

Schizophrenia

The term 'Schizophrenia' introduced to psychiatry at 1911 by Eugen Bleuler (Swiss Psychiatrist).The term schizophrenia comes from the Greek words σχίζω (schizo, split or divide) and φρενός (frenos, mind) and it is best interpreted as a "shattered mind". Therefore, Schizophrenia, is the "splitting of the mind" rather than the popular mistranslation as "splitting of the personality".

Historical background

Recognizable descriptions of schizophrenia are considerably less common in historical medical texts or literatures than those of 'melancholia' or 'mania'. The earliest clear descriptions dated back merely to the end of the 18th century.

In 19th century, one view was that all serious mental disorders were expressions of single disorder which Griesinger called *Einheitpsychose* (unitary psychosis).

The alternative view, which was put forward by Morel in France, was that mental disorders could be separated and classified. In 1852 he gave the name *démence précoce* to a disorder starting in the adolescence and leading first to withdrawal, odd behaviors, and self-neglect, and eventually to intellectual deterioration.

Emil Kraepelin (1855-1926) divided mental illnesses into two types depending on the course and prognosis; 'Dementia Praecox' (Schizophrenia), which runs chronic progressive course, and 'Manic-Depressive Insanity' (Bipolar disorders), which runs a waxing and waning course, with relatively better prognosis.

Eugen Bleuler (1857-1959), however, was concerned less with the prognosis, but more with the mechanisms of symptom formations and psychopathology. Bleuler proposed the name schizophrenia to denote a 'splitting of psychic function', which he thought to be of central importance. He divided the clinical features into two groups (table 2-2).

Table: 2-2: Bleuler's classification of schizophrenia symptoms	
Fundamental Symptoms (the 4 As)	Accessory (Secondary) Symptoms
Loosening of association	Hallucinations
Autistic thinking (Withdrawal)	Delusions
Blunted affect	Catatonia
Ambivalence	Abnormal behavior

Kurt Schneider (1887-1967), unlike Kraepelin and Bleuler, was more concerned with the clinical features. Schneider tried to make the diagnosis more reliable by identifying a group of characteristic symptoms (First Rank Symptoms) of schizophrenia which are rarely found in other disorders (table 2-3).
These symptoms usually present early in the course of schizophrenia and disappear with time. In a study conducted by the author, more than half of first episode psychoses were presented with Schneiderian first rank symptoms. Furthermore, the study concluded that these symptoms co-exist with each other with the priorities to 'Third person' and 'Running commentary' auditory hallucinations as well as 'Passivity phenomena' (Al-Yasiri and Rahim, 2008).

Table 2-3: Schneiderian first and second rank symptoms		
#	First Rank Symptoms	Second Rank Symptoms
1	Thought Echo	Other Disorders of Perception
2	Third Person Auditory Hallucination	Sudden Delusional Ideas
3	Running Commentary Auditory Hallucination	Perplexity
4	Somatic Passivity Experiences	Mood Changes
5	Thought Withdrawal	Feeling of Emotional Poverty
6	Thought Insertion	
7	Thought Broadcasting	
8	Made Impulse	
9	Made Affect	
10	Made volition	
11	Delusional Perception	

DSM-5 criteria and clinical features

Usually there is no classical clinical picture. Symptoms may vary from person to person, and within the same person may vary from

time to time. Age of onset is usually in the adolescence, or early adulthood, but it may occur at any age, particularly from 15-45. The onset is earlier among males.
There are two main groups of symptoms: (table 2-4)
A. **Positive symptoms:** positive means emergence of experiences (symptoms) that do not exist normally. E.g. hallucinations and delusions. Hence, all psychotic symptoms (see above) are regarded as POSITIVE symptoms. Sometimes, psychosis or the positive symptoms named ACUTE symptoms too.
B. **Negative symptoms:** negative means loss of experiences or mental functions that an average human being does have. Example of negative symptoms:

1. **Alogia.**
2. **Flat (blunted) affect.**
3. **Lack of willpower (avolition):** this means the patient may lack the interest to do his own basic needs, lack of enthusiasm, and carelessness. This may be manifested as an extreme self-neglect.
4. **Anergia:** loss of physical and mental energy.
5. **Anhedonia:** loss of interests (similar to depression, though, the mood of the patient is not depression)
6. **Social withdrawal and self-isolation.**

Negative symptoms, sometimes, named Residual symptoms, or Chronic symptoms.
Therefore, we have two groups of symptoms; the patient may suffer from one, or both together throughout the course of the disorder.

Table 2-4: Positive vs negative symptoms of schizophrenia

Symptoms	**Psychosis, Positive, Acute**	**Residual, Negative, Chronic**
Appearance	Odd, disturbed	Apathetic, self-neglect, limited activities
Emotions	Sad, happy, irritable	Flat, or blunted
Thoughts	Delusions + FTD	Alogia
Perception	Hallucination	No clear, or absence of hallucination

DSM-5 criteria for diagnosis

A. Two (or more) of the following, each present for a significant portion of time during a 1-month period (or less if successfully treated). At least one of these must be (1), (2), or (3):
   1. Delusions.

2. Hallucinations.
3. Disorganized speech (Formal Thought Disorders).
4. Grossly disorganized or catatonic behaviors.
5. Negative symptoms.

B. For a significant portion of the time since the onset of the disturbance, level of functioning in one or more major areas, such as work, interpersonal relations, or self-care, is markedly below the level achieved prior to the onset (or when the onset is in childhood or adolescence, there is failure to achieve expected level of interpersonal, academic, or occupational functioning).

C. Continuous signs of disturbance persist for at least 6 months. This 6-month period must include at least 1 month of symptoms (or less if successfully treated) that meet Criterion A (i.e., active-phase symptoms) and may include periods of prodromal or residual symptoms. During these prodromal or residual periods, the signs of the disturbance may be manifested by only negative symptoms or by two or more symptoms listed in Criterion A present in an attenuated form (e.g., odd beliefs, unusual perceptual experiences).

D. Schizoaffective disorder and depressive or bipolar disorder with psychotic features have been ruled out because either 1) no major depressive or manic episodes have occurred concurrently with the active-phase symptoms, or 2) if mood episodes have occurred during active-phase symptoms, they have been present for a minority of the total duration of the active and residual periods of the illness.

E. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition.

F. If there is a history of autism spectrum disorder or communication disorder of childhood onset, the additional diagnosis of schizophrenia is made only if prominent delusions or hallucinations, in addition to the other required symptoms of schizophrenia, are also present for at least 1 month (or less if successfully treated).

Clinical phases

A. Pre-schizophrenia:

   Pre-schizophrenic adolescents may have no close friends, no dating, and may avoid team sports. They may enjoy watching movies and TV or listening to music or playing computer games to the exclusion of social activities. Many times, the adolescent may look shy and sensitive, anxious, always worry, and irritable. Some adolescent patients may show a sudden onset of obsessive-compulsive behavior as a part of the prodromal picture.

B. Prodromal phase:

The presentation is often insidious and hard to date. The person become more withdrawn, doing less, and losing his/er drives. This is reflected in the reduced achievements at school or college, not holding down a job and making fewer contacts with the outside world. During this stage, a patient may begin to develop interests in abstract ideas, philosophy, and the occult or religious questions.

Occasionally, and because of the early primary delusions, especially Delusional Mood, the patient might look confused and doesn't know what happened, or s/he might know that there are some strange issues happened to him/er, yet cannot explain them. Moreover, the patient may appear irritable and anxious due to some early psychotic experiences. Diagnosis at this phase is far difficult, indeed.

C. Acute (Active) phase: (Positive symptoms)

1. Behavior: is broadly variable; may be mute and withdrawn, violent, taking strange postures, or even overactive. They usually neglect their personal hygiene; for instance they bath less frequently.
2. Affect: perplexity, frightened and highly aroused, bemused and bewildered, incongruent affect, or flattening or blunting of the affect.
3. Speech: it reflects patient's thought patterns (Formal Thought Disorders).
4. Thought: delusions of most types can occur but usually persecutory delusion, delusion of reference, grandiosity, religious delusions, and passivity phenomena are the usual delusional themes. Passivity phenomena are specifically characteristic for schizophrenia (i.e., passivity phenomena when present, the most likely diagnosis is schizophrenia).
5. Hallucination: in all five sensory modalities can occur, but auditory type is the commonest type, especially third person and running commentary auditory hallucinations.
6. Insight: Lack of insight occurs in about 97% of cases in the acute phase.

D. Chronic phase: (Negative symptoms)
   In the chronic phase, the features of acute phase will subside by time, and the patient stays with some (negative) or (residual) symptoms (see above).
   Various disorders of movement and behavior occur, additionally, in the chronic phase: like stereotypes, mannerisms, and catatonic symptoms.
   As with acute phase, the symptoms and signs of chronic phase are variable. At any stage, features of acute phase may recur or become exacerbated; this may be in response to life events, or discontinuation of medication.
   Characteristically, more than 90% of schizophrenic patients, in the chronic phase, are nicotine smokers, which may be complicated, in addition to other side effects of smoking, by finger-sites burning or excessive nicotine stains due to self neglects and decreased attention.

Course and prognosis: (table 2-5)
They are rather variable. Some may recover with little or no residual deficits. However, the more common pathway is a chronic running course of exacerbations and remissions throughout the life. Twenty percent attempt suicide; yet, 5-6% do the final act. Co-existence of depression and presence of hallucination, found by the author to be the commonest significant predictors of suicide among a group of Kurdish schizophrenic patients (Rahim and Saeed, 2013).

Epidemiology
Prevalence of schizophrenia is around 3-7/1000 (around 0.6%), life-time risk is around 1%, and the annual incidence is within the figure of 15/100 000. It is distributed equally between both genders; however, the female-to-male ratio is about 9:1 in the late-onset schizophrenic population. Schizophrenia is more prevalent among urban, industrialized, slum areas, among low socioeconomic classes, and those of winter birth dates.

Table 2-5: Prognoses of schizophrenia		
Factor	**Good Prognosis**	**Poor Prognosis**
Age	Late age of onset	Early age of onset
Gender	Female	Male
Occupation	Employed	Unemployed
Address	Rural area	Inner slum city area
Marital Status	Married	Single/Divorced/Separated /Widowed
Onset	Acute	Insidious
Episode	Short	Long
Clinical Features	Few or no negative symptoms	Negative symptoms
Affective symptoms	Present	Absent
Sexual Complaints	Absent	Frequent
Pre-morbid Personality	Stable, well adjusted	Unstable, poorly adjusted
Past Psychiatric History	Negative	Positive
Family history	Negative	Positive
Social Relationship	Good	Isolated
Compliance with Treatment	Good	Poor

Etiology

Till now, the exact cause is unknown. But there are many theories to explain the disorder. The causes of schizophrenia may be due to one or more of the following factors:

A. **Biological factors:**
   1. **Genetic:**
      Twin study shows that the concordance rate is higher in MZ twins (50%) than DZ twins (17%). Moreover, family studies revealed that if somebody is schizophrenic, the life time risk of developing schizophrenia for the relatives will be higher than general population (table 2-6).

Table 2-6: Risk of schizophrenia among family members	
Relationship	**Risk %**
General Population	1
Parents	6
All siblings	9
Siblings (one parent also schizophrenic)	17.2
Children	13
Children(both parents schizophrenics)	48
Grandchildren	5
First cousins	2
Uncles/aunts	2
Nephews and nieces	4

2. **Environmental biological factors;** including:
    a. Obstetrical complications
    b. Maternal influenza and other infections
    c. Rhesus incompatibility
    d. Malnutrition
    e. Winter birth, probably due to prevalent infections at winter
    f. Substance-induced, like cannabis, alcohol, nicotine
3. **Structural brain damage:**
   Prevalence of schizophrenia is higher among subjects with intellectual disabilities (3%), and those with epilepsy. Studies found ventricular enlargement to be evident in many schizophrenic patients, yet nobody can conclude that these changes are causes or consequences of schizophrenia.
4. **Cerebral blood flow:**
   Studies showed that there are hypo perfusions (decreased metabolism) in the frontal lobe of schizophrenic patients, though; nobody can claim that these are causes or consequences yet.
5. **Neuro-chemical findings**; including:
    a. Dopamine: increased dopamine is associated with the positive (psychotic) symptoms. However, decreased dopamine associated with the negative (residual) symptoms. i.e., dopamine has a dual effect.
    b. Glutamate: increased glutamate is associated with the positive symptoms, and its reduction is associated with the negative features.
       Note: the above two findings might explain to some extend why more than 90% of those with long-term schizophrenia do smoke nicotine heavily. This is because

nicotine induces the release of both dopamine and glutamate. Thereby, patients try to overcome the negative symptoms (self-management). On the other hand, excessive nicotine smoking and subsequent increases in dopamine and glutamate secretions are the probable explanations for nicotine as an etiological factor for schizophrenia. In a survey done by the author and a co-investigator, the rate of nicotine smoking was found to be almost the double compared to control group, this is beside that majority of schizophrenic patients started their smoking prior to the illness-onset (Rahim and Muhyadin, 2011).

c. Serotonin: increased serotonin neurotransmission is associated with positive symptoms, this is why hallucinogens and lysergic acid diethylamide (LSD), which are serotonin receptor ($5HT_2$) agonists, do induce hallucinations. Moreover, the new antipsychotics like Olanzapine, Risperidone, etc… are dopamine and serotonin ($5HT_2$) receptors antagonists.

d. Gamma Amino Butyric Acid (GABA): reduction in GABA might be one of the patho-physiological mechanisms of schizophrenia symptoms productions, particularly positive symptoms. Benzodiazepines, which are GABA receptors agonists, are useful in reducing core psychotic and agitation symptoms in schizophrenia, brief psychotic disorder, as well as during the manic episode of bipolar I disorder.

B. Social factors:

Social factors associated with schizophrenia include the following:

1. Residency in inner city, overcrowded, slum areas.
2. Poor socioeconomic status.
3. Social isolation: singles, widows/widowers, or divorced.
4. Disturbed relations among family members.
5. Early childhood trauma; for instance child sexual abuse.

C. **Psychological factors:**

1. Pre-morbid personality traits of schizoid, schizo-typal, and paranoid are risk factors for developing schizophrenia. *(Schizoid traits; include social isolation, preference of loneliness, restricted emotions); (Schizo-typal traits; include*

staying for long time in day dreams and fantasies, those who believe more on extraordinary powers like sixth sense, etc…); (Paranoid traits; include mistrusting others and over suspiciousness).

2. Excessive use of 'projection' as an unconscious defense mechanism. *(Projection is excessive blaming of others for oneself made faults, which leads to suspiciousness and development of paranoid ideations (persecution and reference ideas and delusions).*

Treatment

Treatment of schizophrenia consists of biological, social, and psychological treatments:

A. Biological treatment

1. Pharmacotherapy:

a. Antipsychotic medications reduce the core symptoms and are the cornerstone in the treatment of schizophrenia (see chapter 16).

How to decide which antipsychotic to start with?

The decision depends on:

I. Side effects tolerability of the patient: for instance, if a schizophrenic patient is diabetic at the same time, Olanzapine is relatively contraindicated.

II. Past history of response: if the patient responded well to haloperidol in his previous episode for instance, it's justifiable to keep him/er on the same medication for the current episode.

III. Family history of response: if one of the patient's relatives was schizophrenic and responded well to Trifluperazine for example, its rationale to start with the same drug for the current patient.

IV. Costs of the drug: generally speaking, atypical drugs are more expensive than typical drugs.

V. If there are no any clues about which antipsychotic to start with, then its preferable to start with one of the atypical drugs, like Olanzapine, Risperidone, etc…with the minimum effective dose and increase gradually, if the patient doesn't respond up to 2 months, then shift to a second drug and wait for another 2 months. If still, however, the patient doesn't respond, then shift to Clozapine but with cautions, because of the higher risks of agranulocytosis with Clozapine. Therefore, Clozapine should never be considered as a first line drug.

VI. Depots should not be used until the patient refuses oral medication or shows poor adherence with the oral regimen.

b. **Antidepressants and mood stabilizers** may be beneficial when mood changes accompany the schizophrenic symptoms.

c. **Benzodiazepines**, especially short acting benzodiazepines, like Lorazepam, might be used temporarily, especially for agitations.

2. **Electroconvulsive therapy (ECT):** is rarely used in the treatment of schizophrenia, but may be worthwhile when catatonia or prominent affective symptoms are present (see chapter 17).

B. Social treatments

In conjunction with medications, social therapies are often indicated. Day treatment programs, with emphasis on social skills training, occupational rehabilitation, can improve functioning and decrease relapse. Family therapy and individual supports are also beneficial in relapse prevention.

C. **Psychotherapy**

Recently, many forms of Cognitive-Behavioral Therapy have been tried stressing more on training the patient to deal with his delusions as believes rather than facts, and to decrease stresses associated with the hallucinations (see chapter 18).

Indications for hospitalization

A. In acute stage, when the patient is agitated and difficult to control.
B. Catatonic schizophrenia.
C. Refusal to take food.
D. Suicidal attempts.
E. When the patient is at danger for others.
F. When patients are unable of taking care of their own basic needs: although institutionalization is no more recommended; psychiatrists frequently find long-term admission beneficial, at least to pick up patients out of streets for the homeless patients, or to teach and rehabilitate patients and help them reintegrate into their own societies.

Schizophrenia-Like Disorders

These include a group of mental disorders which share many signs and symptoms of schizophrenia, yet differ on other aspects. These include: Schizophreniform disorder; Brief psychotic disorder; Schizoaffective disorder; Delusional disorders; Catatonia; Schizotypal (Personality) disorder; Substance/Medication-induced psychotic disorders; and Psychotic disorder due to another medical condition.

Schizophreniform Disorder

It is similar to schizophrenia except for the course, including the treatment & improvement, should be shorter than six months. It is prevalent in about 0.2% of population, with equal sex ratios.
Treatment: like schizophrenia.

Brief Psychotic Disorder: (Acute, Brief Reactive Psychosis)

In this disorder, the onset is very rapid, with florid delusions, hallucinations and disorders of speech and behavior emerge quickly, often overnight.
These frequently occur following some intense stressor such as bereavement or involvement in a disaster or war. It usually occurs in young adults, but is relatively rare & it needs to be differentiated from reactions to stress that might be regarded as acceptable in certain cultures.
The course, including the treatment & improvement, should be less than one month.
DSM-5 criteria of diagnosis are similar to that of schizophrenia, except for the absence of the residual (negative) symptoms, and the duration of less than 1 month.

Types

A. With marked stressor (brief reactive psychosis): when symptoms occur in response to markedly stressful events.
B. Without marked stressor: when symptoms occur in absence of any stressful events.
C. With postpartum onset: when symptoms occur during pregnancy or within 4 weeks postpartum.

Epidemiology

The exact prevalence is unknown, but it is thought to be more common in developing countries, and among low socioeconomic groups, with 2:1 female-to-male ratio.

Treatment
Consists of short acting benzodiazepines like Lorazepam in a dose of 0.5 to 2 mg and/or antipsychotics.

Prognosis
Overall, the prognosis is optimistic. 50% of all patients have no further major psychiatric problems, while another 50% may have continued to develop schizophrenia or mood disorders.

Schizoaffective Disorder
In 1933, Jacob Kasanin introduced the term schizoaffective disorder to refer to a disorder with symptoms of both schizophrenia and mood disorders.

DSM-5 Criteria for diagnosis

A. An uninterrupted period of illness during which there is a major mood episode (major depressive or manic) concurrent with Criterion A of schizophrenia.
B. During the same period of illness, there have been delusions and hallucinations for at least 2 weeks without mood changes.
C. Symptoms that meet criteria for a major mood episode are present for the majority of the total duration of the active and residual portions of the illness.
D. The disturbance is not attributable to the effects of a substance (e.g., a drug of abuse, a medication) or another medical condition.

Epidemiology
It accounts for around 0.3% of population. Females slightly outnumber males.

Treatment
Consists of a combination of both antipsychotics and antidepressants/mood stabilizers.

Prognosis
Generally, it's better than that of schizophrenia, but not as good as outcomes of mood disorders.

Delusional Disorders

The person has one or more delusions, but does not have any other symptoms that would lead to a diagnosis of schizophrenia. If hallucinations are present they are very closely related to the delusions and are often not very prominent.

DSM-5 criteria for diagnosis

A. Presence of one (or more) delusions with a duration of at least one month.
B. Criterion A for schizophrenia has never been met.
   Note: Hallucinations, if present, are not prominent and are related to the delusional themes (e.g., the sensation of being infested with insects associated with delusions of infestation).
C. Apart from the impact of delusion(s) or its ramifications, functioning is not markedly impaired, and behavior is not obviously bizarre or odd.
D. The disturbance is not attributable to the physiological effects of a substance or another medical condition and is not better explained by another mental disorder, such as body dysmorphic disorder or obsessive-compulsive disorder.

Types of Delusional Disorder

A. **Somatic (monosymptomatic hypochondriacal delusions)**
   The patient believes that s/he has serious medical condition or his/er physical appearance has been misshapen.
B. **Erotomania (de Clerambault's syndrome)**
   It is a rare disorder in which persons hold a delusional belief that another person, usually of a higher social status, is in love with them. For instance, a girl from rural area or from moderate socioeconomic group claims that the 'Mr. President' has fell in love with her.
C. Grandiose:
   Here, there are delusions of inflations, self-importance, knowledge, talents, or special relationship with famous persons.
D. Persecutory:
   The sufferers believe that they have had some injustice dealt out to them and are being conspired against in a systematic way by some person or organization. This may lead to legal action or direct, sometimes violent, action against those who are believed to be the cause of the person's distress.
E. Jealous (Delusional Jealousy) (Delusion of infidelity) (Othello Syndrome):

Delusional jealousy or **Othello syndrome** is a psychiatric disorder in which a person holds a delusional belief that his/er spouse or sexual partner is unfaithful.

It is named after the character in Shakespeare's play '**Othello'**, who murders his wife based on his false belief that she has been disloyal.

Affected individuals typically make repeated accusations of infidelity based on insignificant or minimal evidence, often citing on everyday events or material to back up their claims. They may also test their partner's fidelity and can go to considerable lengths to monitor their behaviors and movements. This may be taken to extremes, such as waiting outside of the partner's workplace during their working day or even following them into the bathroom in case their partner has an illicit meeting with their perceived lover or further spying on the partner by spreading cameras whole around the house.

Delusional jealousy can occur in both heterosexual and homosexual individuals, however, it is more often found in males than females.

Unlike other delusional disorders, delusional jealousy has a strong association with violence and in some cases stalking behavior and even murder may be the consequences.

At the very least, affected individuals tend to be irritable and confrontational.

Differential diagnoses of jealousy: Jealousy as an idea; may be normal, preoccupied, overvalued, obsessional, or delusional ideas:

1. Normal jealousy
2. Pathological (morbid) jealousies:
    a. Due to medical conditions: any chronic medical condition that associated with chronic disabilities, especially sexual disabilities, might lead to pathological jealousy.
    b. Substance induced: like Alcohol.
    c. Other Psychiatric disorders: like schizophrenia spectrum disorders including delusional disorders, personality disorder particularly paranoid, borderline, histrionic, and narcissism types, and sexual disorders.

Epidemiology

The annual incidence of delusional disorders is around 1-3/100 000. The life-time prevalence, however, is appraised at around 0.2%. Mean age of onset is expected to be above 40s, and persecutory type is thought to be the commonest one. Although the disorder is evenly distributed between both genders, jealous type is thought to be more prevalent among male and erotomania type is more among female populations.

Treatment

It consists of a combination of antipsychotics and psychotherapies in form of cognitive behavioral, supportive, and counseling psychotherapies. In delusional jealousy, separation of the partners might be advantageous to avoid harms to each other.

Prognosis

Overall prognosis is disappointing, mainly due to the poor compliance with treatment.

Catatonia

The term 'catatonia' first coined by Kahlbaum in 1874 to denote a psychotic condition with marked psychomotor disturbances. In DSM-IV, catatonia was mainly categorized as a clinical subtype of schizophrenia. However, in its latest version, DSM-5, American Psychiatric Association (APA) described the condition independently so as to be a possible 'companion' to several other psychiatric, including schizophrenia, as well as non-psychiatric conditions.

DSM-5 Criteria for the diagnosis:

For any condition to be nominated as 'catatonia', there should be at least three out of the following 12 symptoms:

1. Stupor (i.e., no psychomotor activity; not actively relating to environment).
2. Catalepsy (i.e., passive induction of a posture held against gravity).
3. Waxy flexibility (i.e., slight, even resistance to positioning by examiner).
4. Mutism (i.e., no, or very little, verbal response in absence of 'Aphasia').
5. Negativism (i.e., opposition or no response to instructions or external stimuli).
6. Posturing (i.e., spontaneous and active maintenance of a posture against gravity).
7. Mannerism (i.e., odd, circumstantial repetition of normal, goal-directed, actions).

8. Stereotypy (i.e., repetitive, abnormally frequent, non-goal directed movements).
9. Agitation, not influenced by external stimuli.
10. Grimacing.
11. Echolalia (i.e., mimicking another's speech).
12. Echopraxia (i.e., mimicking another's movements).

Differential Diagnoses

A. Medical conditions: like encephalitis, autoimmune conditions, stroke, and metabolic conditions.
B. Substance-induced: like alcohol and benzodiazepine withdrawal, amphetamine-like substances, antipsychotics side effects.
C. Psychiatric disorders: like schizophrenia spectrum and other psychotic disorders, mood disorders, PTSD, neurodevelopmental disorders, and even conversion disorder.

Treatment

A. Benzodiazepines: especially short-acting drugs like Lorazepam in high doses. For instance 4-6 mgs.
B. NMDA antagonists: like Memantine
C. Antipsychotics: but with caution since it may exacerbate the condition.
D. Electro-Convulsive Therapy (ECT): Usually 1-3 sessions may be satisfactory and it's a very efficient mean of rapid resolution of the condition.
E. Treating underlying condition.

03 MOOD DISORDERS

Although the concept of 'Mood Disorders' might be confusing intrinsically, since 'Mood' includes anxiety symptoms additionally, yet 'Anxiety Disorders' classified independently from what are known to psychiatrists as 'Mood Disorders'; but, such a nomenclature may still be more convenient for daily practices, and still keeps its place in most recent writings as well.
In the past, the term 'Affective Disorders' was referred to the same disorders of this chapter, which still is pretty widely used.
However, unlike its previous editions, in DSM-5 there is no place for what is called 'Mood Disorders'. Instead, disorders related to 'Mood' have been described over two different chapters: one for 'Bipolar and Related Disorders', and a second for only 'Depressive Disorders'.
Main disorders in these two chapters of DSM-5 include: Bipolar I, Bipolar II, and Cyclothymic disorders in the first chapter, and Major Depressive, Persistent Depressive, Premenstrual Dysphoric Disorders in addition to a pediatric depressive disorder named Disruptive Mood Dysregulation Disorder in the second chapter.
In this chapter, the concern will be on the first main six 'Mood' disorders.

Bipolar Disorders

Emil Kraepelin (1855-1926) called nowadays bipolar disorder 'manic-depressive insanity' in order to differentiate the condition from the chronic debilitating condition of schizophrenia, 'dementia praecox' (see chapter 2).
Clinically, two kinds of bipolar disorders do exist. Bipolar I and II disorders. For a diagnosis of bipolar I disorder, its necessary for the patient to suffer from at least one 'manic episode'. This episode may have been preceded or followed by 'hypomanic', or 'major depressive' episodes.
For a diagnosis of bipolar II disorder, on the other hand, it is necessary for the patient to suffer from current or past episode(s) of 'hypomania' and current or past 'major depressive' episode(s).
Overall, in mood disorders we have 'mood episodes' through which the psychiatrist determines the 'disorder'. Pathological mood episodes include: Manic, Hypomanic, and Major Depressive episodes. Bipolar disorders are:

A. **Bipolar I** = Manic episode ± (Hypomanic or Major Depressive episodes). Hence, presence of mania alone qualifies the diagnosis of Bipolar I disorder.
B. **Bipolar II** = Hypomanic episode + Major Depressive episode. Therefore, both these mood episodes are necessary for the diagnosis of Bipolar II disorder.

Manic episode

DSM-5 criteria for diagnosis

A. A distinct period of abnormally and persistently elevated, expansive, or irritable mood and abnormally and persistently increased goal-directed activity or energy, lasting at least 1 week and present most of the day, nearly every day (or any duration if hospitalization is necessary).

B. During above period (A), three (or more) of the following symptoms (four if the mood is only irritable) are present to a significant degree and represent a noticeable change from usual behavior:
   1. Inflated self-esteem or grandiosity.
   2. Decreased need for sleep (e.g., feels rested after only 3 hours sleep).
   3. More talkative than usual or pressure to keep talking.
   4. Flight of ideas or subjective experience that thoughts are racing.
   5. Distractibility (i.e., attention too easily drawn to unimportant or irrelevant external stimuli), as reported or observed.
   6. Increase in goal-directed activity (either socially, at work or school, or sexually) or psychomotor agitation (i.e., purposeless non-goal-directed activity).
   7. Excessive involvement in activities that have a high potential for painful consequences (e.g., engaging in unrestrained buying sprees, sexual indiscretions, or foolish business investments).

C. The episode is severe enough to cause marked social or occupational impairments, or necessitate hospitalization, or there are psychotic symptoms.

D. The episode is not secondary to physiological effects of substances or another medical condition.

Hypomanic episode

DSM-5 criteria for diagnosis

A. A distinct period of abnormally and persistently elevated, expansive, or irritable mood and abnormally and persistently increased activity or energy, lasting at least 4 days and present most of the day, nearly every day.

B. Similar to 'manic episode'.

C. The episode is associated with an unequivocal change in functioning that is uncharacteristic of the individual when not symptomatic.

D. The disturbance in mood and the change in functioning are observable by others.

E. The episode is not severe enough to cause marked impairment in social or occupational functioning or necessitate hospitalization. If there are psychotic features, the episode is, by definition, manic.

F. The episode is not secondary to physiological effects of substances or another medical condition.

Major depressive episode

DSM-5 criteria for diagnosis

A. Five (or more) of the following symptoms have been present during the same 2-week period and represent a change from previous functioning; at least one of the symptoms is either (1) or (2).

1. Depressed mood most of the day, nearly every day, as indicated by either subjective report (e.g., feels sad, empty, or hopeless) or observation made by others (e.g., appears tearful).
2. Markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day (as indicated by either subjective account or observation).
3. Significant weight loss when not dieting or weight gain (e.g., a change of more than 5% of body weight in a month), or decrease or increase in appetite nearly every day.
4. Insomnia or hypersomnia nearly every day.
5. Psychomotor agitation or retardation nearly every day (observable by others; not merely subjective feelings of restlessness or being slowed down).
6. Fatigue or loss of energy nearly every day.
7. Feelings of worthlessness or excessive or inappropriate guilt (which may be delusional) nearly every day (not merely self-reproach or guilt about being sick).
8. Diminished ability to think or concentrate, or indecisiveness, nearly every day (either by subjective account or as observed by others).

9. Recurrent thoughts of death (not just fear of dying), recurrent suicidal ideation without a specific plan, or a suicide attempt or a specific plan for committing suicide.

B. The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

C. The episode is not attributable to the physiological effects of a substance or another medical condition.

Clinical types (Specifiers) of bipolar I and II disorders

Following the diagnosis of either bipolar disorder, the clinician then has to specify the clinical type(s) of the bipolarity which include:

A. With **anxious distress:** when at least two of the following symptoms are prominent: feeling tense, unusually restless, difficulty concentrating because of worry, fear that something awful may happen, or feeling of loss of control on oneself. This type has higher risk of suicide, and its commonly noticed in primary care settings.

B. With **mixed features:** when both features of manic or hypomanic and major depressive episodes are concurrently present.

C. With **rapid cycling:** the essential feature of a rapid-cycling bipolar disorder is the occurrence of at least four mood episodes during the previous 12 months in any combination and order.

D. With **Melancholic features:** the term 'Melancholy' is regarded one of the oldest medical terms. Hippocrates was the first who described the term to denote one of his four 'temperaments'. Literally, the term stands for 'sadness'. However, in modern psychiatric textbooks and literatures, the term applied to stances of more sever down mood, mainly with disabling physical signs and symptoms (box 3-1).

E. With **atypical features:** when there are mood reactivity (i.e., mood brightens in response to actual or potential positive events), as well as at least two of the following: significant weight gain, hypersomnia, leaden paralysis, or a long-standing pattern of interpersonal rejection sensitivity.

F. With **psychotic features:** when there are hallucinations or delusions. The psychotic symptoms may be 'mood-congruent' or 'mood-incongruent' (see chapters 1 and 2).

G. With **catatonia:** when there is catatonia (see chapter 2) parallel to the manic or major depressive episodes.

Box 3-1: DSM-5 criteria for the specifying melancholic features

A. One of the following is present during the most severe period of the current episode.
1. Loss of pleasure in all, or almost all, activities.
2. Lack of reactivity to usually pleasurable stimuli (does not feel much better, even temporarily, when something good happens.

B. Three (or more) of the following:
1. A distinct quality of depressed mood characterized by profound despondency, despair, and/or moroseness or by so-called empty mood.
2. Depression that is regularly worse in the morning.
3. Early-morning awakening (i.e., at least 2 hours before usual awakening).
4. Marked psychomotor agitation or retardation.
5. Significant anorexia or weight loss.
6. Excessive or inappropriate guilt.

H. With **peri-partum onset:** when either bipolar disorders' most recent mood episode(s) occur during pregnancy or within 4 weeks following delivery. About 3-6% of women will experience the onset of a major depressive episode during pregnancy or in the weeks or months following delivery. However, the estimate may differ according to culture and the period of assessment. For instance, in a M.Sc. thesis project supervised by the author, an estimate of almost 40% of depressive disorder during the postpartum period was determined (Ramadan and Rahim, 2012).

Several factors might be disposed to depressive disorder during pregnancy or after giving birth; like, anxiety and worry about pregnancy and delivery, marital conflicts, maternity "baby" blue, etc…

Moreover, peri-partum mood episodes might be coupled by psychotic symptoms. When this happens, the condition named 'Postpartum Psychosis'. The associated psychotic symptom may be commanding hallucinations to kill the infant, or delusions that the infant is possessed which may end in 'Infanticide'.

Postpartum psychosis occurs in 1:500 to 1:1000 deliveries, and it is more prevalent among primiparous women. Once a women has had a postpartum psychosis, the risk of recurrence in subsequent deliveries is about 30% to 50%.

I. With **seasonal pattern:** here, at least one mood episode, manic, hypomanic, or the major-depressive, will occur regularly during a particular season for at least 2 subsequent years. Furthermore, full remission also follows regular seasonal pattern. For instance every time the mood episode starts during the fall and ends with the rise of spring. Common seasons noticed for the onset are ‘falls’ and ‘springs’.

Cyclothymic Disorder

DSM-5 criteria for diagnosis

A. For at least 2 years (1 year in children and adolescents) there have been numerous periods with hypomanic symptoms that do not meet criteria for a hypomanic episode and numerous periods with depressive symptoms that do not meet criteria for a major depressive episode.
B. During A, symptoms were persistent for at least half the time, and the patient was never symptoms-free for longer than 2 months at a time.
C. Criteria for mood episodes (manic, hypomanic, or major depressive) have never been met.
D. Symptoms are not better explained by psychotic disorders (see chapter 2).
E. Symptoms are not attributable to the physiological effects of a substance or another medical condition (e.g., hyperthyroidism).

Course and Prognoses of Bipolar and Related Disorders

A. Bipolar and related disorders may start at any age. Nevertheless, peak onset for Bipolar I and Cyclothymic disorders is around 18 year-old, while for Bipolar II disorder is around mid-20s.
B. 5-15% of Bipolar II and 15-50% of Cyclothymic disorders eventually progress to Bipolar I disorder.

Prevalence of Bipolar and Related Disorders

12-month prevalence of Bipolar I and II disorders is around 0.5% to 0.8%. Cyclothymic disorder is slightly commoner.
Conditions are almost equally incident in both genders.

Depressive Disorders

These disorders present only with depressive episodes or only symptoms without the opposite pole of elation of manic or hypomanic symptoms. Depressive disorders include: Major Depressive Disorder, Persistent Depressive Disorder, Premenstrual

Dysphoric Disorder, in addition to a pediatric depressive disorder named Disruptive Mood Dysregulation Disorder.

<u>Major Depressive Disorder</u>

In many literatures, this disorder named 'Unipolar Depression' to denote the presence of the depressive episode "pole" only. Here, there is at least a major depressive episode (see above) without history of manic or hypomanic mood episodes.

Clinical types (Specifiers) of depressive disorder

Same clinical types and specifiers of bipolar disorders, except 'with rapid cycling' applied here too.

Courses and prognoses

Age of onset is variable. Peak of onset is usually in the 20s. The outcome is variable as well; some individuals may not experience remission at all; However, two fifth of sufferers recover within 3 months, and four fifth within 1 year from the onset. Substantial proportion of cases of major depressive disorder progress to future bipolar disorders.

<u>Persistent Depressive Disorder (Dysthymia)</u>

This condition characterized by 'chronic', but less intense, depressive signs and symptoms.

DSM-5 criteria for diagnosis

A. Depressed mood for most of the day, for more days than not, as indicated by either subjective account or observation by others, for at least 2 years (1 year for children).

B. Presence, while depressed, of two (or more) of the following:

1. Poor appetite or overeating.
2. Insomnia or hypersomnia.
3. Low energy or fatigue.
4. Low self-esteem.
5. Poor concentration or difficulty making decisions.
6. Feelings of hopelessness.

C. During the above period, the individual has never been without the symptoms for more than 2 months at a time.

D. Criteria for a major depressive disorder may be continuously present for 2 years.

E. There has never been a manic or a hypomanic episodes, and criteria have never been met for cyclothymic disorder.

F. The disturbance is not better explained by other psychotic disorders (see chapter 2).
G. The symptoms are not attributable to the physiological effects of a substance or another medical condition (e.g., hypothyroidism).
H. The symptoms cause clinically significant distress or impairment in social, occupational, or other important area of functioning.

Course and prognosis

It has an early and insidious, and by definition, a chronic course.

Premenstrual Dysphoric Disorder

DSM-5 criteria for diagnosis

A. In the majority of menstrual cycles, at least five symptoms must be present in the final week before the onset of menses, start to *improve* within a few days after the onset of menses, and become *minimal* or absent in the week post-menses.
B. One (or more) of the following symptoms must be present:
1. Marked affective lability (e.g., mood swings; feeling suddenly sad or tearful; or increased sensitivity to rejection).
2. Marked irritability or anger or increased interpersonal conflict.
3. Marked depressed mood, feelings of hopelessness, or self-deprecating thoughts.
4. Marked anxiety, tension, and/or feelings of being keyed up or on edge.

C. One (or more) of the following symptoms must additionally be present, to reach a total of *five* symptoms when combined with symptoms in B.
1. Decreased interest in usual activities (e.g., work, school, friends, hobbies).
2. Subjective difficulty in concentration.
3. Lethargy, easy fatigability, or marked lack of energy.
4. Marked change in appetite; overeating; or specific food craving.
5. Hypersomnia or insomnia.
6. A sense of being overwhelmed or out of control.
7. Physical symptoms such as breast tenderness or swelling, joint or muscle pain, a sensation of "bloating", or weight gain.

D. The symptoms are associated with clinically significant distress or interference with work, school, usual social activities, or relationships with others.

E. The disturbance is not merely an exacerbation of the symptoms of another disorder, such as major depressive, panic, persistent depressive, or personality disorders (although it may co-occur with any of these disorders).
F. Criterion A should be confirmed by prospective daily ratings during at least two symptomatic cycles.
G. The symptoms are not attributable to the physiological effects of a substance or another medical condition.

Course and prognosis

It may occur at any time after menarche and it ceases completely after menopause. The condition may worsen, however, as the woman approaches menopause.

Prevalence of Depressive Disorders

12-month prevalence of Major depressive disorder is around 7%. The prevalence is three times higher among younger age groups. Females outnumber males by around 2-3 times.
Twelve-month prevalence of Dysthymia is around 1%, whereas12-month prevalence of premenstrual dysphoric disorder among menstruating women is between 1.8% and 5.8%.

Risk of Suicide in Mood Disorders

Bipolar disorders account for nearly one-fifth of all completed suicides. Its estimated that risk of suicide among bipolar patients is 15 times higher than general population. The prevalence of suicide attempts is similar between Bipolar I and II disorders. However, the lethality of attempts is higher in Bipolar II disorder (see chapter 9 for more detail).

Etiology of Mood Disorders

A. Biological factors:

1. Genetic: genetic contributions to mood disorders may be both directly, through affecting the neurotransmitters physiology (e.g., monoamine theory) and Hypothalamic-Pituitary-Adrenal (HPA) axis, or indirectly via modifying the personality and cognition toward predisposition to mood disorders.

Overall, genetic influences appear greater in bipolar more than depressive disorders. For instance, concordance rate for bipolar disorders in mono-zygotic (MZ) twins is between 60-70%, but it's merely 20% for di-zygotic (DZ) twins. Whereas in depressive disorders, the concordance rate for MZ twins is around 46% and for DZ twins is around 20%.

Recently, genetic studies showed evidences for common genetic pools between Schizophrenia and Bipolar I disorders in one hand, and between Major Depression and Generalized Anxiety Disorder on the other hand. These evidences call for replications before considering "Re-classifications" of these disorders.

2. Neuro-pathology of mood disorders:

   a. Mono-amine hypothesis: this hypothesis suggests that depressive disorder is due to under-regular neurotransmission of monoamines in one or more circuits in brain. Three main monoamines have been implicated: Serotonin (5-Hydroxytryptamine), Noradrenaline, and Dopamine. The last two monoamines named (Catecholamines).

   This theory is crucial not only because its an explanation for the signs and symptoms of depressive disorders, but also because, up to this minute, almost all antidepressants act via regulation of these monoamines.

   b. Similar to psychotic disorders, in manic episode there is evidence of dopamine and glutamate over activities (see chapter 2).

3. Hormonal dysregulation: especially thyroid hormone dysregulation disturb the mood of the sufferers. This may happen because hormones play role in regulating neurotransmission.

B. Psychological factors:

1. Cognitive-behavioral theory: this theory views cognitive distortion (illogical thinking) as the central event in maintaining the 'automatic negative thoughts' about 'self' and 'world'. These negative perceptions, in turn, generate negative emotions and depression. Based on animal studies, 'learned helplessness' model suggests that depressive moods arise secondary to individuals' believe of loss of control over stresses and pains.

2. Psychoanalytical theory: here, interpersonal losses (actual or perceived) are the risk factors for depression. Also, Freud proposed that each individual has two primary, but opponents' instincts: Eros; which is the motive of life and manifested as 'love' and 'sex'; and Thanatos; which is the death motive and displayed as aggression. The loss of 'object' may lead to loss of 'libido' which may be complicated by 'aggression' standing. According to Freud, depression emerges when these 'aggressions'

redirected towards oneself which may end up by suicidal thoughts and acts.

Additionally, this theory accuses depressed individuals for their excessive reliance on 'introjection' as a copying defense mechanism.

3. Personality: personality traits of shyness, obsessive, perfectionist, hesitant, histrionic, as well as impulsive are possible predisposing traits for mood disorders. Likewise, cluster B and C personality disorders (see chapter 8) are risk factors for co-morbid mood disorders.

C. Social/environmental factors: adverse life events, particularly 'loss events', might precipitate depressive disorders and to a lesser extend bipolar disorders. For instance, bereavement, divorce, breakup, loss of job, and so forth.

Treatment of Mood disorders

Three therapeutic options are available, thus far, for mood disorders, including pharmacotherapy, psychotherapy, and electro-convulsive therapy (ECT). As in several psychiatric disorders, combination of both pharmaco- and psychotherapies may be the best strategy.

Psychotherapy, is the starting regimen in depressive disorders. However, there are little evidences for the merits of psychotherapy alone for manic episodes. Instead, it may be almost mandatory to prescribe 'mood stabilizers' for bipolar I patients during their manic phase.

ECT is kept for resistant to other therapies patients as well as when there is impending risk of suicide. (See part three for more details)

A. Psychotherapy: different kinds of psychotherapy might be beneficial for depressive disorders; like, supportive, psychodynamic, marital, and interpersonal therapies. However, cognitive-behavioral psychotherapy (CBT) remains the most evidence-based effective mode of psychotherapy for depressed patients.

B. Pharmacotherapy: the table below enumerates the drugs may be prescribed for patients with mood disorders.

Table 3-1: Drugs prescribed in mood disorders	
Group	**Drugs**
Antidepressants	
SSRI	Fluoxetine, Sertraline, Citalopram, Escitalopram, Paroxetine, Fluvoxamine
TCAD	Amitriptyline, Imipramine, Trimipramine, Clomipramine, Nortriptyline
MAOI	Isocarboxazid, Phenelzine, Tranylcypromine, Moclobemide
Others	Duloxetine, Mianserin, Mirtazapine, Venlafaxine, Reboxetine, Trazodone
Mood Stabilizers	
Lithium	Mainly for bipolar disorders, but can be used in resistant depression with or without antidepressants
Anticonvulsants	Valproate, Carbamazepine, Lamotrigine, Levetiracetam, Phenytoin, Topiramate
Other Groups of drugs	
Antipsychotics	For both bipolar and depressive disorders
Benzodiazepine	Especially short acting groups for rapid control of bipolar disorders
Tri-iodothyronine (T3)	For resistant depression, or when there is evidence of thyroid dysfunctioning

In depressive disorders, SSRIs are the first-line drugs. Once there are no responses, then gradual increase in doses should be considered. Should the patient shows no response or couldn't tolerate side effects, then another drug from other groups should be tried. Antipsychotic augmentation, Lithium, two antidepressants at the same time, or T3 augmentation are alternative strategies as well for refractory patients.

In Bipolar disorders, Lithium is the best starting mood stabilizer. However, there is risk of lithium toxicity which calls for frequent serum level assessment. Alternatives to Lithium are anti-epileptic drugs; among which Valproate is the best studied drug. Nevertheless, cautions are required for women who have plan for pregnancy due to the risk of neural tube defects.

Antipsychotics are beneficial as adjuvant to mood stabilizers or even alone in reducing core psychotic symptoms during manic episodes. Additionally, benzodiazepines, particularly short-acting drugs, like Lorazepam, are helpful in calming down agitated and aggressive patients.

04 ANXIETY DISORDERS

Anxiety disorders are a group of one of commonest psychiatric disorders. Although anxiety symptoms may be normal responses to dangers, distressing anxiety symptoms are companions to almost all psychiatric morbidities including psychotic disorders, mood disorders, neurodevelopmental, neurocognitive conditions, etc…

In anxiety disorders, however, morbid anxiety symptoms are the most severe and protruding, and, perhaps, the sole complaints.

Before introducing the main anxiety disorders, it's indispensable to define a number of anxiety-related terms, shed light on how one can distinguish between 'normal' and 'morbid' anxiety response, and finally spell out the anxiety symptoms:

Terminology

Anxiety: is feeling of apprehensions caused by anticipation of danger, which may be internal or external.

Free-floating anxiety: is a pervasive, unfocused fear not attached to any idea.

Fear: is an anxiety caused by consciously recognized and realistic danger.

Agitation: is the subjective feeling of being upset, angry, disturbed, or unable to rest.

Tension: is an increased, unpleasant, motor and/or psychological activity.

Panic: is an acute, episodic, intense attack of anxiety associated with overwhelming feelings of dread and autonomic discharge.

When anxiety is regarded to be abnormal?

Anxiety is a normal response to danger. Mild-to-moderate anxiety enhances most kinds of performance, but high levels interfere with it.

Nonetheless anxiety turns out abnormal when its severity is out of proportion to the threat of danger, or when it outlasts the threat. Consequently, an anxiety symptom is judged to be a 'morbid' response when:

A. It's persistent.
B. It's irrational (out of reason, out of proportion to the danger).
C. It interferes with the normal social and/or occupational activities of the individual.
D. It roots personal distress.

As a conclusion, any anxiety symptom may be:

A. A normal response.

B. Pathological:
   1. Medical illness.
   2. Substance induced.
   3. Psychiatric disorders other than anxiety disorders.
   4. Anxiety Disorders.

Anxiety symptoms

Anxious mood is associated with psychological, autonomic, and somatic arousals (table 4-1).

Anxiety Disorders

Anxiety disorders are abnormal states in which the most striking features are mental and physical symptoms of anxiety, occurring in the absence of organic brain disease or another psychiatric disorder. Although all the anxiety symptoms (listed above) can occur in any type of the anxiety disorders, there is a characteristic pattern in each disorder. The disorders share many features of their clinical picture and etiology but there are also differences. For instance, in generalized anxiety disorder (GAD) the anxiety is continuous but it may vary in intensity in Phobia the anxiety is intermittent, arising in response to the phobic stimuli. Likewise, in Panic disorder the anxiety is intermittent, but its occurrence is unrelated to any particular circumstances or stimuli.

Classification of anxiety disorders

American Psychiatric Association (APA) in its latest edition of DSM (DSM-5) lists 11 Anxiety Disorders

A. Generalized Anxiety Disorder (GAD)
B. Specific Phobia
C. Social Anxiety Disorder (Social Phobia)
D. Agoraphobia

$\}$ Phobic Anxiety Disorders

E. Panic Disorder
F. Separation Anxiety Disorder
G. Selective Mutism
H. AD due to General Medical Conditions
I. AD Substance-induced
J. Other Specified Anxiety Disorder
K. Unspecified Anxiety Disorder

Since this textbook addresses undergraduate students, the chapter will focus on the main anxiety disorders: GAD, Phobic Anxiety Disorders, and Panic Disorder.

Table 4-1: Anxiety Symptoms	
Psychological Arousal	Fearful anticipation Irritability Sensitivity to noise Restlessness Poor concentration Worrying thoughts Insomnia Nightmare and terror
Autonomic Arousal	**Gastro Intestinal:** Dry mouth, Difficulty in swallowing, epigastric discomfort, excessive wind, frequent or loose motion. **Respiratory Tract:** Constriction in the chest, Difficulty inhaling. **Cardiovascular:** palpitation, discomfort in the chest, awareness of missed heartbeats. **Genitourinary:** frequent micturition, failure of erection, menstrual discomfort, amenorrhea.
Somatic Arousal	**Muscle tension:** tremor, headache, aching muscles. **Hyperventilation:** dizziness, tingling in the extremities, numbness, paresthesia.

Etiology of anxiety disorders

A. Biological factors

1. Genetic: Family studies show that the rate of anxiety disorder is more among first-degree relatives of patients than that of control. Twin studies, additionally, showed a higher concordance rate of anxiety disorders among monozygotic than dizygotic twins.
2. Neurotransmitters' imbalances:
   a. Gama Amino Butyric Acid (GABA) vs. Glutamate imbalance: under-activation of GABA and/or over-activation of Glutamate.
   b. Serotonin (5HT) transmission disturbances.
   c. Increased release of noradrenaline in response to stress.

B. Psychological factors

1. Psychoanalytic theory**:** anxiety, in general, arises when Ego is overwhelmed by excitation from any of outside world stimuli (reality anxiety), Id's neurotic desires (sexual and/or aggression impulses), or Superego's ethical pressures (moral anxiety). (**Reality anxiety** is any tension arises from our daily living, for instance examination, financial, marital, or occupational stresses and so forth. **Neurotic anxiety** is the unconsciously-derived impulses like sexual desires which arouse the Ego. While **Superego or moral anxiety** is the

feelings of guilt and remorse get up by our conscience, ethical attitudes, religious beliefs and so forth).

2. Cognitive-Behavioral Theories**:** Cognitive theories propose that Anxiety Disorders arise as result of a tendency to worry unproductively about problems and to focus attention on threatening situations and failure to recognize the neutral or the positive situations. These thoughts strengthen by process of learned behavior and conditioning of (Avoidance).
3. Personality Traits: Anxiety as a symptom is associated with neuroticism as a trait. Anxiety Disorders are more among those with Cluster C Personality Disorder (Chapter 8).

C. Social factors

1. Stressful daily life events: like 'threat event' might lead to anxiety unlike 'loss events' which may import grief and depression.
2. Early childhood experiences: objective studies suggest that early adverse experiences are causes of anxiety disorders rendering to accounts given by patients.

<u>Generalized Anxiety Disorder</u>

DSM-5 criteria for diagnosis

A. Excessive anxiety and worry (apprehensive expectation), presenting most days for at least six months period and involves a number of life events.

B. The anxiety is difficult to control.

C. At least three of the following:

1. Restlessness or feeling on edge.
2. Easy fatigability.
3. Difficulty concentrating.
4. Irritability.
5. Muscle tension.
6. Sleep disturbance.

D. The anxiety or physical symptoms cause significant distress or impairment in socio-occupational functioning.

E. Symptoms are not caused by substance use or a medical condition (e.g. Hyperthyroidism).

F. Symptoms are not better explained by another Anxiety Disorder or other Mental Disorders.

Epidemiology

Lifetime risk of GAD is around 9%, whereas12-month prevalence ranges from 0.4% to 3.6%. The female-to-male sex ratio for GAD is

2:1. Most patients report excessive anxiety during childhood or adolescence; however, onset after age 20 may sometimes occur. The condition is more prevalent in high-income countries.

Co morbidity

GAD commonly coexist with other Anxiety and Mood Disorders. Two Third of those with GAD have additional diagnoses, most commonly of phobias, or depression.

Treatment

Best strategy is the combination of both Psychotherapy and Medications:

Patients should be directed to minimize drinking coffee and other caffeinated beverages, as well as avoiding excessive alcohol consumption. Patients should also get adequate sleep, with the use of medication if necessary. Moderate exercise each day may help reduce the intensity of anxiety symptoms.

Psychotherapies in forms of counseling, relaxation training, or cognitive behavioral therapy are favorable over medications.

Nevertheless, should the anxiety symptoms be sternly distressing, or psychotherapies are of little worth, **medications** like Benzodiazepines, particularly long acting drugs like (Chlorodiazepoxide, Diazepam, Clonazepam are valuable options. However, clinicians should pay attentions to the hazards of substance misuse and dependence.

Buspirone, a non-GABA agonist anxiolytic, is an alternative selection. Buspirone has the advantage of no risk of dependence and it is as effective as Benzodiazepines.

Antidepressants, additionally, have anxiolytic properties and relatively lack the risk of misuse or dependence. However, they require more times for their anxiolytic effects to be apparent. Preferable antidepressants for the treatment of GAD include Venlafaxine (SNRI), and Paroxetine (SSRI).

Prognosis

It is overall unenthusiastic; more than 30% fail to respond to treatments.

<u>Phobias</u>

As clarified above, fear is an anxiety caused by consciously recognized and realistic danger. However, phobia is an irrational, exaggerated, and invariably pathological dread of a specific

stimulus or situation; results in a desire to avoid the feared stimulus.
Phobic Anxiety Disorders have the same ‘anxiety symptoms’ of GAD, but these symptoms occur only in the presence or the anticipation of the feared stimuli. Another characteristic feature of phobia is the (Avoidance) of feared stimuli and situations.

Specific phobia

DSM-5 criteria for diagnosis

A. Marked fear or anxiety about specific object or situation (e.g., flying, heights, animals, receiving an injection, seeing blood).
B. The phobic object or situation almost always provokes immediate fear or anxiety.
C. The phobic object or situation is actively avoided or endured with intense fear or anxiety.
D. The fear or anxiety is out of proportion to the actual danger posed by the specific object or situation and to the sociocultural context.
E. The fear, anxiety, or avoidance is persistent, typically lasting for 6 months or more.
F. The fear, anxiety, or avoidance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
G. Symptoms are not caused by another mental disorder (e.g., fear of dirt in someone with Obsessive Compulsive Disorder).

Common types of specific phobias

A. Animal (e.g., dogs).
B. Natural Environmental (e.g., heights, storms, floods).
C. Blood-injection injury.
D. Situational (e.g., airplanes, elevators, enclosed places, dentist’s clinic).

Epidemiology

12-month prevalence of phobias is around 10%. However, its highly cultural as well as age dependent. Age of onset is variable, and females with the disorder far outnumber males.

Treatment

The primary treatment is behavioral therapy. A commonly used technique is Systemic Desensitization, consisting of gradually

increasing exposure to the feared situation, combined with a relaxation technique such as deep breathing.
Beta-blockers, or Benzodiazepines may also be useful prior to confronting the specific feared situation. SSRI (Paroxetine) can be used with Psychotherapy.

Prognosis

If started in adults it carries good prognosis, while if the onset is in childhood, it may continue for many years.

Social Anxiety Disorder (Social Phobia)

DSM-5 criteria for diagnosis

A. Marked fear or anxiety about one or more social situations in which the individual is exposed to possible scrutiny by others. Examples include social interactions (e.g., having a conversation, meeting unfamiliar people), being observed (e.g., eating or drinking), and performing in front of others (e.g., giving a speech).
B. The individual often fears that s/he will act in a way or show anxiety symptoms that will be negatively evaluated (i.e., will be humiliating or embarrassing; will lead to rejection or offend others).
C. The social situations almost always provoke fear or anxiety.
D. The social situations are avoided or endured with intense fear or anxiety.
E. The fear or anxiety is out of proportion to the actual danger posed by the social situation and to the sociocultural context.
F. The fear, anxiety, or avoidance is persistent, typically lasting for 6 months or more.
G. The fear, anxiety, or avoidance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
H. The fear, anxiety, or avoidance is not better explained by effects of substances or another medical condition.
I. The fear, anxiety, or avoidance is not better explained by the symptoms of another mental disorder (e.g., Body Dysmorphic Disorder, or Autism Spectrum Disorder).
J. If another medical condition (e.g., Parkinson's disease, obesity, disfigurement from burns or injury) is present, the fear, anxiety, or avoidance is clearly unrelated or is excessive.

Epidemiology

Lifetime prevalence is 2-12%. It is more frequent (up to six fold) in first-degree relatives of patients with social phobia. Onset usually occurs in adolescence, with a childhood history of shyness. Social Phobia is slightly more prevalent in males.

Treatment

SSRIs, such as paroxetine or sertraline, are first-line medications for social phobia. Venlafaxine may also be used.
Benzodiazepines, such as clonazepam, may be useful if antidepressants are ineffective. However, they carry the risk of dependence.
Social phobia with performance anxiety (for specific situations known to be anxiety provoking) responds well to beta-blockers, such as propranolol. The effective dosage can be very low, such as 10-20 mg. It may also be used prior to the performance; 20-40 mg given 30-60 minutes before the anxiety provoking event.
Cognitive/behavioral therapies are effective and should focus on cognitive retraining, desensitization, and relaxation techniques.
Combined pharmacotherapy and cognitive or behavioral therapies is the most effective strategy.

Prognosis

Social phobia is often a lifelong problem, but the disorder may remit or improve in adulthood.

<u>Agoraphobia</u>

DSM-5 criteria for diagnosis

A. Marked fear or anxiety about two (or more) of the following five situations:
    1. Using public transportation like buses.
    2. Being in open spaces like marketplaces or bridges.
    3. Being in enclosed places like shops, theatres, or cinemas.
    4. Standing in line or being in a crowd.
    5. Being outside home alone.
B. The individual fears or avoids these situations because of the thoughts that escape might be difficult or help might not be available in the event of developing panic-like symptoms or other embarrassing symptoms (e.g., fear of falling or incontinence).
C. Situations in A almost always provoke fear or anxiety.

D. The situations in A are actively avoided, require the presence of a companion, or are endured with intense fear or anxiety.
E. The fear or anxiety is out of proportion to the actual danger posed by the situations in A and to the sociocultural context.
F. The fear, anxiety, or avoidance is persistent, typically lasting for 6 months or more.
G. The fear, anxiety, or avoidance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
H. If another medical condition (e.g., Parkinson's disease or inflammatory bowel disease) is present, the fear, anxiety, or avoidance is clearly excessive.
I. The fear, anxiety, or avoidance is not better explained by the symptoms of another mental disorder (e.g., Body Dysmorphic Disorder, Specific Phobia, or Obsessive Compulsive Disorder).

Epidemiology

12-month prevalence is about 1.7%. Female to male ratio is about 2:1. Incidence peaks at late adolescence or early adulthood.

Treatment

Combination of both CBT and Antidepressant drugs is the best strategy.

Prognosis

The condition is chronic and its not uncommon to be complicated by depression, other anxiety disorders, or substance dependence.

Panic Disorder

The term Panic Disorder introduced to psychiatry in 1980 in DSM-III. Before that time similar conditions were known as (Irritable Heart) or (Effort Syndrome).

The word 'panic' derived from experience suffered by travelers through the woods in ancient Greek mythology who experienced acute attacks of anxiety when the mischievous god, Pan, jumped out in front of them or teased them in other frightening ways.

What is Panic Attack?

It is an acute, unexpected, attack of severe anxiety during which at least four of the following symptoms begin abruptly and reach a peak within minutes in the presence of intense fear:

1. Palpitations, increased heart rate.
2. Sweating.

3. Trembling or shaking.
4. Sensation of shortness of breath.
5. Feeling of choking.
6. Chest pain or discomfort.
7. Nausea or abdominal distress.
8. Feeling dizzy, lightheaded or faint.
9. Chills or hot flushes.
10. Paresthesias.
11. Derealization (feelings of unreality) or depersonalization (being detached from oneself).
12. Sense of losing control or going crazy.
13. Sense of dying.

Panic attack is not a mental disorder per se; instead, it may occur in the perspective of several other mental disorders like other anxiety disorders, PTSD, or depressive disorders, as well as non-psychiatric medical conditions like cardiac, respiratory, vestibular, or gastrointestinal conditions.

DSM-5 criteria for diagnosis

A. Recurrent unexpected panic attacks (the core feature).
B. At least one of the attacks has been followed by one month (or more) of one or both of the following:
    1. Persistent concern about having additional attacks or their consequences (e.g., losing control, having a heart attack, "going crazy").
    2. A significant maladaptive change in behavior related to the attacks (e.g., behaviors designed to avoid having panic attacks, such as avoidance of exercise or unfamiliar situations).
C. Panic attacks are not due to the effects of a substance (e.g., Cocaine, Caffeine) or medical condition (e.g., Hyperthyroidism, Cardiac Arrhythmias).
D. The panic attacks are not better explained by another mental disorder (e.g., panic attacks on exposure to social situations in social phobia; in response to specific phobic stimulus, or stimuli of a severe stressor such as with post-traumatic stress disorder).

Epidemiology

The lifetime prevalence of panic disorder is between 1.5% and 3.5%. The female-to-male ratio is 2:1. Up to one-half of panic disorder patients have agoraphobia as well.

Panic disorder usually develops in early adulthood with a peak onset in the mid-twenties. Onset after age 45 years is unusual.

Treatment

Mild cases of panic disorder can be effectively treated with cognitive behavioral psychotherapy with an emphasis on relaxation and instruction on misinterpretation of physiologic symptoms.

Pharmacotherapy is indicated when patients have marked distress from panic attacks or are experiencing impairment in work or social functioning:

A. SSRI and TCAD are most often used. The initial dose should be small.
B. Benzodiazepines may be used adjunctively with TCADs or SSRIs during the first few weeks of treatment.
C. Buspirone is not effective for panic disorder. Instead, it may exacerbate the condition due to its sympathomimetic properties.
D. MAOIs may be the most effective agents available for panic disorder, but these agents are not often used because of concern over hypertensive crisis.

For optimal responses, medications are preferable to be combined with cognitive behavioral therapy.

Prognosis

The course of the illness is often chronic, but symptoms may wax and wane depending on presence of stressors. 50% of panic disorder patients are only mildly affected, and 20% have marked symptomatology. The suicide risk is markedly increased, especially in untreated patients. Substance abuse, especially of alcohol, may occur in up to 40% of patients.

05 OBSESSIVE COMPULSIVE AND RELATED DISORDERS

Obsessive-Compulsive Disorder (OCD) was classified with the Anxiety Disorders in the prior versions of DSM. American Psychiatric Association (APA), however, incorporated new independent chapter for OCD and related disorders in its latest version of DSM, DSM-5. The inclusion of this new chapter reflects the increasing evidence of these disorders' relatedness to one another.

These disorders include: OCD, Body Dysmorphic Disorder, Hoarding Disorder, Trichotillomania (Hair-Pulling Disorder), Excoriation (Skin-Picking Disorder), Substance/Medication-induced obsessive-compulsive and related disorder, obsessive-compulsive and related disorder due to another medical condition, and other specified obsessive-compulsive and related disorder and unspecified obsessive-compulsive and related disorder (e.g., obsessional jealousy).

Clinicians are encouraged to screen for these conditions in individuals who present with one of them and be aware of overlaps between these conditions.

For the sake of relevancy to undergraduate medical students, this current chapter will focus, only, on the main disorders including: OCD, Body Dysmorphic Disorder, and Trichotillomania.

Etiology

A. Biological factors:
   1. Genetics: Rates of 2 to 10 times among first-degree relatives of patient with OCD were detected. Similar findings were repeated for most of the rest disorders. Furthermore, the rates of both Body Dysmorphic Disorder and Trichotillomania are increased among first-degree relatives of OCD.
   2. Separately, dysfunctions in orbitofrontal as well as anterior cingulate cortices have been strongly implicated.

B. Psychological factors: childhood negative emotions, excessive repression, behavioral inhibition, and indecisiveness are possible psychological factors

C. Childhood histories of neglect or abuse as well as stressful life events have been associated with an increased risk for developing OCD and related disorders.

<u>Obsessive-Compulsive Disorder</u>

Definitions

Obsession: is defined by the presence of both (1) and (2):

1. Recurrent and persistent thoughts, urges, or images that are experienced as intrusive and unwanted, and causing marked anxiety or distress.
2. The individual attempts to ignore or suppress (1), or to neutralize them with some other thought or action (i.e., by performing a compulsion).

Compulsion: is defined by the presence of both (1) and (2):

1. Repetitive behaviors (e.g., hand washing, ordering, checking) or mental acts (e.g., praying, counting, repeating words silently) that the individual feels driven to perform in response to an obsession or according to rules that must be applied rigidly.
2. These (1) are aimed at preventing or reducing anxiety or distress, or preventing some dreaded event or situation; however, these (1) are not connected in a realistic way to what they are attempting to prevent, or they are clearly excessive.

DSM-5 criteria for diagnosis

A. Obsessions and/or Compulsions are present.

B. The A are time-consuming (e.g., take more than 1 hour per day) or cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

C. The symptoms are not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition.

D. The disturbance is not better explained by the symptoms of another mental disorder (e.g., preoccupation with food in an eating disorder, preoccupation with drugs in a substance abuse disorder, body shape in Body dysmorphic disorder, or hair pulling in trichotillomania).

Common presentations: (box 5-1)

Epidemiology

Obsessions and compulsions as isolated symptoms may occur in up to 10% of population at any time. However, the lifetime prevalence of OCD is approximately 2%. There is no sex difference, but the age of onset is earlier in males.

Box 5-1: Common Presentation of OCD	
Common Obsessions	**Common Compulsions**
Contamination	Washing and Cleaning
Doubt (Rumination)	Checking
Symmetry/Exactness	Ordering/Arranging
Fears	Avoidance
Losing things	Saving
Others: Sexual, Religious, etc…	Others: Counting, Repeating, etc…

Treatment

A. Pharmacotherapy is almost always indicated.
   1. Clomipramine (Anafranil), sertraline (Zoloft), paroxetine (Paxil) fluoxetine (Prozac), and fluvoxamine (LuVox) are effective.
   2. Standard antidepressant doses of clomipramine are usually effective, but high doses of SSRI's maybe required, such as 60-80 mg of fluoxetine (Prozac), 40-60 mg of paroxetine (Paxil) or 200 mg of sertraline (Zoloft).
   3. In resistant OCD, Antipsychotics, Lithium, Sodium Valproate may be effective joint therapies.

B. Psychotherapy: like Exposure to obsessional stimuli with response (compulsion) prevention.

C. Other Forms of Therapy:
   1. ECT, occasionally used.
   2. Trans cranial Magnetic Stimulation
   3. Neurosurgery, like cingulotomy, limbic leucotomy, subcaudate tractotomy, and many other surgical procedures have been tried, but they are associated with several serious complications.

Prognosis

OCD usually begins in adolescence or early adulthood, but it may occasionally begin in childhood. The onset is usually gradual and most patients have a chronic disease course with waxing and waning of symptoms in relation to life stressors.

Fifteen percent of patients have a chronic debilitating course with marked impairment in social and occupational functioning. More than 65% of cases show some improvement by the end of the first year of therapy.

Body Dysmorphic Disorder

*Dys*morphic = Ugliness

It is over concern about one or more of body shape or body appearance. This disorder was classified with Somatoform Disorders in DSM-IV. However, as the main psychopathology is 'Obsession' and 'Compulsion', and since researches concluded common genetic background with OCD, in its latest edition, DSM-5, APA categorized it with OCD and related disorders.

DSM-5 criteria for diagnosis

A. A preoccupation with one or more perceived defects or flaws in physical appearance that are not observable or appear slight to others.

B. At some point during the course of the disorder, the individual has performed repetitive behaviors (e.g., mirror checking, excessive grooming, skin picking, reassurance seeking) or mental acts (e.g., comparing his/er appearance with that of others) in response to the appearance concerns.

C. The preoccupation causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.

D. The appearance preoccupation is not better explained by concerns with body fat or weight in an individual whose symptoms meet diagnostic criteria for an eating disorder.

Clinical presentation

Facial features, hair, and body build are the most frequent 'defective' features. Multiple visits to surgeons and dermatologists are common.

Major depressive disorder and anxiety disorders frequently coexist with body dysmorphic disorder.

Epidemiology

Point prevalence is around 2% in general population. However, it's noticeably more common in dermatological, cosmetic surgery, orthodontia, and maxillofacial surgery outpatient clinics with pictures of up to: 15%, 16%, 8%, and 10% respectively.

The disorder is most common between the ages of 15 and 20 years, with women affected as frequently as men. However, in USA and Germany females slightly outnumber males.

Family history reflects a higher incidence of mood disorders and obsessive-compulsive disorder (OCD).

Treatment
SSRI antidepressants and clomipramine are effective in reducing symptoms in 50% of patients. Cognitive behavioral therapy may have some efficacy.
Coexisting psychiatric conditions, such as a mood disorder, should be treated.
Surgical repair of the "Defect" is rarely successful.

Trichotillomania
Definition
The name, coined by French dermatologist François Henri Hallopeau, derives from the Greek: *trich-* (hair), *till (en)* (to pull), and *mania* (madness). Therefore, it's a compulsive urge to pull out (and in some cases, eat) one's own hair, leading to a noticeable hair loss, distress, and social or functional impairment.
It was classified as an Impulse Control Disorder in DSM-IV. Nevertheless, due to similar reasons mentioned above for Body Dysmorphic Disorder, APA classified it with OCD and related disorders in DSM-5.

DSM-5 criteria for diagnosis
A. Recurrent pulling out of one's hair, resulting in hair loss.
B. Repeated attempts to decrease or stop hair pulling.
C. The condition causes significant distress or impairment in social, occupational, or other important areas of functioning.
D. The condition is not attributable to another medical condition (e.g., a dermatological condition).
E. The condition is not better explained by the symptoms of another mental disorder (e.g., attempts to improve a perceived defect in appearance in body dysmorphic disorder).

Clinical presentation
Hair pulling may occur from any region of the body in which hair grows. But most common sites are scalp, eyebrows, and eyelids. Hair pulling may occur in brief episodes throughout the day or during less frequent, yet more sustained, periods that can continue for hours.
Patients may cover the hair loss by using makeup. Scarves, or wigs. There might be some rituals accompanied the process of hair-pulling, for instance looking for particular hair, pulling from specific area only, pulling the hair in a certain way, or examining or even tasting the hair after pulling.

Epidemiology

12-month prevalence is around 1%-2%, and female-to-male ratio is about 10:1.

Treatment

Psychotherapy like Habit Reversal Training and Cognitive Behavioral Therapy are valuable. Pharmacotherapies like SSRIs have limited usefulness. However, there are evidences for the effectiveness of TCADs like clomipramine. Naltrexone may be a worthwhile drug as well.

06 TRAUMA AND STRESSOR-RELATED DISORDERS

Stressful events range from hearing bad personal news like failure in examinations to more severe one like loss of job, changing place of residency, car accident witnesses, death of a loved body, to worse events like being a victim of rape, earthquake, a terroristic act, and so forth.

How an individual, normally, does respond to a stressful event?

Human being comes across different kinds of stressors on a daily base. Hence, evolutionally, it was necessary for the mankind to adapt to these stressors for the sake of survival. These adaptations include one or more of the following:

A. Emotional response; like 'anxiety' for 'threat' events, and 'depression' for 'loss' events.
B. Coping strategy; like seeking helps from friends or relatives, religious faiths, obtaining information and advices, solving problems, etc…
C. Defense mechanisms; denial, rationalizations, etc…

The pivotal question is 'When reaction to stressful events is regarded to be pathological?' The answer may be; when it occurs in an exaggerated intensity, persists for a long time, or manifests in a distorted manner.

Then, what are the stress-related disorders?

They are disorders (abnormal reaction to stress) that occur particularly after experiencing a traumatic event. These include: Reactive attachment disorder, Disinhibited social engagement disorder, Post-traumatic stress disorder (PTSD), Acute stress disorder (ASD), Adjustment disorders, as well as other specified and unspecified trauma disorders.

Because the main focus of this book is to deliver psychiatric disorders to undergraduate students, the chapter will only tackle main trauma and stressor related disorders like PTSD, ASD, and Adjustment disorders. Additionally, disorders related to grief will be mentioned in this chapter since they occur in response to the death trauma of beloved persons.

Etiology

The fundamental query is why not all victims of any kind of traumatic event, regardless of its severity, do not develop these disorders? The answer may be due to one or more of the following risk factors:

A. Biological factors: overall, the risk of trauma-related disorders is more among female gender. This fact might reflect hormonal, sex gene, as well as feminine personality phenotypes and copying strategies as contributors to this gender-difference. Furthermore, certain genotypes may either be protective or increase the risk of these disorders.

B. Psychological factors: the higher prevalence of these disorders among younger age group might reflect improper personality development and lack of problem solving strategies Also, early childhood emotional problems, prior mental disorders like depression, panic disorder, or OCD, as well as neurotic personality trait are among other risk factors.

C. Social factors: history of prior trauma beside absences of supportive family members or friends may prone victims to these disorders.

<u>Acute Stress Disorder</u>

DSM-5 criteria for diagnosis

A. Exposure to actual or threatened death, serious injury, or sexual violation in one (or more) of the following ways:
   1. Directly experiencing the traumatic event(s).
   2. Witnessing, in person, the event(s) as it occurred to others.
   3. Learning that the event(s) occurred to a close family member or close friend.
   4. Experiencing repeated or extreme exposure to aversive details of the traumatic event(s) (e.g., first responders collecting human remains, police officers repeatedly exposed to details of child abuse).

B. Presence of nine (or more) of the following symptoms:

   Intrusion symptoms
   1. Recurrent, involuntary, and intrusive distressing memories of the traumatic event.
   2. Recurrent distressing dreams (nightmares) in which the content of the dream related to the event.

3. Dissociative reactions (e.g., flashbacks) in which the individual feels or acts as if the events were recurring (Such reactions may occur on a continuum, with the most extreme expression being a complete loss of awareness of present surroundings).
4. Intense or prolonged psychological distress or marked physiological reactions in response to internal or external cues that symbolize or resemble an aspect of the traumatic event(s).

Negative mood

5. Persistent inability to experience positive emotions (e.g., inability to experience happiness, satisfaction, or loving feelings).

Dissociative symptoms

6. An altered sense of reality of one's surroundings or oneself (e.g., seeing oneself from another's perspective, being in daze, time slowing).
7. Amnesia of an important aspect of the event(s) (typically due to dissociative amnesia and not to other factors such as head injury, alcohol, or drugs).

Avoidance symptoms

8. Efforts to avoid distressing memories, thoughts, or feelings about or closely associated with the traumatic event(s).
9. Efforts to avoid external reminders (people, places, conversations, activities, objects, situations) that arouse distressing memories, thoughts, or feelings about or closely associated with the traumatic event(s).

Arousal symptoms

10. Sleep disturbance.
11. Irritable behavior and angry outbursts.
12. Hyper-vigilance.
13. Problems with concentration.
14. Exaggerated startle response.

C. Duration of B is 3 days to 1 month after trauma exposure.
D. The disturbance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
E. The disturbance is not attributable to the physiological effects of substances (e.g., medication or alcohol) or another medical condition (e.g., mild traumatic brain injury) and is not better explained by brief psychotic disorder.

Epidemiology

The exact rate in community is unknown. However, its more prevalent among female gender. The rate inflates as the traumatic event is more personal. For instance, among survivors of car accidents is around13-20%, among victims of violent crimes is around19%, among the witnesses of mass shooting is about 33%, and among victims of rape is up to 50%.

Treatment Strategy for ASD

Psychotherapy in forms of counseling (Debriefing), supportive psychotherapy, as well as Cognitive Behavioral Therapy are helpful. Benzodiazepines, nevertheless, can be used for short periods of time.

Prognosis

78% of those with ASD continue to develop PTSD.

<u>Post-Traumatic Stress Disorder (PTSD)</u>

DSM-5 criteria for diagnosis

A. Exposure to actual or threatened death, serious injury, or sexual violation (see relevant criterion of ASD).
B. Presence of one (or more) of the following intrusion symptoms associated with the traumatic event(s), beginning after the traumatic event(s) occurred:
   1. Recurrent, involuntary, and intrusive distressing memories of the traumatic event(s).
   2. Recurrent distressing dreams (nightmares) in which the content and/or affect of the dream are related to the event(s).
   3. Dissociative reactions (e.g., flashbacks) in which the individual feels or acts as if the event(s) were recurring (Such reactions may occur on a continuum, with the most extreme expression being a complete loss of awareness of present surroundings).
   4. Intense or prolonged psychological distress at exposure to internal or external cues that symbolize or resemble an aspect of the traumatic event(s).
   5. Marked physiological reactions to internal or external cues that symbolize or resemble an aspect of the traumatic event(s).

C. Persistent avoidance of stimuli associated with the traumatic event(s), beginning after the traumatic event(s) occurred, as evidenced by one or both of the following:
   1. Avoidance of or efforts to avoid distressing memories, thoughts, or feelings about or closely associated with the traumatic event(s).
   2. Avoidance of or efforts to avoid external reminders (people, places, conversations, activities, objects, situations) that arouse distressing memories, thoughts, or feelings about or closely associated with the traumatic event(s).

D. Negative alterations in cognitions and mood associated with the traumatic event(s), beginning or worsening after the traumatic event(s) occurred, as evidenced by two (or more) of the following:
   1. Amnesia of an important aspect of the traumatic event(s) (typically due to dissociative amnesia and not to other factors such as head injury, alcohol, or drugs).
   2. Persistent or exaggerated negative beliefs or expectations about oneself, others, or the world (e.g., "I am bad," "no one can be trusted," "the world is completely dangerous," "My whole nervous system is permanently ruined").
   3. Persistent, distorted cognitions about the cause or consequences of the traumatic event(s) that lead the individual to blame oneself or others.
   4. Persistent negative emotional state (e.g., fear, horror, anger, guilt, or shame).
   5. Markedly diminished interest or participation in significant activities.
   6. Feelings of detachment or estrangement from others.
   7. Persistent inability to experience positive emotions (e.g., inability to experience happiness, satisfaction, or love feelings).

E. Marked alteration in arousal and reactivity associated with the traumatic event(s), beginning or worsening after the traumatic event(s) occurred, as evidenced by two (or more) of the following:
   1. Irritable behavior and angry outbursts.
   2. Reckless or self-destructive behavior.
   3. Hyper-vigilance.
   4. Exaggerated startle response.
   5. Problems with concentration.
   6. Sleep disturbance.

F. Duration of B, C, D, and E is more than 1 month.
G. The disturbance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
H. The disturbance is not attributable to the physiological effects of substances (e.g., medication, alcohol) or another medical condition.

Epidemiology
10% of those experiencing a significant traumatic event actually go on to develop PTSD. Female-to-male ratio is around 2:1.

Treatment
It's similar to that of ASD in addition to trials of antidepressants specially SSRIs, particularly (Sertraline or Paroxetine).
Other agents also can be used like: TCAD, MAOI, Lithium, Anticonvulsants, Clonidine, or Antipsychotics.

Prognosis
More than 50% recover within first year.

Adjustment Disorder
It refers to the psychological reactions in adapting to new circumstances like divorce, shift from school to university, change of role like become a mother or father, migration, and so onward.
It may present with any of the Anxiety Symptoms (see chapter 4) or with depression.
Sometime the condition may present with aggressive behaviors, self-harm, alcohol or other substances abuse.

DSM-5 criteria for diagnosis
A. The development of emotional or behavioral symptoms in response to an identifiable stressor(s) occurring within 3 months of the onset of the stressor(s).
B. These symptoms or behaviors are clinically significant, as evidenced by one or both of the following:
   1. Marked distress that is out of proportion to the severity or intensity of the stressor, taking into account the external context and the cultural factors that might influence symptom severity and presentation.
   2. Significant impairment in social, occupational, or other important areas of functioning.

C. The stress-related disturbance does not meet the criteria for another mental disorder and is not merely an exacerbation of a pre-existing mental disorder.
D. The symptoms do not represent bereavement.
E. Once the stressor or its consequences have terminated, the symptoms do not persist for more than an additional 6 months.

Epidemiology

The prevalence rate in community is unknown. 5-20% of patients attending a psychiatric out-patient could be diagnosed with Adjustment Disorder.

Treatment

The treatment is aimed at decreasing denial and avoidance of the stressful events, in addition to encouraging problem solving strategies as well as discouraging maladaptive coping responses like uses of alcohol. Nonetheless, occasionally anxiolytics can be used.

Grief and Bereavement

Bereavement is the loss through death of a loved person; **grief,** however, is the involuntary emotional and behavioral response to bereavement. Though, **mourning** is the voluntary expression of behaviors and rituals that are socially suitable responses to bereavement.
These behaviors and rituals differ in different societies and religious groups both in their forms and durations.

Abnormal or pathological grief: (Box 6-2)

Grief is regarded as abnormal if it is unusually:

A. Intense
B. Prolonged
C. Delayed
D. Inhibited or distorted

Abnormally intense grief

Here, the intensity of the sadness increased to reach the diagnosis of depressive disorders. About 35% of bereaved people meet the criteria of depressive disorders which resolve within 6 months. Nevertheless, more than 20% of those with depressive disorders exceed 6-month duration.

While assessing for depression, particular attention should be paid for suicidal ideations.

Stages of grief: (Box 6-1)

Box 7-1: Stages of normal grief*
Stage I hours to days
Denial, disbelief
Numbness, lack of emotional response
Feeling of unreality
Sometime restlessness, searching for the dead person
Stage II weeks to 6 months
Sadness, weeping waves of grief
Somatic symptoms of anxiety
Restlessness
Poor sleep
Poor appetite
Guilt, blame of oneself
Experience of a presence
Illusions, vivid imagery
Hallucinations of the dead voices
Preoccupation with memories of deceased
Social withdrawal
Stage III weeks to months
Symptoms resolve
Social activities resumed
Memories of good time

*adapted from Gelder, M. Harrison, P. Cowen, P. (2006)

Box 6-2: Etiology of pathological grief:
The death was sudden and unexpected.
The bereaved person had a very close, dependent, or ambivalent relationship with the deceased.
The survivor is insecure.
The bereaved person has history of psychiatric illnesses.
The survivor has to care for dependent children and cannot show grief easily.

Abnormally prolonged grief

Grief is considered abnormally prolonged if stage II persists for more than 6 months. It may associate with depressive disorders.

Delayed grief

Grief is regarded to be delayed if the first stage does not occur up to more than 2 weeks after the death. It is more frequent after sudden, traumatic, or unexpected deaths.

Inhibited or distorted grief

Inhibited grief refers to a reaction which lacks some normal features. Whereas distorted grief refers to features other than

depression that are either unusual in degree (e.g., marked hostility, over activity, or extreme social withdrawal), or unusual in kind (e.g., physical symptoms that were part of the last illness of the deceased).
Persistent avoidance of situations and of other reminders of deceased is common in all forms of abnormal grief.

Treatment

Counseling: Through letting the bereaved to talk about the loss and guide him/er to accept that the loss is real as well as readjusting the sufferer to life without the deceased.
Sometimes ‘guided mourning’, a kind of exposure behavioral therapy, aiming at reducing avoidance of memories of the dead one may shorten the duration of grief.
Pharmacotherapy cannot remove the distress of normal grief; however, anxiolytics may be needed for few days in stage I, and antidepressants in stage II if criteria for depressive disorders are met.

07 PSYCHOSOMATIC MEDICINE

Psychosomatic medicine is a branch of medicine that deals with the effects of human being's mind on the body and the effect of the body on mind.

Disorders with prominent somatic symptoms constitute a new category in DSM-5 named 'Somatic Symptoms and Related Disorders'.

Generally speaking, the relation between medical 'body' symptoms or disorders and mental state can be classified like the following:

A. **Specific mental disorders with mainly somatic presentations** like Somatic symptom disorder, Illness anxiety disorder, Conversion disorder, and Factitious disorder.
B. **Psychological Factors Affecting Other Medical Conditions.**
C. **Medical Conditions Causing Psychological Symptoms.**

A distinctive characteristic of these kinds of disorders is not the somatic symptom per se, but, instead, the way they present and interpret them. Therefore, incorporating affective, cognitive, and behavioral components into the criteria for these disorders provide a more comprehensive and accurate understanding of the actual clinical picture than can be achieved by assessing the somatic symptoms alone.

Almost all individuals with these disorders, remarkably, make their first contacts with non-psychiatric health services.

Etiology

A. Biological factors: for instance increased sensitivity to pain, or genetic predisposition.
B. Psychological factors:
   1. Cognitive interpretation:
      Body perception (e.g., chest pain, palpitation, etc…) in addition to patient's health knowledge and beliefs, all together lead to misinterpretations of the body symptoms, henceforth production of more symptoms which end by the believe that s/he is suffering from an identifying or non-identifying illness(s).
   2. Uses of Defense Mechanisms:
      Whenever a person is subjected to excessive daily social or personal stressors, the person's mind try to deal with these stressors by using one or more of the coping mechanisms. One

of these coping mechanisms is the use of unconsciously or consciously determined defense mechanisms (Freudian) in order to cope or escape from the stressful situations. Among the defense mechanisms involved in the emergence of Somatoform Disorders are: Somatization, Conversion, Hypochondriasis, Dissociation, etc…

C. Social Factors:
   1. **Illness Behavior:** any behavior related to the illness done by the subject. For example, doctor visit, taking pills, and so forth might maintain the somatic complaints.
   2. **Sick Role:** Is the role given to the 'sick' individual by the society (for instance, attention obtained from the illness).

      Lack of reinforcement of non-somatic expressions of distress as well as cultural/social norms that devalue and stigmatize psychological suffering as compared to physical suffering might, altogether, slope susceptible individuals toward these disorders.

<u>Somatic Symptom Disorder</u>

DSM-5 criteria for diagnosis

A. One or more somatic symptoms that are distressing or result in significant disruption of daily life.
B. Excessive thoughts, feelings, or behaviors related to the somatic symptoms or associated health concerns as manifested by at least one of the following:
   1. Disproportionate and persistent thoughts about the seriousness of one's symptoms.
   2. Persistently high level of anxiety about health or symptoms.
   3. Excessive time and energy devoted to these symptoms or health concerns.
C. Although any one somatic symptom may not be continuously present, the state of being symptomatic is persistent (typically more than 6 months).

Prevalence

Exact prevalence is unknown. However, in adult population it may be around 5-7%. Females tend to report more somatic symptoms than do males. Consequently, the prevalence is likely to be higher among females.

Treatment

The physical complaints that occur in somatic symptom disorder are an expression of emotional issues. Psychotherapy is beneficial to help the patient find more appropriate ways of expression. The patient should have a primary care physician, and should be seen at regular intervals.

Illness Anxiety Disorder

In previous editions of DSM, this condition was known as 'Hypochondriasis'. The term 'Hypochondriasis' is one of the oldest medical terms, originally used to describe disorders believed to be due to diseases of the organs situated in the hypochondrium. However, the term, in psychiatry, used to denotes (disease conviction).

Also this condition might be known to several medical professionals as 'Health Anxiety', 'Health Obsession', or even 'Disease Phobia'. Nevertheless, one should not misinterpret this disorder as Anxiety, Obsessional, or Phobic disorders.

DSM-5 criteria for diagnosis

A. Preoccupation with having or acquiring a serious illness.
B. Somatic symptoms are not present or, if present, are only mild in intensity. If another medical condition is present or there is a high risk for developing a medical condition, the preoccupation is clearly excessive or disproportionate.
C. There is high level of anxiety about health, and the individual is easily alarmed about personal health status.
D. The individual performs excessive health-related behaviors (e.g., repeatedly checks his/er body for signs of illness) or exhibits maladaptive avoidance (e.g., avoids doctor appointments and hospitals).
E. Illness preoccupation has been present for at least 6 months, but specific illness that is feared may change over that period of time.
F. The illness-related preoccupation is not better explained by another mental disorder, such as somatic symptom disorder, panic disorder, generalized anxiety disorder, body dysmorphic disorder, obsessive-compulsive disorder, or delusional disorder, somatic type.

Specify whether:

Care-seeking type: medical care, including physician visits or undergoing tests and procedures, is frequently used

Care-avoidant type: medical care is rarely used.

Prevalence

The prevalence rate ranges from 4 - 9%. It is most frequent between ages 20 to 30 years and there is no sex predominance.

Treatment

Improvement usually results from reassurance through regular physician visits. Group rather than individual therapy is most helpful. Coexisting psychiatric conditions, likewise, should be treated.

<u>Conversion Disorder:</u> Functional Neurological Symptom Disorder

Definition

The term 'Conversion' introduced by Freud for the sudden 'conversion' of psychological distresses into physical (neurological) symptoms.

Previously, conversion disorder was known as 'Hysteria' = (Hystero) (Uterus- Greek). They named the condition like that because ancient physicians believed that this condition is due to mal-position of the uterus. Hence, they misleadingly believed that the condition is only occurring in female.

DSM-5 criteria for diagnosis

A. One or more symptoms of altered voluntary motor or sensory function.
B. Clinical findings provide evidence of incompatibility between the symptom and recognized neurological or medical conditions.
C. The symptom or deficit is not better explained by another medical or mental disorder.
D. The symptom or deficit cause clinically significant distress or impairment in social, occupational, or other important areas of functioning or warrants medical evaluation.

Clinical types

A. With weakness or paralysis
B. With abnormal movement (e.g., tremor, dystonic movement, myoclonus, gait disorder)
C. With swallowing symptoms
D. With speech symptom (e.g., dysphonia, slurred speech)
E. With attacks of seizures (called: Pseudo seizure)
F. With anesthesia or sensory loss
G. With special sensory symptom (e.g., visual, olfactory, or hearing disturbance)
H. With mixed symptoms

In these situations it's clinically important to differentiate between the conversional status and the true one. For example, the differences between true and pseudo seizures sound tricky; however, few clinical notes might be helpful (table 7-1).

Prevalence

Conversion disorder occurs in 2-5/100,000/year in the general population, and it constitutes up to 3% of psychiatric outpatients, and up to 5% of neurological outpatients' referrals. The disorder is more common in lower socioeconomic groups.

Table 7-1: True vs. Pseudo Seizures

Features	True Seizure	Pseudo Seizure
Typical tonic, clonic, tonic-clonic seizures	Present	Absent
Self-Harm	Present	Absent
Incontinence of urine	Present some time	Absent
Aura	Present	Absent
Post ictal confusion and headache	Present	Absent
Post ictal absence of corneal reflex	Present	Absent
Post ictal extensor planter response	Present	Absent
Post ictal raised prolactin level	Present	Absent
Sleep attacks	Present	Absent
Attacks even when the patient is alone	Present	Absent

Treatment

Symptoms classically last for days to weeks and typically remit spontaneously. Supportive, insight oriented or behavioral therapy can facilitate recovery.
Anxiolytics may also be helpful in some cases.

<u>Factitious Disorder:</u> Munchausen Syndrome

The name 'Munchausen' used for the first time to describe what is called todays 'Factitious Disorder' by Richard Asher in the journal of 'The Lancet' at 1951. He described a kind of self-harm where an individual, deliberately, assembles histories, and signs and symptoms of an illness so as to draw the attention of his/er surroundings.
The syndrome's name derived from Baron Münchhausen (Karl Friedrich Hieronymus Freiherr von Münchhausen, 1720–1797), a German nobleman working in the Russian army, who allegedly told

many fantastic and impossible stories about himself, which Rudolf Raspe later published as *The Surprising Adventures of Baron Münchhausen*.

DSM-5 criteria for diagnosis

A. Falsification of physical or psychological signs and symptoms, or induction of injury or disease, associated with identified deception.
B. The individual presents him/er self to others as ill, impaired, or injured.
C. The deceptive behavior is evident even in the absence of obvious external rewards.
D. The behavior is not better explained by another mental disorder, such as delusional disorder or another psychotic disorder.

Clinical presentation

Unlike 'Malingerers', individuals with this condition would not seek an extrinsic rewards like financial compensation, exemption from duties including military duties, school examinations, and so forth. Instead, the key feature of this condition is the intention of the sufferers to draw the attention of medical staffs, friends, or family members to their selves. I.e., they look after the 'Sick Role'.

The condition is not limited to manipulating minor medical conditions. In its place, its not uncommon for the sufferer to present a slide of 'leukemia', for instance, to convince the hematologist that s/he is a victim of this kind of tumor. Alternatively, s/he may inject urine or whatever another septic fluid in his/er knee joint so as to get real knee joint inflammation, or inject insulin in order to get 'hypoglycemia'.

Patients are able to provide a detailed history and describe symptoms of a particular disease. Common coexisting psychological symptoms include depression or factitious psychosis.

Furthermore, there is a sort of this disorder whereas the individual imposes an illness on another one, possibly under his/er care, like his/er children, or on his/er pet. In such an instance, the condition named 'Factitious disorder imposed on another', or 'Factitious Disorder by proxy', or 'Munchausen' Syndrome by proxy'. This kind is considered to be a criminal child or animal abuse.

Prevalence

It begins in early adulthood. More frequent in men and among health-care workers.

Exact prevalence is unknown yet. However, it is estimated to accounts for around 1% of hospital patients.

Treatment

No specific treatment exists, and the prognosis is generally poor. The condition should be recognized early, and needless medical procedures should be prevented. Close collaboration between the medical staff and psychiatrist is recommended.

Psychological Factors Affecting Other Medical Conditions

Psychosomatic medicine is not only limited to specific psychiatric disorders presented mainly as body-like symptoms. Rather, almost all medical conditions may be influenced by different psychological states.

How psychological factors affect other medical conditions? The impact of these factors may be through one or more of the following:

A. Affecting the course of the illness like precipitating, exacerbating, or maintaining the medical condition. For example, an individual with ischemic heart disease and anxiety symptoms. Anxiety symptoms like panic, fears, tachycardia, or palpitation may precipitate the ischemic changes, exacerbate, or delay recoveries as well.
B. Interfering with the treatment. For instance, an individual with malignant tumor and depressive symptoms. The depressive disorder may lead to 'hopelessness' and denial to seek professional helps.
C. Constitute additional health risks as in an individual with HIV infection and suicidal ideations.

Medical Conditions Causing Psychological Symptoms

Likewise, majority of medical conditions associated, more or less, with psychological signs and symptoms ranging from a mild 'flu-like illness' which may present with low mood, fatigue, reduced concentration, nightmares, and semi-delirious state due to possible associated fever, to the most sever life-threatening conditions like malignant tumors or HIV infections which commonly coupled with depressive clinical features.

Table (7-2) enumerates few medical conditions frequently presented with psychiatric signs and symptoms:

Table 7-2: medical conditions causing psychological symptoms	
Medical Conditions	Examples
Endocrine Disorders	Thyroid dysfunctions, Cushing syndrome, Addison disease, Diabetes Mellitus, etc…
Neurological Disorders	Stroke, Parkinson disease, Epilepsy, Huntington Disease, etc…
Nutritional Disorders	Folic Acid Deficiency, Thiamine Deficiency, etc…
Rheumatologic Disorders	Systemic Lupus Erythematosis, Rheumatoid Arthritis, etc…
Metabolic Syndromes	Hepatic Encephalopathy, Uremic Encephalopathy
Infections	HIV infection, Tuberculosis, etc…
Tumors	Any Malignant or even Benign Tumors

08 PERSONALITY DISORDERS

The term PERSONALITY derived from the Latin word PERSONA, which means mask.

Persona as in the everyday usage is a social role or a character played by an actor in a masked ball theatre. This **is** an Italian word that derives from the Latin for "mask" or "character”.

Moreover, the term persona derives from Latin {"per" meaning "through"} and {"sonare" meaning "to sound"}, meaning "through which the actor speaks,” i.e. a mask.

In the theatre of the ancient Latin-speaking world, the mask was not used as a plot device to disguise the identity of a character, but rather was employed to represent or typify that character.

Therefore, persona is the mask or the appearance one presents to the world. Importantly, the persona, used in this meaning, is not a pose or some other intentional misrepresentation of the self to others. Rather, it is the self as self-construed, and may change according to situation and context.

Therefore, Personality is the continuing pattern of thinking, feeling and behaviour which makes one individual distinguishable from another. Personality is stable – or at least relatively stable, i.e. we do not change dramatically from week to week.

General characteristics of personality traits or elements

A. Present since teenage years.
B. Consistent overtime.
C. Recognized by friends and relatives.
D. Stable in different situations.

Everyone has a collection of personality traits but if these traits lead to a personal distress or to problems in the social or occupational functioning, the person then may suffer from what is named personality disorder.

General diagnostic criteria for Personality Disorder according to DSM-5

A. An enduring pattern of inner experience and behavior that deviates markedly from the expectations of the individual's culture. This pattern is manifested in two (or more) of the following areas:

1. Cognition (i.e., ways of perceiving and interpreting self, other people, and events).
2. Affectivity (i.e., the range, intensity, liability, and appropriateness of emotional response).
3. Interpersonal functioning.
4. Impulse control.

B. The enduring pattern is inflexible and pervasive across a broad range of personal and social situations.
C. The enduring pattern leads to clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The pattern is stable and of long duration, and its onset can be traced back at least to adolescence or early adulthood.
E. The enduring pattern is not better explained as a manifestation or consequence of another mental disorder.
F. The enduring pattern is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition (e.g., head trauma).

The general characteristics of Personality Disorders

A. Personality Disorders are remarkably stable and "orderly"; i.e., predictable and unmalleable, so the term "disorder" is somewhat misleading.
B. They tend to be rigid & inflexible, show a restricted range of traits with a dominant single trait.
C. They have low grade chronic problems with no insight, and often no enough pain to seek help themselves; therefore they often forced to treatment, but difficult to be assessed.
D. An individual may suff**er** from more than one Personality Disorder.

Classification of personality disorders

Cluster A: they are odd and/or eccentric. This cluster comprises: Paranoid, Schizoid, and Schizotypal Personality disorders. This cluster of disorders shares features with Psychotic disorders. Most recently, in DSM-5, Schizotypal Personality Disorder is classified with Schizophrenia-spectrum disorders.

Cluster B: they are dramatic and / or erratic. This cluster includes: Antisocial, Borderline, Histrionic, and Narcissistic Personality disorders. This set of disorders resembles Bipolar Mood disorders.

Cluster C: they are anxious and / or fearful. This cluster embraces: Avoidant, Dependent, and Obsessive-compulsive Personality disorders. This group of disorders mimics Anxiety disorders.

Paranoid Personality Disorder

Clinical features

A. Highly suspicious and preoccupied with distrust toward others but no delusions or hallucinations.
B. Often hostile, irritable, argumentative and envious.
C. No confidants; the surrounding people have sinister motives.
D. Sense of exploitation.
E. Sensitivity to any feeling for insult, and develop grudges.

Prevalence

It is around 2.5% of general population.

Schizoid Personality Disorder

Clinical features

A. They are aloof.
B. Keep a distance from social relationships (withdrawn and reserved).
C. Few interests and pleasures in life. Takes pleasure in few activities.
D. Lacks close friends and aside from relatives.
E. Emotional coldness, detachment or flattened affect with little interest in sexual experiences.
F. Chooses solitary activities.
G. Indifferent to praise or criticism and they do not secretly wish for popularity.

Prevalence

It is prevalent in around 4% of general population.

Schizotypal Personality Disorder

Clinical features

A. Eccentric as well as odd thinking, behavior and appearance.
B. Speaks in unusual vague and circumstantial ways.
C. Ideas of reference (incorrect interpretations).
D. Unconventional beliefs as having extrasensory abilities, or excessive magical thinking as about superstition or telepathy.
E. Suspiciousness and paranoid ideation.
F. Inappropriate or constricted affect.
G. Lack of close friends.
H. Excessive social anxieties and Problems with close personal relationships.

Prevalence
It is prevalent in around 1.9% of general population.

Anti-social Personality Disorder
Clinical features
A. Failure to conform to social norms with respect to lawful behaviors as indicated by repeatedly performing acts that are grounds for arrest
B. Deceitfulness, as indicated by repeated lying, use of aliases, or conning others for personal profit or pleasure.
C. Impulsivity or failure to plan ahead.
D. Irritability and aggressiveness, as indicated by repeated physical fights or assaults
E. Reckless disregard for safety of self or others
F. Consistent irresponsibility, as indicated by repeated failure to sustain consistent work behavior or honor financial obligations
G. Lack of remorse (regret), as indicated by being indifferent to or rationalizing having hurt, mistreated, or stolen from another.

Prevalence
It constitutes around 3% of males and 1% of females in community samples. However, it accounts for up to 30% in substance abuse settings and a total of 40-75% of prisons population. Most of the male cases have history of conduct disorder (see chapter 12).

Borderline Personality Disorder
Clinical features
A. Feelings of emptiness and boredom.
B. Fragile identity with poor self-image.
C. Unpredictable, impulsive, argumentative, irritable with bouts of anger when their expectations are not met. Anger may directed toward themselves (self-mutilating behavior). Suicide threats as well as actions may happen.
D. Fear of rejection and loss.
E. Rapidly changing mood and behavior with intense unstable relationships.
F. Brief psychotic episodes may occur.
G. Splitting: divide persons into those who like & those who hate them.
H. Turning against the self (self-hate is prominent).

I. In their thirties or forties usually gain more stability.

Prevalence

It forms around 2% of general population. 75% are female and 90% have other psychiatric diagnosis.

Histrionic Personality Disorder

Clinical features

A. Wants to be the center of attention, and often interrupts others to be the center of conversation.
B. Rapidly shifting and shallow with excessive expression of emotion.
C. Very seductive "life of the party", but shows difficulty in relationships and may be frigid sexually.
D. Shows self-dramatization and exaggerate illnesses and friendships.
E. Suggestible; i.e., easily influenced by others or circumstances.
F. Consider relationships to be more intimate than they actually are.

Prevalence

It is around 1.84% of general population.

Narcissistic Personality Disorder

Clinical features

A. Inflated sense of their own importance with boastfulness.
B. Believe that they should associate with high-status people (or institutions) as they perceive themselves to be.
C. Unable to see that others may not appreciate them, but hyper-sensitive to criticism.
D. Offer help or facilities to others but to exploit them in the future.
E. Lack of empathy and litigious in the quarrels.
F. Grandiosity, egocentricity, vengeful, but low self-esteem.

Prevalence

It is around 3% of general population.

Avoidant Personality Disorder

Clinical features

A. Extreme sensitivity to criticism and disapproval, so avoid situations in which criticism may be given since they feel anxious, get tachycardia, sweating, and flushing.
B. Consider themselves to be socially incompetent or personally unappealing, and avoid social interaction for fear of being ridiculed or humiliated.
C. Inhibited, introverted with low self-esteem.
D. Shy and restrained.
E. Avoidance of intimacy.
F. Only involved if sure of praise.

Prevalence

It is around 2% of general population.

Dependent Personality Disorder

Clinical features

A. Pervasive psychological dependence on other people.
B. Has difficulty making everyday decisions without an excessive amount of advice, direction and reassurance from others.
C. Low opinion of their own abilities with fear of disapproval.
D. Fears of being alone & being unable to cope.
E. Not uncommon for such a patient to be living with a controlling, domineering, or overprotective person.

Prevalence

It is around 0.5% of general population.

Obsessive-compulsive Personality Disorder

Clinical features

A. Inflexibility, stubbornness, and desire for perfection.
B. Tend to stress perfectionism above all else, and feel anxious when they perceive that things aren't "right".
C. A sense of availability of only one way to do things.
D. Failure to appreciate creative solutions.
E. Over-meticulous attention to the system and details.
F. Failure to get things done as a result of excessive attention to detail.
G. A tendency to check and re-check, but absence of obsessional thoughts and compulsive behaviors.

Prevalence
It is around 5% of general population.

Management of Personality Disorder
Personality Disorder patients have no insight and lack awareness that they are the cause of their own relationship problems, so they rarely seek psychological help unless
compelled by others.

General principles in the management
A. Be Respectful with a non-judgmental attitude.
B. Acknowledge affect (emotion).
C. Be aware of the countertransference.
D. Set realistic expectations.
E. Set limits and provide consistent structure.
F. Maintain professional boundaries.
G. Label the maladaptive behaviors of the patient.
H. Seek consultation from a psychiatrist.

Medication
Pharmacological treatment has no proven usefulness. Antipsychotics, antidepressants as well as antianxiety drugs can, however, be used when there are psychotic, depressive or anxiety features or disorders. Nevertheless, severe behavioral dys-control as sometimes seen in anti-social and borderline types may respond to carbamazepine, impulsivity to SSRIs, affective dys-regulation to SSRI or mood Stabilizers.
Thus, medication choice based more on the target symptom less than the diagnosis.

Psychotherapy
Individual, family, and group psychotherapies, in addition to self-help groups may benefit patients with personality disorders.
Overall, treatment of Cluster A disorders is the most difficult and treatment of the Cluster C disorders seems most promising.
For borderline personality disorder, cognitive behavioral therapy shows promise through building tolerance for distresses and negative affects without being self-destructive, decrease self-harm behaviors (suicide, mutilation, substance abuse), decrease behaviors that interfere with therapy (missing sessions, lying), and learn to regulate emotions and improve interpersonal skills.

In anti-social personality disorder, there are difficulties in establishing therapeutic alliance. Treatment approaches are not very successful, and more success occur with prevention and early intervention.

09 PSYCHIATRIC EMERGENCIES

In recent years, psychiatry has been moving closer to middle-of-the-road of medicine. This development raises the needs of more proper recognition and management of acute emergency presentations of mental pathological signs and symptoms by entire physicians, not only the psychiatrists.
Any psychiatric disorder might turn out to an acute life-threatening condition; nevertheless, one can enumerate the common, yet not all, psychiatric emergencies to the following conditions:

A. Suicide
B. Agitation
C. Loss of consciousness
D. Catatonia
E. Refusal to drink or take food
F. Substances withdrawal or intoxication
G. Side-effects of psychotropic medicines
H. Severe anxiety like acute panic attack
I. Somatic-like presentation, like conversional seizure

This chapter will focus more on the most serious and fatal psychiatric presentation, the suicide, at first. Then after, definition and management of agitation is undertaken since agitation is frequently faced in the psychiatric emergency as well as ordinary wards.
For the rest of psychiatric emergencies like catatonia, substance withdrawal and intoxication, severe anxiety, somatic-like emergency presentations, and serious side effects of psychotropic drugs, the reader can refer to the relevant chapters.

Epidemiological Data
Psychiatric emergency rooms are used equally by men and women, more by single than married. About 20% of these patients are suicidal and 10% of are violent or agitated.
The most common diagnoses are Mood Disorders, Schizophrenia, and Alcohol Dependence. About 40% of all patients seen in psychiatric emergency rooms require hospitalization. Most visits occur during the night hours. Contrary to popular belief, studies have not found that use of psychiatric emergency rooms increases during a full moon, Christmas, Bairams, or holiday seasons.

Suicide

Definitions

Suicide, as a term, derived from the Latin (Suicidium) which came from (sui caedere); i.e., 'to kill oneself'.
Therefore, suicide can be defined as an act with a fatal outcome that is deliberately initiated and performed by the person in the knowledge or expectation of its fatal outcome.
Death wishes: is one's desire for death which is not necessary to ends up by suicide. Death wishes are common presentation of depressive disorders.
Suicidal thought (ideation): is one's thinking about attempting suicide which is not necessary to be preceded neither by death wishes nor by depressive disorders. Moreover, it's not mandatory to pave the road toward the act of suicide.
Suicidal plan: is the blueprint of the suicidal act. The "where", "when" and "how" to attempt the act. Suicidal plan usually preceded by 'thoughts', but it's not necessary to be practiced out. Additionally, it's not obligatory for the 'act' to be preceded by the 'plan'.
Suicidal attempt: is the trial. In other words, it's the actual suicidal challenge that may, however not always, ends up by the death. The 'attempt' might be virtuously 'impulsive' without any former 'plan'.
Committed suicide: is the death due to suicide.

Epidemiology

Suicide is the tenth leading cause of death. Approximately 0.5-1.4% of people die by suicide, about 800,000-1 million people die annually by suicide. Hence the mortality rate is about 11.6/100,000/year. Rates of suicide inflated by 60% from 1960s to 2012. For each 'committed suicide' there are 10-40 'attempts'.
Suicide is more common in India, China, Northren and Eastern Europe especially in Lithuania and Hungary, USA, and Canada. Females attempt four times more often than males; however, males commit four times more frequently than females.
'Attempt' is more prevalent among younger age groups. Whereas 'Commit' tends to be predominant among mid and older age groups.

Risk factors

A. **Mental disorders:** up to 90% of those who complete suicide had a mental disorder at the time of their death:
    1. Mood disorders: 10-20%
    2. Schizophrenia: 5%
    3. Substance and Alcohol misuse: 15%

4. Personality disorders: 10-15%

B. **Medical conditions:** any chronic disabling, or life threatening conditions that are complicated by "hopelessness" might lead to suicide. For instance: end stage tumors, AIDS, stroke, etc…
C. **Occupation:** unemployment is a risk factor for suicide. Among employed groups, it is found to be more prevalent among professionals who have frequent contact with life-threatening instruments like farmers, doctors, particularly dentists, surgeons, anesthetists, and even psychiatrists, veterinary surgeons, and veterans.
D. **Gender:** females attempt more, but males die more by suicide.
E. **Age:** younger groups attempt more, but older groups die more by suicide.
F. **Marital status:** singles attempt more, but widow/ers, and divorcees die more by suicide. Marriage and children are protective factors.
G. **Social classes:** suicide is more common among both social class V (non-skilled workers) and class I (professionals).
H. **Previous suicide attempt(s)** raise the risk of future suicides.
I. **Family history of suicide** increases the risk of suicide even in the absence of any family history of mental disorders.
J. **Race:** the rate is higher among Caucasians and aboriginal people.
K. **Life stresses,** especially loneliness, isolation, and loss experiences.

Means of suicide

It differs across different countries depending on the availability of different methods as well as the gender and the cultural background. In general, men attempt suicide by more violent means. In USA, firearm is the commonest way, whereas pesticides poisoning is the commonest mean in farming populations like Latin America and Pacific region. In Kurdish society, despite the lack of clear-cut scientific data, burning seems to sound the commonest way of suicide, principally among females (**Note:** the author acknowledges the necessity of approval/disapproval of this claim by scientific epidemiological data, which is lacking in Kurdistan up to the time of this publication). In one of the local studies, however, Mohammad Amin et al (2012) showed that 27% of burn victims in Sulaimani province were intentionally self-attempted burnings (Self-immolitation).

Recorded means of suicide globally include: firearm, hanging, car exhaust poisoning, drug poisoning, drowning, jumping, pesticides consumption, wrist cutting, and burning.

Explanations of suicide

Several theories have been postulated to explain why human species do kill themselves. Best way to summarize all justifications is the 'Biopsychosocial' model of approach. According to this model, suicide might be attributed to one or more of Biological, Psychological, or Social factors:

A. **Biological factors**
   1. **Genetic:** although genetic studies failed in detecting particular gene responsible for suicidal behaviors, there are several genetic evidences supporting the 'genetic hypotheses of suicide. These hypotheses are reinforced by both presence of family history of suicide in the absence of other mental disorders and by particular geographic distributions of suicide worldwide as well; for instance the 'J' distribution of suicide from North-Eastern Europe downward.
   2. **Neurotransmitters:** postmortem studies of suicide victims revealed back low levels of serotonin and serotonin metabolites in the cerebrospinal fluid (CSF) as well as low level of Brain-Derived Neurotrophic Factor (BDNF) in the hippocampus and prefrontal cortex.

B. **Psychological Factors**
   1. **Freudian School:** Freud proposed that every human being born with two main unconsciously determined motives (desires),Thanatos (the god of death), and Eros (the god of love). Hence, according to Freud, every human being has the desire of 'death'. However, such a desire is, unconsciously, counteracted by the opposite desire, 'love'. These two desires are manifested as 'aggression' and 'sex' correspondingly.
      In their conflicts with each other, a desire might overcome the next. Then, according to Freud, the power of 'Thanatos' may be behind our 'suicide' wishes.
   2. **Cognitive Theory:** here, it's the inability of the mind to find a solution of what s/he is concerning about. Then, it's the failure of the 'problem-solving strategy' which ended up by suicidal thoughts as the best solution.
   3. **Behavioral Theory:** according to this theory, 'Modelling' and imitation of others' behaviors, attitudes, and thoughts may distribute the suicidal behavior. For instance, there are records

of several cases of suicide which followed the suicide of a celebrity. The obvious example is the suicide case of Marilyn Monroe. It's not uncommon for media to exacerbate the pandemic of suicide by romanticizing the 'Fashion' of suicide.

C. **Social Factors**

1. **Durkheim's Theory:** Durkheim, a French sociologist, attempted to explain suicide upon sociological backgrounds. He divided suicide in to four social categories:
   a. **Egoistic suicide**: applies to those who are not strongly integrated into any social group. The lack of family integration explains why unmarried persons are more vulnerable to suicide than married, and why couples with children are the best protected group for instance.
   b. **Altruistic suicide**: applies to those who their susceptibility to suicide is stemming from their excessive integration into a group. For example, a soldier who sacrifices his life in battle, or an Islamic soldier who bombs oneself. The most famous example in history is the 'Kamikaze' group during World War II.
   c. **Anomic suicid**e: applies to persons whose integration into society is disturbed so that they cannot follow customary norms of behavior.
   d. **Fatalistic suicide**: applies to those who stick to excessive rules, orders, and society regulations.

2. **Suicide and Religions:** Suicide is generally considered equally sinful as murdering another person in contemporary Hindu, Christian, Islamic, and Jewish societies. However, Hinduism accepts a man's right to end one's life through the non-violent practice of fasting to death, termed '*Prayopavesa*'. But Prayopavesa is austerely restricted to people who have no desire or ambition left, and no responsibilities remaining in this life. Also, the norm of 'self-immolation' by widows after the death of their husbands was prevalent in Hindu society during the middle Ages. Recently, during late 20th and 21st centuries, suicide bombing became a religious and political cult of both civilian and military protests.

3. **Suicide and Philosophy:** despite the strong opposition of suicide by several philosophical disciplines, there are philosophies which advocate for suicide. Supporters of this position maintain that no one should be forced to suffer against

their will, particularly from conditions such as incurable disease, mental illness, and old age that have no possibility of improvement. They reject the belief that suicide is always irrational, arguing instead that it can be a valid last resort for those enduring major pain or trauma. A stronger stance would argue that people should be allowed to autonomously choose to die regardless of whether they are suffering or not. Noteworthy supporters of this thought discipline include Scottish empiricist David Hume and American bioethicist Jacob Appel.

Management of Suicide

A. Assessment

It's the duty of every physician to be capable of assessing the risk of suicide. The first requirement is a willingness to make tactful but direct enquiries about the patient's **intentions.** Unlike the popular believe, asking a patient about suicidal inclinations does not make suicidal behaviors more likely. On the contrary, the patient who has already thought of suicide will feel better understood when the doctor raises the issue: (Note: the assessing questions posted down are only examples, the assessor may adopt other more culturally suitable questions)

1. **Begin with questions that address the patient's feeling about living:**
   Have you ever felt life was not worth living?
   Did you ever wish you could go to sleep and just not wake up?
2. **Follow on with specific questions that ask about thoughts of death, self-harm, or suicide:**
   Is death something you have thought about recently?
   Have things reached the point that you have thought of harming yourself?
3. **For individuals who have thoughts of self-harm or suicide:**
   When did you first notice such thoughts?
   What led up to the thoughts (e.g., interpersonal and psychosocial precipitants, including losses, specific symptoms such as mood changes, anhedonia, hopelessness, psychotic symptoms).
   How often have thoughts occurred?
   How close have you come to acting on those thoughts?
   Have you ever started to harm yourself but stopped before doing something?

Have you made a specific plan to harm or kill yourself? If so what does the plan include?
(Note: The most obvious warning sign is a direct statement of intent.)

There is no truth in the idea that people who talk about suicide won't enact it. On the contrary, two-thirds of those who die by suicide have told someone about their intentions. Risk is also assessed by considering the presence of **risk factors** for suicide that outlined above.

4. **Completing the psychiatric history:**
   The interview should be conducted in an unhurried and sympathetic way that allows the patient to admit any despair or self-destructive intentions.
   Ask about current problems and the patient's reaction to them.
   Ask about physical illness.
   Ask about family history of suicide.
   Assess previous personality; the important points include mood swings, impulsive or aggressive tendencies.

B. Treatment of suicidal people

1. **General issues:** decide whether the patient should be admitted to hospital or treated as an out -patient. If suicidal risk is judged to be high, in-patient care is nearly always required. An occasional exception may be made when the patient lives with reliable relatives, but only if those relative understand their responsibilities, and are able to fulfill them. If outpatient treatment is chosen, patients should be given a telephone number with which they can, at all times, obtain help when feeling worse.
2. **Care of suicidal patient in the hospital:** The first requirement is to prevent patients from harming themselves. These include:
   a. Safe ward environment by minimizing the availability of means of self-harm. For instance, preventing access to open windows and other areas of buildings where jumping could lead to serious injury, as well as removal of potentially dangerous objects such as belts and razors.
   b. Special nursing arrangements may be needed, at times, so that the patient would never be left alone.
   c. Agree level of observation and if patient leaves ward without notice, take immediate action.

d. Treatment of any associated mental illness.
e. On discharge, prescribe adequate but non-dangerous amount of drugs, with frequent follow up.

3. **Care of suicidal patient in community:** requires the following actions:
   a. Full range of assessment should be carried out properly.
   b. Organization of adequate social support.
   c. Full dosage of safe psychiatric treatments.
   d. Choose less toxic drugs.
   e. Small prescriptions.
   f. Involve relatives in care of tablets.
   g. Arrange immediate access to extra help for patients and relatives.

C. Suicide prevention

Prevention of suicide may be the best strategy. However, it is the hardest applicable step. Anticipation and prevention of suicide might be secured through the following steps:

1. Better and more available psychiatric services.
2. Restricting means of suicide; reducing the availability of methods of suicide.
3. Restricting opportunities for imitation: the evidence on the importance of imitation as a factor in precipitating suicide suggests the need to persuade the media to take a responsible view of the reporting and portrayal of suicide.
4. Providing educational programs tackling the issue of suicide, yet not in a 'fashionable' way, and how to seek helps when it raised in one's mind.
5. Availability of crisis centers and 'hot lines'.

Agitation

Definitions

Agitation is the subjective feeling of being upset, angry, disturbed, or unable to rest. Whereas **Aggression**, is a destructive or a punitive behavior directed toward people or objects. However, **Violence** is an aggressive behavior that transgresses social norms (e.g., boxing is an aggression, but street fighting is violence).

When you receive an agitated patient in the emergency room try to:

A. Ensure your own security via:
1. Set your seat between the patient and the door.

2. Keep alarms by hand.
3. Let your colleagues know where and when you are going to interview the patient.

B. Think about the possible causes for his/er agitation from the beginning:
1. Medical conditions like delirium.
2. Substance/ drug misuse and dependency.
3. Psychiatric Disorders; for example, Acute Psychosis, Mood Disorders, Anxiety Disorders, Grief Reaction, Personality Disorders; etc…

C. Never confront the patient.

D. Try to calm the patient down verbally by reassuring sentences (de-escalation).

E. Offer help and agreement; offer food, a glass of water, etc…

F. Medications:
1. Try oral medications at beginning; if refused then parenteral.
2. Benzodiazepine is beneficial: lorazepam 1-2 mgs, midazolam 7.5-15 mgs, and diazepam 5-10mgs; repeat as it is neccessary after 20- 30 minutes and for maximum of three times. If you decide to give intravenous diazepam, try to deliver it slowly over 5 minutes to decrease the risk of respiratory depression. Keep flumazenil (a benzodiazepine antidote) ampule by hand, which should be given if the rate of respiration falls below 10/minute (ampule of Flumazenil contains 0.5mgs/5ml; give 0.2mgs over 30 seconds, if the consciousness does not return after 30 seconds, then give the rest of the ampule over another 30 seconds. If the patient stayed comatosed, then give another 0.5 mgs and repeat up to a maximum of 3 mgs. If after 5 minutes the patient is not awakened, then the cause of unconsciousness is not caused by benzodiazepine).
3. Antipsychotics: haloperidol 5-10mgs, olanzapine 5-10mgs, ensure that anticholinergics or antihistamines are available before giving antipsychotics so as immediately treat quickly occurring dystonia.
4. Check vital signs each 5 minutes for the first hour and then half an hourly until the patient become ambulant.

G. Restraints:
1. Preferable to be done by at least 4 well trained persons.
2. Explain to the patient why s/he is going to be restrained.
3. Reassure the patient continuously.
4. Use Leather restraints, but not shackles.
5. Try to elevate the head of the patient slightly, to prevent aspiration and to make the patient more comfortable.

6. The hands should be restrained in a manner that intravenous routes could be accessible if needed.
7. After restraining the patient, start treatment.

When the patient become under control, remove the restraints by removing each one after 5 minutes interval, until only two remained, and then remove the remaining two restraints at once.

PART II

Specialties In PSYCHIATRY

10 SUBSTANCE-RELATED AND ADDICTIVE DISORDERS

The impact of alcohol and substances is not only limited to a particular syndrome of 'dependence' or 'addiction. Additionally, alcohol and substances might predispose to almost all kinds of psychiatric disorders, like psychotic, mood or anxiety disorders. Furthermore, substances may maintain and even prolong the course of a preexisting psychiatric disorder. These, undoubtedly, beside their medical complications.
APA lists 10 separate classes of substances:

A. Alcohol
B. Opioids
C. Cannabis
D. Stimulants
E. Caffeine
F. Tobacco
G. Sedatives, Hypnotics, and Anxiolytics
H. Hallucinogens
I. Inhalants
J. Other (unknown) substances

There are several other kinds of 'addiction', especially behavioral addictions like Gambling, Internet Gaming, Sex Addiction, etc…This chapter, however, will only deal with substance-related disorders. In this scene, APA recognized two groups of disorders: substance-use disorders and substance-induced disorders. The latter include: intoxication, withdrawal, and substances-induced mental disorders (see relevant chapters).
Therefore, this chapter begins with a general discussion of features of the main three problems: substance-use disorder, substance intoxication, and substance withdrawal.
The remainder of the chapter is ordered according to the class of substances and labels their unique features. At the end, the chapter will close up by exploring general epidemiological data, etiology, and management of substance-related disorders.

Pathophysiology of dependence

All drugs that are taken in excess share the stimulation of the brain reward system, which is, more or less, dopamine-dependent

and it's the base for 'behavior' reinforcement and memory-pathway formation. The intensity of stimulation done by these drugs overrides those of other rewarding activities. Therefore, normal, adaptive activities like one's job or social life might be left behind unkempt for the sake of looking after the targeted substance.
Additionally, predisposed Individuals may show lower levels of self-control; i.e., more impulsive reflecting possible impairments of brain inhibitory mechanisms.

Substance Use Disorders

The corner-stone feature of substance-use disorder is the continuous use of the substance by an individual despite a group of psychological and medical signs and symptoms and regardless of evidences of problems related to that substance's usage.

DSM-5 criteria for diagnosis

DSM-5 identified four groups of criteria to diagnose substance-use disorder:

Impaired control over the substance use:

1. The individual may take the substance in larger amounts or over a longer period than was originally intended.
2. Persistent desire to cut down or regulate substance use and may report multiple unsuccessful attempts to reduce or take off the substance.
3. The individual may spend a great deal of time obtaining, using, or recovering from effects of the substance.
4. Craving: its manifested by an intense desire or urge for the substance that may occur at any time, but is more likely when in an environment where the substance was previously obtained or used. Craving is, every so often, used to measure the treatment outcome.

Social impairment:

5. Recurrent use of the substance my lead to failure in fulfilling major role obligations at work, school, or home.
6. Continuous use of the substance despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of the substance.
7. Important social, occupational, or recreational activities may be given up or reduced because of substance use. For instance, the individual may withdraw from family activities and hobbies in order to use the substance.

Risky use:

8. Recurrent substance use in situations in which it is physically hazardous.
9. Continuous substance use despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the substance.

Pharmacological criteria:

10. Tolerance: defined as necessity of a markedly higher dosages of the substance to attain the desired effect; or, a noticeably reduced effect when the ordinary dose is consumed. Features of tolerance vary depending on the substance used.
11. Withdrawal: is a syndrome occurs as a consequence of declination in tissue/blood concentrations of the substance that has been used over a prolonged period of time. Similar to 'tolerance', features of 'withdrawal' depend on the kind and property of the misused substance.

Its not mandatory for all the above 11 criteria to be evident for diagnosing substance-use disorder. Evidences of two (or more) are satisfactory to label the diagnosis. Furthermore, neither tolerance nor withdrawal is necessary for the diagnosis.

Substance Intoxication

Intoxication is a transient syndrome happens due to recent substance ingestion that produce clinically significant psychological and physical impairment. Frequently, intoxication referred to as 'drunken' among lay people.

Each substance has its own specific 'intoxication' features depending on its pharmacological properties. However, DSM-5 recognized general prototype criteria for intoxication:

A. Development of a reversible substance-specific syndrome due to recent ingestion of a substance.
B. The clinically significant problematic behavioral or psychological changes (A) are attributable to the physiological effects of the substance on the central nervous system (CNS) and develop during or shortly after use of the substance.
C. Description of more substance-specific signs and symptoms, and/or significant socio-occupational impairments.
D. The symptoms are not due to another medical condition and are not better explained by another mental disorder.

Tobacco is the only listed substance which lacks specific intoxication syndrome.

Notably, presence of intoxication doesn't necessarily verify presence of 'substance-use disorder'.

Substance Withdrawal

In the vein of 'intoxication', there are specific withdrawal features for each substance. Nevertheless, DSM-5 acknowledged an overall criteria pattern for withdrawal:

A. Development of substance-specific problematic behavioral changes, with physiological and cognitive concomitants, that is due the cessation of, or reduction in, heavy and prolonged substance use.
B. Description of more substance-specific signs and symptoms of withdrawal.
C. The syndrome causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The symptoms are not due to other medical or mental disorders.

DSM-5 doesn't recognize specific withdrawal features for both 'Hallucinogens' and 'Inhalants'.

Unlike intoxication, withdrawal is usually, but not always, associated with a substance use disorder. Indeed, withdrawal distress may be the most influential factor behind the individual's urge to re-administer the substance and, henceforth, 'substance-use disorder' would be the outcome.

Alcohol

Although alcohol was used for several reasons thousands years ago, the first alcohol was only discovered by the Persian chemist, Al-Razi, in the tenth century. The term 'alcohol' derived from the Arabic term 'Al-Kuhl' which is a powder used as an eyeliner for cosmetic reasons and as an antiseptic.

Chemically, there are around 20 alcohol substances. Nevertheless, the kind present in beverages is 'Ethanol'. Despite presence of hundreds of marketing alcohol beverages, the percent of ethanol within drinks ranges from 3% to 48% which roughly classified to three classes: beer, cidar, and soft vodka (3-10%); wines (10-20%); and spirits (>20%) like whisky, rum, vodka, and so forth.

Alcohol is regarded to be the third commonest substance of misuse next to Caffeine and Tobacco. Despite different drinking-legislations across different states, alcohol drinking is legal,

currently, in almost all countries excluding a number of Islamic countries.
The term ‘Alcoholism’ was widely used in the earlier medical writings to denote mental, social, as well as medical harmful and habitual use of alcohol. However, the pejorative meanings of ‘Alcoholism’ let professionals to be hesitant for its technical usage.

Alcohol’s unit

The medically accepted safe level of alcohol consumption is expressed in ‘unit’. Several health authorities, worldwide, advice not to regularly drink more than 3-4 units a day for men, and 2-3 units for women. However, with so many kinds of alcohol drinks from lager, wine, to spirit, as well as different size glasses from shots, pints, to bottles, one may easily get confused about how many units the drink contains.
Basically, each ‘alcohol unit’ is equal to a pure 8 grams (g) or 10 milliliters (ml) of alcohol. Hence, one should not habitually drink more than 24-32 grams or 30-40 ml of pure alcohol a day.
To calculate the units of alcohol per beverage, one should know both the size of the beverage and the strength of alcohol within it. The latter is expressed in percent (%), Alcohol by Volume (ABV), or only Volume (Vol). Both the size of the beverage and the strength of alcohol are written on labels of the cans or bottles. Therefore, a bottle of wine that says 12% alc or 12 ABV means that 12% of the volume of that drink is purely alcohol.
One can work out to calculate the units of a drink by multiplying the total size of the beverage in (ml) by its alcohol strength in % or ABV and dividing the result by 1000:
{A unit of alcohol = Volume (ml) x Strength (%) / 1000}
For example, one liter of 43% Whisky contains 43 units of alcohol which is equivalent to 430 ml or 344 g of pure alcohol. Whereas a pint (568ml) of 5.2 ABV lager contains 2.95 units of alcohol which is equal to 29.5 ml or 23.6 g of pure alcohol.

Alcohol’s energy

Alcohol may be a direct reason of obesity, apart from its indirect effects through associated decreased physical activities and increased appetite. Each gram of alcohol provides around 6.5-7 kcals; hence, one alcohol unit may provide the body with up to 56 kcals. If an individual drinks 4 units daily, then s/he receives a 224 kcal beside the kcal of the beverage. For instance, a 350 ml beer

contains 10-15 grams carbohydrates which are equivalent to 40-60 kcals, per se, in addition to the kcals of the alcohol content.
Consequently, alcohol is an extra, unnecessary, source of energy which should be taken in account, especially among those who attempt to reduce weight and among diabetic patients.

Alcohol intoxication: DSM-5 criteria

A. Recent ingestion of alcohol.
B. Clinically significant problematic behavioral or psychological changes (e.g., inappropriate sexual or aggressive behavior, mood lability, impaired judgment) that developed during, or shortly after, alcohol ingestion.
C. One (or more) of the following signs or symptoms developing during, or shortly after, alcohol use:
   1. Slurred speech.
   2. Incoordination.
   3. Unsteady gait.
   4. Nystagmus.
   5. Impairment in attention or memory (blackout: it's a short-term amnesia frequently reported after heavy drinking when the individual forgot what happened last night during drinking despite meantime intact consciousness).
   6. Stupor or coma.
D. The signs and symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Alcohol withdrawal: DSM-5 criteria

A. Cessation of (or reduction in) alcohol use that has been heavy and prolonged.
B. Two (or more) of the following, developing within several hours to a few days after the cessation of (or reduction in) alcohol use described in criterion A:
   1. Autonomic hyperactivity (e.g., sweating or pulse rate greater than 100 bpm).
   2. Increased hand tremor.
   3. Insomnia.
   4. Nausea or vomiting.
   5. Transient visual, tactile, or auditory hallucinations or illusions.
   6. Anxiety.
   7. Generalized tonic-clonic seizures.

C. The signs and symptoms in criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The signs and symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Delirium Tremens (DT)

DT is a kind of toxic delirium occurs as a consequence of withdrawal from prolonged and substantial alcohol drinking. Here, there is marked tremor in addition to the cardinal features of any delirium (see chapter 11). It is a life threatening with a mortality rate of almost 5%.

Symptoms of DT include impaired consciousness, almost all sensory modalities hallucinations, delusions, insomnia, agitation, autonomic hyperactivity (tachycardia, hypertension, sweating, and fever), as well as noticeable tremor which usually reach peak after 72-96 hours of last drink. DT is more common among heavy drinkers, older age groups, and individuals with underlying medical conditions. DT is a medical emergency and calls immediate hospitalization.

Complications of alcohol-use disorder

Habitual consumption of relatively large amounts of alcohol may affect almost all body organs, especially gastrointestinal, cardiovascular, and nervous systems as well as mental states:

A. Gastrointestinal tract:
1. Gastritis.
2. Peptic ulcer.
3. Liver cirrhosis and/or pancreatitis in about 15% of heavy users.
4. Cancer.

B. Cardiovascular system:
1. Hypertension.
2. Increased risk of ischemic changes secondary to hyperlipidemia.
3. Cardiomyopathy.

C. Nervous system:
1. Myopathies.
2. Peripheral neuropathies.
3. Cerebellar ataxia.

D. Psychiatric complications: repeated alcohol intakes might contribute to the emergence or maintenance of almost all psychiatric disorders; yet, mostly to:
1. Neurodegenerative disorders: including Wernicke-Korsakoff syndrome and dementia due to the direct effect of alcohol, head injury, or thiamine deficiency as well as delirium.
2. Mood disorders, especially depressive disorders.
3. Psychotic disorders: especially during withdrawal.
4. Anxiety disorders: especially during withdrawal.
5. Sleep disturbances: especially during withdrawal.
6. Sexual disorders: secondary to vascular disorders, neuropathies, as well as mood changes and diminished libido.
7. Personality disorders: personality disorders, particularly cluster B disorders are risk factors for alcohol use disorder. However, continuous alcohol consumption may be a maintaining factor for these disorders.
8. During alcohol intoxication, the risk of suicide may be high even without clear mood changes.

Laboratory tests

Although laboratory tests do not establish a diagnosis of an alcohol-related disorder, yet these measures can be useful in emphasizing individuals who need further assessments. These laboratory findings include:

A. Elevation of gamma-glutamyltransferase (GGT) level.
B. Elevation of carbohydrate-deficient transferrin (CDT) level.
C. Elevated mean corpuscular volume (MCV).
D. Impairment of liver function test.
E. Disturbed lipid profile.

Management of alcohol-related disorders

Alcohol intoxication require no more than supportive measures until alcohol level lessened, spontaneously, in the body. Nevertheless, should the individual be agitated or disinhibited, then relevant measures must be applied.

For alcohol use disorder, particularly for withdrawal syndrome, a comprehensive biopsychosocial strategy is required to help the individual overcomes the condition.

Alcohol withdrawal and detoxification can be done safely at home providing that there are regular supervision. Not all patients require pharmacological intervention.

Before starting any alcohol withdrawal and detoxification programs, a target point should be addressed and agreed upon by both therapists and the patient.

Gradual reduction in the frequency and amount of alcohol drinking toward the safe alcohol levels according to daily units (see above) is one of the therapeutic approaches. However, several studies showed evidences of poor responses and possible relapse of heavy drinking. Instead, complete abstinence may be the best strategy. Nevertheless, there might be severe withdrawal syndrome especially during the first couple of weeks.

Meanwhile, pharmacological coverage may be necessary to prevent sufferings as well as fatal outcomes of alcohol withdrawal.

Indications of in-patient detoxification

A. Severe, heavy, and prolonged drinking.
B. Previous history of DT or seizure.
C. Previous history of failed community detoxification.
D. Concurrent substance misuse (polydrug use).
E. Concurrent psychiatric disorders.
F. Concurrent chronic medical illnesses.
G. Poor social and family supports.

Pharmacotherapy

A. Since alcohol is Gamma Amino Butyric Acid (GABA) receptor agonist, its preferable during withdrawal to replace the brain with GABA receptor agonists too. For instance, benzodiazepines (BZN) particularly long-acting BZNs like Chlorodiazepoxide is advisable. However, short-acting BZNs like Oxazepam is preferable if there are clinical and laboratory evidences of liver failure. If the withdrawal syndrome complicated by seizure, then Diazepam is more efficient in controlling the seizure than Chlorodiazepoxide.

   The dosage of Chlorodiazepoxide and duration of therapy depend on the severity of withdrawal syndrome and alcohol use disorder. Overall, a starting dose of 100 mgs over 4 divided doses is satisfactory, with gradual reduction of the dosage by one quarter daily. The clinician may manipulate the doses according to response and withdrawal signs and symptoms. Nevertheless, a starting dose of more than 250 mgs is almost always unnecessary.
B. Thiamine supplementation: in the form of parenteral vitamin B-compound instead of oral tablets since thiamine is not

effectively absorbed. Thiamine is useful as both prophylaxis against and treatment of existing Wernicke Korsakoff Syndrome (WKS) secondary to alcohol misuse. 100 to 300 mgs of thiamine daily for 3-5 days or as long as improvement continues are worthwhile.

C. Relapse prevention: after detoxification, the next aim would be to prevent relapse of drinking. During this stage, both psychological and social therapies inform of cognitive behavioral, group, and supportive therapies (see chapter 18) as well as biological therapy might be necessary.

Acamprosate, which acts on glutamate and GABA neurotransmitter systems has been found to have modest effect in supporting alcohol abstinence. Disulfiram (Antabuse), which inhibits aldehyde dehydrogenase leading to acetaldehyde accumulation after drinking and an unpleasant physical effects including vomiting, also found to be, more or less, useful in controlling drinking. Recently, Naltrexone which is an opioid receptor antagonist, and Topiramate which is anticonvulsant have been recorded to be useful in reducing alcohol craving.

Opioid

There are different nomenclatures unaccountably used by both publics and professionals. These include: Narcotic, Opioid, Opiate, and Opium.

Narcotic: This term basically referred medically to any substance with sleep-inducing properties. In the USA, the term has been associated with opioids, commonly morphine and heroin. The term is, today, loosely defined and typically has negative connotations. When used in a legal context, a *narcotic* drug is simply one that is totally prohibited. The term has no biological, pharmacological, or medical ground. Therefore, its better to be avoided.

Opium (poppy tears): is the dried latex obtained from the opium poppy (*Papaver somniferum*) plant. It contains alkaloids like morphine, codeine, and thebaine in addition to non-analgesic alkaloids such as papaverine and noscapine. The old-fashioned method of obtaining the latex is to cut the immature seed pods (fruits) by hand; the latex leaks out and dries to a sticky yellowish residue that is later scraped off, and dehydrated.

Main producers of this plant are Afghanistan, Pakistan, India, Turkey, Laos, Burma, Mexico, Colombia, and Hungary. However, USA and Europian countries are the main consumers. The price of one kg active substance might reach 16,000 US dollar.

Both the medical and non-medical uses of opium are dated back to thousands years ago since Sumerian era. The primary medical usage of opium was in anesthesia.

Opiate: In medicine, the term **opiate** describes any of the alkaloids found as natural products in the opium poppy plant (see above). This term is often mistakenly used to describe all drugs with opium- or morphine-like pharmacological action, which are more properly classified under the broader term *opioid*.

Opioid therefore is a broad term comprises natural alkaloids (opiate or opium), semi-synthetic and synthetic opioids (see table 10-1).

Opioid receptors

Opioid receptors are a group of G protein-coupled receptors distributed widely throughout nervous system and gastrointestinal tract (see table 10-2).

Table 10-1 Opioids

Name	**Trade name**
Natural (opiate)	
Morphine	Hundreds of names
Codeine	Hundreds of names
Thebaine	Not used therapeutically
Semi-synthetic	
Diacetylmorphine	Heroin, Diamorphine
Hydromorphone	Dilaudid
Oxymorphone	Numorphan
Hydrocodone	Hycodan
Oxycodone	Roxicodone
Synthetic	
Meperidine (Pethidine)	Demerol
Methadone	Dolophine
Propoxyphene	Darvon
Tramadol	Tramal, Ultram
Synthetic-antagonists	
Naloxone	Narcan
Naltrexone	ReVia
Levallorphan	Lorfan
Synthetic-partial agonists	
Pentazocine	Sosegon, Talwin
Butorphanol	Stadol
Buprenorphine	Buprenex
Nalorphine	Lethidrone
Endogenous opioids	
Dynorphin	
Enkephalin	
Endorphin	
Endomorphin	
Nociceptin	

Table 10-2: Opioid receptors			
Receptor	Subtypes	Location	Functions
Delta	δ_1, δ_2	Nervous system	Physical dependence, analgesia, euphoria, convulsion
Kappa	κ_1, κ_2, κ_3	Nervous system	Psychological dependence, analgesia, dysphoria, anticonvulsant effect, miosis, sedation
Mu	μ_1, μ_2, μ_3	Nervous system, GIT	Physical dependence, analgesia, euphoria, miosis, sedation, reduced bowel motility and gastric empty time (constipation, nausea, and vomiting)
Nociceptin	ORL_1	Central nervous system only	Development of tolerance to Mu receptors, mood and appetite changes

Opioid's usages

A. Medical indications:

1. Analgesia: especially for post-operative pains, end-stage tumors, rheumatic problems where other analgesics are useless or contra-indicated. However, its not advisable to prescribe opioids for headache since it may exacerbate the pain and impair alertness. Also, opioids are not beneficial for the neuropathic pains.
2. Anti-diarrhea, since it reduces bowel motility. Nonetheless, constipation, nausea, and vomiting may be complicated side effects.
3. Sometimes smaller doses of codeine or similar opioids may be used as a symptomatic treatment of flu-like illnesses since they reduce respiratory and eyes congestions, in addition to relieving associated fatigue, malaise, and aches.
4. Cough suppressant in forms of syrup or tablet for upper respiratory tract infections.
5. Treatment of shortness of breath in end-stage medical conditions like lung cancer.
6. Treatment of opioid use disorders as a substitution therapy.
7. In the past, opioids were in place for anesthesia and as an antidepressant agent.

B. Illicit usages:

Since opioid induces a strong sense of euphoria "high", analgesia, sedation, and a robust sense of wellbeing, it was subjected to illicit uses (misuses) throughout history. However, its not only the psychological wellbeing behind it's use disorder (dependence); rather, the severity of withdrawal downside signs and symptoms as well as the immediacy of pharmacological tolerance development are all behind habitual opioid use disorders.

Traditionally, opium was smoked via relatively large pipe named “dream stick”. Colloquial speech terms for opium include: “midnight oil”, “tar”, “dope”, and “Big O”. Though “tar” and “dope” may refer to heroin among different users as well.

Nowadays different slangs applied to the illicit uses of opioids depending on the country and the routes of administration. For instance, “chasing the dragon” is a slang for smoking heroin. Whereas “chasing the white dragon “is methamphetamine smoking. Furthermore, morphine known as “M”, “Vitamin M”, “morpho”, “sister morphine”, “Miss Emma”, “Blue velvet which is a mixture of morphine and antihistamine” and several others.

Besides smoking, injection is also a common route of opioid administration, particularly heroin. Its injection is also known under different names among its users. Take for example "slamming", "banging", "shooting up", "digging", "mainlining", “spike up” or “Speedball which is injection of a mixture of heroin and cocaine”. This method of usage carries relatively greater risks than other methods of administration. Heroin base (in Europe), mixed with acids like lemon juice and dissolved in water then heated slowly. However, Heroin (in USA) is found in the hydrochloride salt form, requiring just water (and no heat) to dissolve.

Furthermore, insufflation (snorting) by powdering the heroin and then snorting it with a straw or a rolled up banknote is another route of administration beside occasional uses of suppositories.

Finally, there are several other slangs to denote the act, type of opioid, the route of administration, as well as adjectives for the state effects of opioids, like: “Push off (verb): to do illegal drugs specially heroin”, “Nod (verb): to fall asleep due to effects of illicit drugs specially opioids”, “Jammed (adjective): irritable or mad”, “Smacked (adjective): highly intoxicated by the substance specially heroin and marijuana”. For instance, one might say “Alex got jammed last night, yet not smacked”.

Opioid intoxication: DSM-5 criteria

A. Recent use of an opioid.
B. Clinically significant problematic behavioral or psychological changes (e.g., initial euphoria followed by apathy, dysphoria, psychomotor agitation or retardation, impaired judgment) that developed during, or shortly after, opioid use.
C. Pupillary constriction (or pupillary dilatation due to anoxia from severe overdose) and one (or more) of the following signs or symptoms developing during, or shortly after, opioid use:
   1. Drowsiness or coma.
   2. Slurred speech.
   3. Impairment in attention or memory.
D. The signs and symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Opioid withdrawal: DSM-5 criteria

A. Presence of either of the following:
   1. Cessation of (or reduction in) opioid use that has been heavy and prolonged (i.e., several weeks or longer).
   2. Administration of an opioid antagonist after a period of opioid use.
B. Three (or more) of the following developing within minutes to several days after criterion A:
   1. Dysphoric mood.
   2. Nausea or vomiting.
   3. Muscle aches.
   4. Lacrimation or rhinorrhea.
   5. Pupillary dilation, piloerection, or sweating.
   6. Diarrhea.
   7. Yawning.
   8. Fever.
   9. Insomnia.
C. The above signs and symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The signs and symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Management of opioid use disorders

Treatment of intoxication:

In cases of mild intoxication, supportive measures might be enough until the body metabolize the opioid. Notwithstanding, in severely intoxicated individuals with disturbed consciousness or even coma, pupil constriction, and respiratory depression, admission to intensive care unit is almost a priority providing mechanical ventilation, oxygen, in addition to Naloxone intravenous infusion at a rate of 0.8 mgs per 70 kg of body weight.

With Naloxone infusion, signs of improvement occur within minutes. Nowadays, in several emergency rooms and intensive care units, injection of a combination of hypertonic glucose, thiamine, and naloxone is practiced even if the cause of coma is not opioid intoxication.

Detoxification, treatment of withdrawal syndrome, and maintenance therapy:

A. Stoppage of the illicit drug or substance.
B. Symptomatic treatment of:
   1. Pain; by non-steroidal anti-inflammatory drugs.
   2. Diarrhea; by loperamide.
   3. Nausea or vomiting; by metoclopramide.
   4. Insomnia; by sedating agents like benzodiazepine, but with caution.
C. Prescribing substitution therapy:
   1. Short-term treatment:
      a. Alpha-2 receptor agonist like.
      b. Partial agonist opioid like Buprenorphine has better outcome than Clonidine.
   2. Maintenance treatment:
      a. Methadone, a synthetic opioid, is a beneficial substituent from the first day of detoxification and can be used for longer period of time, especially during pregnancy. However, it should be administered slowly. The therapist should notice that Methadone is at least potent as heroin, but with longer half-life.
      b. Opioid receptor antagonist like Naltrexone likewise can be used as a maintenance therapy, though it has less promising effects than Methadone.

Cannabis

The term cannabis is Greek in origin and its related to the Persian term 'Kanab'. This substance is in use since 2000 BC. It's a derivative of the flowering plant 'Cannabis Sativa' which is mainly produced in central and southern Asia.

Pharmacologically, the main active ingredient is Tetrahydrocannabinol (THC) which acts on two cannabinoid receptors in the nervous system. Additionally, THC also exerts agonistic properties on serotoninergic receptor (5HT A1) which may explain its anxiolytic and antidepressant properties, as well as enhancing dopamine neurotransmission which may explain the high mood associated with its usage; yet, its downside side effect of inducing psychosis, particularly schizophrenia when used for prolonged period of time as well.

Cannabis is known among publics and professionals under different names including: "marijuana", "grass", "Mary Jane", "Ganja", "Weed", "Hashish", "Hash Oil", "Kief" and so forth. Commonly, the plant is cut, dried, sliced, and rolled into cigarettes, merely like tobacco, named "Joint".

Cannabis is the commonest illicit substance of misuse. Although its forbidden in several countries, few other states regulated its usage under different legislations. However, it is used medically to overcome nausea, vomiting, pain, and even depression among end-stage tumor patients in some countries.

Effects

The effects of cannabis is similar, more or less, to those of alcohol. That is to say, effects are dose, mood, person's expectation, and social setting dependent. Cannabis appears to amplifies the preexisting mood whether elation (high spirits) or depression. Moreover, it may distort time sensation, perception, and sleep besides increasing appetite, obesity, dry mouth, and tachycardia.

Side effects

Although cannabis is regarded to be safer than tobacco, it has been reported to be teratogenic, and carcinogenic, especially for the respiratory tract. These in addition to its psychological adverse effects like perceptual disturbances, which turn driving to be risky, anxiety, and transient paranoid ideations. But, established psychoses are reported after chronic usage.

Cannabis intoxication: DSM-5 criteria

A. Recent use of cannabis.
B. Clinically significant problematic behavioral or psychological changes (e.g., impaired motor coordination, euphoria, anxiety, sensation of slowed time, impaired judgment, social withdrawal) that developed during, or shortly after, cannabis use.
C. Two (or more) of the following signs or symptoms developing within 2 hours of cannabis use:
   1. Conjunctiva injection.
   2. Increased appetite.
   3. Dry mouth.
   4. Tachycardia.
D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Cannabis withdrawal: DSM-5 criteria

A. Cessation of cannabis use that has been heavy and prolonged (i.e., usually daily or almost daily use over a period of at least a few months).
B. Three (or more) of the following signs or symptoms develop within approximately one week after criterion A:
   1. Irritability, anger, or aggression.
   2. Nervousness or anxiety.
   3. Sleep difficulty (e.g., insomnia, disturbing dreams).
   4. Decreased appetite or weight loss.
   5. Restlessness.
   6. Depressed mood.
   7. At least one of the following physical symptoms causing significant discomfort: abdominal pain, shakiness/tremors, sweating, fever, chills, or headache.
C. The signs or symptoms in B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Treatment

There is no specific pharmacotherapy for cannabis-related disorders. Only general measures, symptomatic fixation, psychotherapies and supports applied to any substance-related disorder (see below) can be practiced for cannabis users as well.

Stimulants

The term 'stimulant' or 'psychostimulant' refers to any substance, including prescribing drugs, that brings temporary enhancement in physical and/or mental abilities like feeling of wellbeing, increased attention and concentration, improved recall and memory, and reduced feelings of tiredness.
Pharmacologically, stimulants exert their effects through activation of several neuronal circuits and neurotransmission systems, namely Dopaminergic, Adrenergic, Serotoninergic, and Glutaminergic systems.

Effects and medical indications of stimulants

A. Increase in attention and goal directed activities: treatment of Attention Deficit Hyperactivity Disorder.
B. Sleep disturbances particularly insomnia: like treatment of Narcolepsy and Hypersomnia as well as to prolong wakefulness by long-distance truck drivers and students.
C. Appetite suppressant: treatment of obesity.
D. Anxiolytic and euphoric effects: treatment of resistant anxiety and depressive disorders.
E. Libido enhancer: treatment of sexual dysfunctions.
F. Treatment of headache, like combination of caffeine and painkillers.
G. Treatment of nicotine dependence, like Bupropion and nicotine replacing therapies.
H. Vasoconstriction effects: treatment of congestion-induced disorders as decongestants in forms of eye or nasal drops, or in combination with temperature-lowering agents (e.g., Paracetamol) as over-counter pills to treat flu-like illnesses.
I. Sympathomimetic effects like increase in heart rate and blood pressure: treatment of bradycardia or hypotension. For example Pseudoephedrine uses for post-anesthesia hypotension.
J. Treatment of neuro-medical movement disorders, for instance L-dopa for the treatment of Parkinson's disease.

Side effects

A. Cardiovascular: tachyarrhythmia, hypertension, ischemic heart diseases, and stroke.
B. Obstetric: reduced fetal growth, premature delivery, and miscarriage.
C. Psychological: anxiety, irritability, agitation, psychosis, insomnia, and depression on withdrawal.
D. Neuromedical: seizure disorders.
E. Metabolic: weight loss and its consequences.
F. Side effects of it's illicit uses: injection site's infection, septicemia, embolism, transmission of infections like HIV and hepatitis, dental problems, in addition to prompt and rapid development of stimulant-related disorders; for instance, one week continuous use of cocaine might be enough to induce cocaine-related disorders.

There are several drugs, natural substances, and recreational illicit drugs within the 'Stimulant Pool':

A. Amphetamines: examples include, Methylphenidate (Ritalin), Dextroamphetamine (Dexedrine), Methamphetamine (Desoxyn). Street slang names for their illicit uses include: "ice", "crystal", "crystal meth", and "speed".
B. Amphetamine-like substances: examples include: ephedrine and pseudoephedrine, Phenylpropanolamine (PPA), 3,4-methylenedioxyamphetamine (MDMA) which is known as "ecstasy", "XTC", and "Adam".
C. Naturally existing stimulants: like Cocaine, Khat, Caffeine, and Nicotine.
D. Non-amphetamine stimulants: like Modafinil which acts one histamine receptors in addition to enhancing dopamine and nor-epinephrine neurotransmissions.

Cocaine

Cocaine is obtained from the Coca plant which is mainly flourished in South America. It is a potent dopamine, noradrenaline, and serotonin reuptake inhibitor, for which it sometimes named 'Triple Reuptake Inhibitor (TRI)'. It's the most dangerous stimulant of abuse and might lead to death if taken in high doses.

Its usage dated back to thousands years ago by aboriginal Americans in form of chewing the leaves of the plant. Nevertheless, medically cocaine was used as local anesthetic agent for dental pain and minor operations, particularly ophthalmic surgeries.

Illicitly, it's administered in several ways, including oral, sniffing, injection, and as suppositories. Cocaine commonly misused with Heroin (see above), and it owns several slang names likewise. For instance, "crack", "rocks", "speedball", "chiclet", "coke", and so forth.

Khat

Catha edulis (Khat) is a flowering plant cultivated in the Horn of Africa and Arabian Peninsula. It contains cathinone, an amphetamine-like stimulant, which may induce high mood, energy, and loss of appetite.

Although its regarded as a substance of misuse by WHO since 1980, yet its production, sale, and uses are legal in many states of the above mentioned areas, including Yemen, Somalia, Ethiopia, and Djibouti. What's more, its usage is part of social tradition and norm among societies of these countries since thousands of years. Its chewing is more common than drinking coffee and its used in a similar social context of coffee and tea drinking among mentioned countries.

Its known as "qaat", "gaat", or "jaat" among local people. Sometime named "Arabian tea" by western societies.

Beside general side effects of stimulants in general, Khat has also local irritating effects on teeth and gum and might darken the teeth.

Although it's not a common substance of misuse, its estimated that around 10 million inhabitants consume this substance in the cited countries. Several studies revealed that almost all male Yemenis use Khat at least once in life.

Caffeine

Although caffeine has been discovered only at 1821 by a French chemist, Pierre Jean Robiquet, its consumption is dated back to thousands years ago. The Chinese legend says: 3000 BC, tea discovered accidentally by an emperor when he noticed that when certain leaves fall into a boiling water, an aromatic and soothing drink results.

The earliest knowledge of coffee drinking, however, is dated back to the mid fifteenth century from the 'Sufi' monasteries of Mocha, a coastal city of Yemen, where coffee distributed to North Africa, Middle East, Turkey, and the rest of the world.

Its less known on the custom of Kola nut consumption. Nevertheless, there are historical evidences of its usage to overcome hunger among Western African ancient societies.

Caffeine is the commonest substance used globally as prescribing medicines as well as beverages and foods. Since it has no dangerous side effects, besides evidences for its protective properties, when consumed in modest amount, against heart diseases, Parkinson disease, and few tumors, its usage is legal worldwide.
The active substance of caffeine named xanthine exerts its stimulant properties by blocking adenosine (an inhibitory neurotransmitter) receptors in the nervous system, in addition to enhancement of dopamine-serotonin-noradrenaline neurotransmissions.
Sources of caffeine in nature includes: coffee plant, tea bush, and Kola nut. Beverages containing caffeine include: tea, coffee, soft drinks, energy drinks, beside chocolates.

Nicotine

The name derived from the tobacco plant 'Nicotiana tabacum' which further named after the French ambassador in Portugal 'Jean Nicot de Villemain' who exported tobacco to Paris in 1560 and encouraged its medical uses.
It was used as an insecticide in the past until the end of World War II. Although its smoking is almost legal throughout the world, nicotine is regarded as the most dangerous substance on health. Several researchers and health agencies overestimate its harmful effects to exceed those of Heroin and Cocaine. Its damaging effects surpass its haste dependence; in addition, it contains several carcinogens parallel to the carcinogenic properties of nicotine. These are alongside the harmful effects of paired carbon dioxide smoking on the respiratory system.
Nevertheless, there are evidences of its protective properties against several medical conditions including ulcerative colitis, Parkinson's disease, Alzheimer dementia and some inflammatory conditions.
So far, nicotine used medically only in aiding smoking cessation in forms of skin patches, gum, tablets, and electronic cigarettes.

Stimulant intoxication: DSM-5 criteria

A. Recent use of a stimulant.
B. Clinically significant problematic behavioral or psychological changes (e.g., euphoria or affective blunting; changes in sociability; hypervigilance; interpersonal sensitivity; anxiety, tension, or anger; stereotyped behaviors; impaired judgment) that developed during, or shortly after, use of a stimulant.

C. Two (or more) of the following signs or symptoms, developing during, or shortly after, stimulant use:
   1. Tachycardia or bradycardia.
   2. Pupillary dilatation.
   3. Elevated or lowered blood pressure.
   4. Perspiration or chills.
   5. Nausea or vomiting.
   6. Evidence of weight loss.
   7. Psychomotor agitation or retardation.
   8. Muscular weakness, respiratory depression, chest pain, or cardiac arrhythmias.
   9. Confusion, seizures, dyskinesias, dystonias, or coma.
D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Stimulant withdrawal: DSM-5 criteria

A. Cessation of (or reduction in) prolonged stimulant use.
B. Dysphoric mood and two (or more) of the following physiological changes, developing within a few hours to several days after criterion A:
   1. Fatigue.
   2. Vivid, unpleasant dreams.
   3. Insomnia or hypersomnia.
   4. Increased appetite.
   5. Psychomotor retardation or agitation.
C. The signs and symptoms in B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Caffeine intoxication: DSM-5 criteria

A. Recent consumption of caffeine (typically in high dose well in excess of 250 mg).
B. Five (or more) of the following signs and symptoms developing during, or shortly after, caffeine use:
   1. Restlessness.
   2. Nervousness.
   3. Excitement.

4. Insomnia.
5. Flushed face.
6. Diuresis.
7. Gastrointestinal disturbance.
8. Muscle twitching.
9. Rambling flow of thought and speech.
10. Tachycardia or cardiac arrhythmia.
11. Periods of inexhaustibility.
12. Psychomotor agitation.

C. The signs and symptoms in B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Caffeine withdrawal: DSM-5 criteria

A. Prolonged daily use of caffeine.
B. Abrupt cessation of or reduction in caffeine use, followed within 24 hours by three (or more) of the following signs and symptoms:
    1. Headache.
    2. Marked fatigue or drowsiness.
    3. Dysphoric mood, depressed mood, or irritability.
    4. Difficulty concentrating.
    5. Flu-like symptoms (nausea, vomiting, or muscle pain/stiffness).
C. The signs and symptoms in B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Nicotine withdrawal: DSM-5 criteria

A. Daily use of tobacco for at least several weeks.
B. Abrupt cessation of tobacco use, or reduction in the amount of tobacco used, followed within 24 hours by four (or more) of the following signs and symptoms:
    1. Irritability, frustration, or anger.
    2. Anxiety.

3. Difficulty concentrating.
4. Increased appetite.
5. Restlessness.
6. Depressed mood.
7. Insomnia.

C. The signs and symptoms in B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Treatment

There is no specific pharmacological intervention for stimulants detoxification or as a substitution therapy. Most stimulants are short half-lived and the features subsides quickly. Nevertheless, psychotherapy is recommended in addition to symptomatic treatment during these periods.

Nicotine and smoking cessation

Smoking cessation may be the most difficult substance-abstinence therapy due to several reasons, including upsetting withdrawal signs and symptoms, availability of cigarettes, and relative social-acceptability of its usage.

There are three main treatments for discontinuing tobacco smoking, including nicotine-replacement therapy (NRT) in forms of tablet, gum, skin patch, nasal spray, or inhaler; as well as Bupropion, which is atypical antidepressant inhibits dopamine reupatake, and Varenicline titrate which is a partial nicotinic receptors agonist.

Although trials showed optimistic outcomes comparing to placebo, their long-term effects remain, relatively, controversial.

<u>Sedative-, Hypnotic-, or Anxiolytic-Related Disorders</u>

This group of substances includes: Benzodiazepine (e.g., Diazepam, Chlorodiazepoxide, Midazolam, Oxazepam, Lorazepam, Alprazolam, etc…), Benzodiazepine-like drugs (e.g., Zolpidem, Zaleplon), Carbamates (e.g., Glutethimide, Meprobamate), Barbiturates (e.g., Secobarbital), and Barbiturate-like hypnotics (e.g., Methaqualone) (see chapter 16).

Similar to alcohol, these substances are GABA receptor agonists; hence, they are brain depressants. Non-Benzodiazepine anxiolytics (e.g., Buspirone) are not grouped with this class since they are not brain depressants and have no sedative properties (see chapter 16). Medically, this class of substances used to promote sleep, as muscle relaxants, as anesthetic agents, to release tensions and anxieties, or as anticonvulsants. Though, they carry high risks of tolerance and misuse, besides other depressant side effects like respiratory depression, diaphragm paralysis and even respiratory failure, impaired attention and concentration which might turn down driving to a risky task, as well as impaired memory.
Because of similarities in pharmacological effects of these substances to alcohol, APA applied similar criteria of both ‘intoxication’ and ‘withdrawal’ to those of alcohol (see alcohol section).

Treatment
Individuals who are misusing drugs of this group apart from benzodiazepines should be switched to diazepam-equivalent doses of benzodiazepines. If an individual was misusing benzodiazepines, then its preferable to switch to longer acting benzodiazepines like diazepam or chlorodiazepoxide depending on diazepam-equivalent doses. For both groups, then after, its advisable to gradually reduce the dose of the long-acting benzodiazepine over a recognizable long period of time depending on the clinical situation of that individual.

Hallucinogens
A substance sorted as ‘Hallucinogen’ when it has the capability of inducing subjective changes in perception, emotions, thoughts, and even consciousness. Unlike other psychoactive substances (like alcohol, opioid, cannabis, stimulants), hallucinogens induce ‘qualitative’ changes rather than magnifying the preexisting states of mind.
The experiences hallucinogens make are relatively comparable to the unusual states of ‘trance’, ‘dreams’, or ‘psychosis’. These effects, although not well understood, may originate from the disruption of neurotransmission systems. Basically serotonin, but also glutamate and dopamine neurotransmissions as well might be involved.
Furthermore, hallucinogens own stimulants properties to some extent. For instance, MDMA (ecstasy) is classified both with stimulants and hallucinogens substances.

Hallucinogens have long history of medical as well as nonmedical religious usages. They are experimented, up to this date, in several psychiatric disorders like depression, PTSD, alcohol and other substance-use disorders, and headache. Outside medicine, hallucinogens used in shamanic forms of ritual healing, divination, and several other religious rituals, particularly in the American continents. Non-religious experimentations of hallucinogens was also in place, especially after World War II, by artists and writers for creativeness, imagination, recreational, and even hedonistic reasons.

Nevertheless, despite the relative lack of physical negative consequences, hallucinogens have several, terrible, psychological side effects due to their formation of toxic free radicals in the synaptic vesicles. For instance, its chronic use may prime brain atrophy, dementia, Parkinson's disease, schizophrenia, in addition to risky, impulsive suicidal behaviors.

Classification of Hallucinogens

APA classified hallucinogens into two main groups:

A. Phencyclidine- like substances: examples include: Phencyclidine (PCP) which commonly called "Angel dust"; Ketamine; Cyclohexamine; and Dizocilpine. These substances were used first as anesthetics. They are commonly taken orally or smoked, but may also be snorted or injected.
B. Other hallucinogens: examples include:
    1. Phenylalkylamines (e.g., mescaline, DOM [2,5-dimethoxy-4-methylamphetamine], and MDMA [3,4-methylenedioxymethamphetamine; commonly named "ecstasy"]).
    2. Indoleamines (e.g., psilocybin 'psilocin', and dimethhyltryptamine 'DMT').
    3. Ergolines (e.g., lysergic acid diethylamide 'LSD', also known as "Classical Hallucinogen"; and morning glory seeds.

Phencyclidine intoxication: DSM-5 criteria

A. Recent use of phencyclidine (or a pharmacologically similar substance).
B. Clinically significant problematic behavioral changes (e.g., hostility, assaultiveness, impulsiveness, unpredictability, psychomotor agitation, impaired judgment) that developed during, or shortly after, phencyclidine use.
C. Within one hour, two (or more) of following signs or symptoms:

1. Vertical or horizontal nystagmus.
2. Hypertension or tachycardia.
3. Numbness or diminished responsiveness to pain.
4. Ataxia.
5. Dysarthria.
6. Muscle rigidity.
7. Seizures or coma.
8. Hyperacusis.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Other Hallucinogens intoxication: DSM-5 criteria

A. Recent use of a hallucinogen (other than phencyclidine).

B. Clinically significant problematic behavioral or psychological changes (e.g., marked anxiety or depression, ideas of reference, fear of "losing one's mind," paranoid ideation, impaired judgment) that developed during, or shortly after, hallucinogen use.

C. Perceptual changes occurring in a state of full wakefulness and alertness (e.g., subjective intensification of perceptions, depersonalization, derealization, illusions, hallucinations, synesthesias) that developed during, or shortly after, hallucinogen use.

D. Two (or more) of the following signs developing during, or shortly after, hallucinogen use:
1. Pupillary dilatation.
2. Tachycardia.
3. Sweating.
4. Palpitations.
5. Blurring of vision.
6. Tremors.
7. Incoordination.

E. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Treatment

Since intoxication syndromes of hallucinogens mimic 'acute psychosis', similar pharmacological interventions may be beneficial in reducing core symptoms like antipsychotics or benzodiazepines. However, the least anticholinergic antipsychotic should be tried, like haloperidol, since the anticholinergic property may worsen the delirious and psychotic symptoms.

Inhalants

Intoxicative inhalants are a broad range of intoxicative substances in which the volatile vapors are taken via the nose and trachea. They are taken by the usual room temperature volatilization or from a pressurized container (e.g., nitrous oxide), and hence, do not include substances that are sniffed after burning or heating. For example, amyl nitrite and toluene are considered inhalants, on the other hand tobacco, marijuana, and crack are not, even though the latter are also inhaled as smokes.

Inhaling volatile toxic substances is called **huffing**. There are five groups of inhalants depending on the chemical properties (see table 10-3).

Table 10-3: classification of inhalants	
Chemical class	Examples
aliphatic hydrocarbons	petroleum products (gasoline and kerosene), propane, butane
aromatic hydrocarbons	toluene (used in paint thinner and model glue), xylene
Ketones	acetone (used in nail polish remover)
Haloalkanes	hydrofluorocarbons, chlorofluorocarbons, trichloroethylene, 1,1,1-Trichloroethane (including many aerosols and propellants)
Nitrites	alkyl nitrites (poppers such as amyl nitrite), nitrous oxide (found in whipped cream canisters)

Inhalant intoxication: DSM-5 criteria

A. Recent intended or unintended short-term, high-dose exposure to inhalant substances, including volatile hydrocarbons such as toluene or gasoline.
B. Clinically significant problematic behavioral or psychological changes (e.g., hostility, assaultiveness, apathy, impaired judgment) that developed during, or shortly after, exposure to inhalants.
C. Two (or more) of the following signs or symptoms developing during, or shortly after, inhalant use or exposure:
   1. Dizziness.
   2. Nystagmus.
   3. Incoordination.
   4. Slurred speech.
   5. Unsteady gait.
   6. Lethargy.
   7. Depressed reflexes.

8. Psychomotor retardation.
9. Tremor.
10. Generalized muscle weakness.
11. Blurred vision or diplopia.
12. Stupor or coma.
13. Euphoria.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Treatment

Similar to stimulant, there is no specific pharmacological intervention for Inhalant detoxification or as a substitution therapy. Most inhalants are short half-lived and the features subsides quickly. Nevertheless, psychotherapy is recommended in addition to symptomatic treatment during these periods.

Epidemiology of Substance-Related and Addictive Disorders

Prevalence of alcohol and other substances-related disorders is several factors-dependent, including gender, age, occupation, culture, availability of a particular substance, comorbid mental or medical conditions, as well as the kind of the substance.

Generally speaking, alcohol and substance-use disorders, including intoxication and withdrawal syndromes, are roughly at least as twice prevalent among male as compared to female populations. The estimate in particular cultures and for particular substances might hit 4:1 male: female ratio.

Experimentations of alcohol, drug, or other substances begin, usually, during adolescence and early adulthood eras. Likewise, substances intoxications are more prevalent early in the course, when teens and young adults tend to consume substance in peer-groups while still their bodies do not develop tolerances to these substances. However, withdrawal syndromes and, consequently, substance-related disorders prone to emerge after chronic consumptions. It is estimated that substances withdrawals and related disorders appear to clinical attentions during second-half of 20s onward.

Occupation on the other hand has its role in distribution of substance-related disorders, possibly due to job-related stresses as well as drug availabilities. Many population studies showed an overall higher prevalence of substance-related disorders among blue-collar workers. In a cross-sectional study conducted by the author, alcohol dependence was significantly higher among journalists in Erbil (Rahim, 2010a).
Doctors, pharmacists, and barmen are among other professions with greater risk of substance-related disorders.

Among other factors which govern the dissemination of substance-related disorders are culture, tradition, legislations, in addition to acceptability and availability of the substance. For instance, Khat consumption is neither stigmatized nor its usage is regarded illegal in Yemen and parts of Africa. Instead, its consumption might be part of social lives traditions.
Correspondingly, usages of some sorts of hallucinogens are part of religious rituals among different states of American continents. Caffeine beverages and foods, alike, are almost globally legal, may be due to lack of serious adverse effects.

Medical conditions, also, may contribute to the illicit uses of several substances, particularly prescribing drugs. For example, morbid conditions associated with chronic pain, like rheumatic problems and headaches, might end up with opioids, anxiolytics, or sedative misuses. Equally, several mental disorders may be accused to their contributions to substances-related disorders like personality, mood, psychotic, eating, or sleep disorders.

Accurate estimation of the incidence and prevalence of each substance-related disorder is rather a hard task since suffered individuals usually tend to use more than one drug at once, and they lean towards under-estimation of their usages. This ambiguous picture is growing darker by the fluidity of each single substance-use over different periods of time and across different nations. Nevertheless, caffeine and tobacco are expected to be the commonest used substances. Though, cannabis is likely the commonest ‘illicit’ substance used worldwide.

Etiology of Substance-Related and Addictive Disorders

A. Biological factors:

1. Genetic: alcohol and substance-related disorders may run in families, particularly alcohol. The rate might be 3-4 folds higher in close relatives. MZ twins are affected more than DZ twins. This familial aggregation of the condition may account for possible shared social venues as well. However, studies revealed back genetic predisposition of altered liver metabolism of alcohol and certain substances. Furthermore, genetic may predispose an individual toward substance misuse through it's predisposition to the personality traits of novelty seeking and impulsivities.
2. Hormone: the higher rates of alcohol and substance-related disorders among male population, raised the possibilities of an altered level of testosterone hormone as a predisposing factor. Likewise, testosterone may be seen as a biological determinant for all male-kind behaviors like aggression, impulsivity, and lack of control on delaying gratifications.
3. Physiological dependence to prescribing medicines like opioids, sedatives, hypnotics, or anxiolytics may result from chronic prescriptions of above drugs.

B. Psychological factors:

1. As stated above, several personality traits may pave the road for substance misuses including impulsivity, lack of control, aggression, or anxiousness, in addition to lack of appropriate, adaptive, copying strategies to deal with day-to-day life stressors.
2. Alcohol and substance related disorders are more prevalent among psychiatric disorders, especially personality, schizophrenia-like, mood, anxiety, eating, or sleep disorders. In some of these conditions, the co-morbidities may be due to shared predisposing factors, whereas in others may be an etiological factor, a presenting feature, or consequences of these psychiatric disorders.

C. Social factors:

These include availabilities as well as acceptability of the substance in the immediate venue or in the larger culture (see epidemiology section). Additionally, social stressful events like loss of job, marital discomfort, financial difficulties, or bereavement, may all predispose, precipitate, or maintain alcohol or other substances-related disorders.

Management of Substance-Related and Addictive Disorders

It's not necessary for the therapist to be a psychiatrist to come across individuals with substance-related disorders. Every physician, nowadays, is likely to be involved in the management of these conditions including general practitioners.

Assessment

In addition to the basic principles of assessment in psychiatry (see chapter 1), particular attention should be paid to the following points while assessing for alcohol or substance use disorders:

A. Points in history taking

1. Individuals with substance use disorders tend to underestimate their problems and may conceal significant information from therapists (denial). Nevertheless, individuals dependent on opioids, particularly heroin, may overstate their heroin uses in order to obtain larger amounts of substituted methadone therapy.
2. Most individuals are misusers of more than one substance at once. Hence, while the therapist is attempting to assess opioid use, for instance, s/he might forget to assess for other substances commonly taken with opioid like alcohol, sedatives, cocaine, cannabis, or at minimum tobacco.
3. Drug history: for each single drug, the assessor should ask the following questions in sequences:
   a. Last time the drug was taken.
   b. The typical drug-using day: the type, amount, and routes of administration of the drug; the venue of taking the drug, whether alone or within peer-groups, the time of starting within the 24 hours, whether early morning, mid-day, or at night; and any sort of associated food consumption, especially in case of alcohol use. Do not forget asking for tobacco smoking.
   c. Symptoms and signs when the drugs taken: the "high", sedation, or other intoxication features.
   d. Symptoms when the drugs are not available (withdrawal features).
   e. Chronological history of substance misuse since the start.
   f. Abstinence and re-use triggers.

g. Any risky behavior to obtain the drugs.
h. Social adverse consequences of substances consumptions.
i. Legal problems secondary to drugs uses.
j. Medical and psychiatric complications.

B. Physical examinations

Since substance misusers tend to hide their behaviors, its advisable for the physician to inspects for possible physical signs of substances uses like: vein thrombosis at the site of injection, subcutaneous abscesses, scars, needle tracks, features of hepatitis, breath odors, in addition to specific signs of recent uses of particular substance (see each class of substance above).

C. Laboratory investigations

Its wise to confirm the diagnosis by laboratory evidences of drugs uses when its applicable. Body fluids like urine, blood, and saliva, as well as hair are proper pools for detection of substances in certain circumstances. Urine is the commonest medium and the easiest to look after drugs in.

The therapist may provide the laboratory with a complete list of drugs for investigation.

Treatment

Treatment of any drug-related disorder requires a comprehensive bio-psycho-social collective program. The optimal goal of any kind of therapy is the total abstinence; however, this goal may not be achievable among several sufferers. Therefore, alternative targets should be taken in regard like: controlled drinking (for alcohol), replacement therapy (for opioids), or 'Harm Reduction' (for opioids and several other substances) (see below).

Harm reduction program:

Although this program raised a great deal of debates, it remains beneficial in keeping illicit drugs users under medical supervisions and control. The essences of harm reduction are supplying illicit drug users with regular doses of replacement therapies (e.g., methadone in case of heroin) so as keeping them away from drug dealers, as well as supplying suffered individuals with syringes and condoms in order to reduce the risk of HIV and viral hepatitis infections.

There might be mandatory accompanied educational programs and psychotherapeutic sessions during each visit to the care centers each time.

Biological therapy in the form of pharmacotherapy should be initiated when applicable to detoxify, substitute, prevent re-use, or treat the complications of the substances (see each class of substance).

Several kinds of psychotherapies are in place for drug users ranging from education, to behavioral, cognitive and even group therapies (see chapter 18).

Socially, its advisable to recruit family members or friends in the therapeutic system once this attempt is agreed upon by the patient. Nevertheless, there are several societies which might incorporate both the role of social supports as well as kind of psycho-educational therapies. Best example of the latter is Alcoholics Anonymous (AA).

11 NEUROCOGNITIVE DISORDERS

Neurocognitive disorders (NCDs)' category covers a group of disorders were called in the previous version of DSM, DSM-IV-TR, 'Dementia, Delirium, Amnestic, and Other Cognitive Disorders'. The major psychopathology of these disorders is the impairment of cognitive functions (see chapter 1).

Cognitive impairments are not limited to NCD per se, almost all psychiatric disorders may presented with cognitive deficits including schizophrenia, anxiety, or mood disorders. However; in NCDs, cognitive deficits are the core clinical features.

NCDs do not include neurodevelopmental conditions, like intellectual disability, but they cover acquired conditions for which impaired cognitive functions were previously intact.

NCDs are exceptional among psychiatric disorders in that there are clear evidences of brain pathology or presence of underlying medical pathology behind these disorders.

NCDs include: Delirium, Major NCD, and Mild NCD. The last two disorders may be due to several causal syndromes; however, since this book addresses medical students, the focus will be more on well-known syndromes, namely: Alzheimer's disease, Fronto-temporal NCD, NCD with Lewy bodies, and Vascular NCD.

Delirium

Among lay peoples, delirium may denote a state of drowsiness, disorientation, or hallucination. Nevertheless, as a medical term, delirium stands for disturbance of consciousness and reduced attention. 'Acute confusional state' and 'toxic brain syndrome' are among other names of delirium. Disorientations and hallucinations may be associated symptoms, though none of these symptoms are necessary for the diagnosis to be made.

DSM-5 criteria for diagnosis

A. A disturbance in attention (i.e., reduced ability to direct, focus, sustain, and shift attention) and awareness (reduced orientation to the environment).
B. The disturbance develops over a short period of time (usually hours to few days), represents a change from baseline attention and awareness, and tends to fluctuate in severity during the course of a day.

C. An additional disturbance in cognition (e.g., memory deficit, disorientation, language, visuospatial ability, or perception).
D. The disturbances in A and C are not better explained by another preexisting, established, or evolving neurocognitive disorder and do not occur in the context of a severely reduced level of arousal, such as coma.
E. There is evidence from history, physical examination, or laboratory findings that the disturbance is a direct physiological consequence of another medical condition, substance intoxication or withdrawal (i.e., due to a drug of abuse or to a medication), or exposure to a toxin, or is due to multiple etiologies.

Prevalence

Overall, community prevalence may be as low as 1%. However, the rate increases with increase in age, and it's more prevalent among hospitalized patients, particularly post-operative surgical patients.

Etiology

A. Substances, medications, or alcohol induced: both during intoxication or withdrawal syndromes (see chapter 10). In essence, any substance leads to impairment of consciousness, like drugs with anticholinergic properties, may end up with delirium.
B. Medical conditions: several medical conditions, especially those associated with high fever, may complicated by delirium. Noticeable medical conditions associated with delirium are brain infections like encephalitis and meningitis, and metabolic disturbances like hepatic or renal encephalopathies.

Treatment

The key treatment is the detection and treatment of the accusing underlying medical condition(s). Meanwhile, the following supportive measures are mandatory to be applied for all delirious patients:
A. Keep the patient in a well-illuminated and calm ward room. Its advisable to keep lights on during night hours. These are necessary in order to increase perceptual clarities.
B. To reduce disorientation, its preferable to practice the following points:
    1. Only one or two medical staff should follow-up patient's condition.

2. Its preferable to keep only very-limited number of family members during the hospital stay for whom its well-known by the patient.
3. During each medical visit, the medical staff should re-orient the patient by letting the patient to know his/er name, what time it is, and what place the patient is staying in.

C. Low doses of antipsychotics of lower anticholinergic properties like Haloperidol, or short-acting Benzodiazepines like Oxazepam or Lorazepam may be useful in reducing agitation and the core delirious symptoms.
D. Packing or giving antipyretics are beneficial if there is fever.
E. Comprehensive investigations should be carried out, meantime, to uncover the underlying responsible conditions.

<u>Major and Mild Neurocognitive Disorders</u>

Major NCD is the DSM-5's expression of what was named 'Dementia' in previous editions of DSM. Whereas Mild NCD stands for less severe cognitive impairments, yet may be a focus of clinical attention, called in previous editions 'Cognitive Disorders Not Otherwise Specified'.

Major Neurocognitive Disorder

DSM-5 criteria for diagnosis:

A. Evidence of significant cognitive decline from a previous level of performance in one or more cognitive domains (complex attention, executive function, learning and memory, language, perceptual-motor, or social cognition) based on:
   1. Concern of the individual, a knowledgeable informant, or the clinician that there has been a significant decline in cognitive function; and
   2. A substantial impairment in cognitive performance, preferably documented by standardized neuropsychological testing or, in its absence, another quantified clinical assessment.
B. The cognitive deficits interfere with independence in everyday activities (i.e., at a minimum, requiring assistance with complex instrumental activities of daily living such as paying bills or managing medications).
C. The cognitive deficits do not occur exclusively in the context of a delirium.
D. The cognitive deficits are not better explained by another mental disorder (e.g., major depressive disorder, schizophrenia).

Mild Neurocognitive Disorder

DSM-5 criteria for diagnosis

The diagnostic criteria are similar to major neurocognitive disorder, though the cognitive impairments are modest, and the deficits do not interfere with the individual's independency in everyday activities.

Alzheimer's Disease

Alzheimer's disease (AD) is the commonest cause of NCD. It was first described by a German neuro-psychiatrist Alois Alzheimer in 1906 and was named after him.

DSM-5 criteria for diagnosis

A. The criteria are met for major or mild neurocognitive disorder.
B. There is insidious onset and gradual progression of impairment in one or more cognitive domains (for major neurocognitive disorder, at least two domains must be impaired).
C. Criteria are met for either probable or possible Alzheimer's disease as follows:

   For major neurocognitive disorder:

   Probable Alzheimer's disease is diagnosed if either of the following is present; otherwise, **possible Alzheimer's disease** should be diagnosed.

   1. Evidence of causative Alzheimer's disease genetic mutation from family history or genetic testing.
   2. All three of the following are present:
      a. Clear evidence of decline in memory and learning and at least one other cognitive domain (based on detailed history or serial neuropsychological testing).
      b. Steadily progressive, gradual decline in cognition, without extended plateaus.
      c. No evidence of mixed etiology (i.e., absence of other neurodegenerative or cerebrovascular disease, or another neurological, mental, or systemic disease or condition likely contributing to cognitive decline).

   For mild neurocognitive disorder:

   Probable Alzheimer's disease is diagnosed if there is evidence of a causative Alzheimer's disease genetic mutation from either genetic testing or family history.

   Possible Alzheimer's disease is diagnosed if there is no evidence of a causative Alzheimer's disease genetic mutation from either genetic testing or family history, and all three of the following are present:

1. Clear evidence of decline in memory and learning.
2. Steadily progressive, gradual decline in cognition, without extended plateaus.
3. No evidence of mixed etiology (i.e., absence of other neurodegenerative or cerebrovascular disease, or another neurological, mental, or systemic disease or condition likely contributing to cognitive decline).

D. The disturbance is not better explained by cerebrovascular disease, another neurodegenerative disease, the effects of a substance, or another mental, neurological, or systemic disorder.

Fronto-Temporal NCD

This kind of NCD was first described by Arnold Pick, a professor of psychiatry in Prague, in 1892 and was initially called 'Pick Disease'. The main neuronal losses are distributed to frontal and temporal lobes, the reason for which its called 'Fronto-Temporal' NCD. Pick identified silver-stained round collections he named "Pick bodies". Later, these Pick bodies were recognized to be collections of a kind of protein named 'Tau Protein'.

DSM-5 criteria for diagnosis

A. The criteria are met for major or mild NCD.
B. The disturbance has insidious onset and gradual progression.
C. Either (1) or (2):
   1. Behavioral variant:
      a. Three or more of the following behavioral symptoms:
         i. Behavioral disinhibition.
         ii. Apathy or inertia.
         iii. Loss of sympathy or empathy.
         iv. Perseverative, stereotyped or compulsive/ritualistic behavior.
         v. Hyperorality and dietary changes.
      b. Prominent decline in social cognition and/or executive abilities.
   2. Language variant: prominent decline in language ability, in the form of speech production, word finding, object naming, grammar, or word comprehension.
D. Relative sparing of learning and memory and perceptual-motor function.

E. The disturbance is not better explained by cerebrovascular disease, another neurodegenerative disease, the effects of a substance, or another mental, neurological, or systemic disorder.

NCD with Lewy Bodies

Although Friedrich Heinrich Lewy, a German neurologist, was the first who discovered the abnormal protein deposits 'Lewy Bodies' in early 1900s, dementia or NCD with Lewy bodies was first described by a Japanese psychiatrist, Kenji Kosaka, in 1976. This kind of dementia started to be diagnosed only in the mid-1990s after discovering a kind of protein named 'alpha-synuclein'. Features of this kind of NCD bridge both NCD and Parkinson's disease.

DSM-5 criteria for diagnosis

A. The criteria are met for major or mild NCD.
B. The disorder has an insidious onset and gradual progression.
C. The disorder meets a combination of core diagnostic features and suggestive diagnostic features:
   1. Core diagnostic features:
      a. Fluctuating cognition with pronounced variations in attention and alertness.
      b. Recurrent visual hallucinations that are well formed and detailed.
      c. Spontaneous features of Parkinsonism, with onset subsequent to the development of cognitive decline.
   2. Suggestive diagnostic features:
      a. Meets criteria for rapid eye movement sleep behavior disorder.
      b. Severe neuroleptic sensitivity.

      Probable NCD with Lewy bodies = (at least 2 core features) or (at least one of core and one of suggestive diagnostic features)

      Possible NCD with Lewy bodies = (only one core feature) or (at least one suggestive feature).
D. The disturbance is not better explained by cerebrovascular disease, another neurodegenerative disease, the effects of a substance, or another mental, neurological, or systemic disorder.

Vascular NCD

Vascular NCD or dementia also called sometime 'multi-infarct dementia'. Its caused mainly by a series of small infarctions across the entire cortex.

DSM-5 criteria for diagnosis

A. The criteria are met for major or mild NCD.
B. The clinical features are consistent with a vascular etiology, as suggested by either of the following:
   1. Onset of cognitive deficits is temporally related to one or more cerebrovascular events.
   2. Evidence of decline is prominent in complex attention (including processing speed) and frontal-executive function.
C. There is evidence of the presence of cerebrovascular disease from history, physical examination, and/or neuroimaging considered sufficient to account for the neurocognitive deficits.
D. The symptoms are not better explained by another brain disease or systemic disorder.

Associated Clinical Features of Major and Mild NCD

Although cognitive deficits are the core signs and symptoms, hardly ever are the reasons of consultations however. Instead, several other more frustrating symptoms may be behind first consultation, as follows:

A. In Alzheimer's disease, up to 80% of patients may suffer from distressing psychological and behavioral symptoms ranging from mild-to-moderate mood swings to more distressful irritability and agitation, paranoid ideas and delusions, wandering, or even to more complicated presentations of gait disturbances, incontinence, or seizures late in the illness.
B. In frontotemporal NCD, there might be associated extrapyramidal features as well as features of convoluted neuromedical disorders such as motor neuron disease, or supranuclear pulsy. Furthermore, since the main neuropathology is located in both frontal and temporal lobes, both motor and sensory aphasias may be common presentations.
C. In NCD with Lewy bodies, psychotic symptoms, particularly hallucinations are common, besides frequent episodes of syncope, falls, and loss of consciousness, as well as Parkinson's disease-like features.

D. In vascular NCD, in addition to other features of predisposing stroke or ischemic heart diseases, depression and personality changes are common.

Epidemiology of NCD

NCDs, in general, account for around 1-5% of 65-70 year-old population. The prevalence increases with aging, reaching a peak of around 45-50% among those older than 90 years. Alzheimer's disease is the commonest kind of NCD among all age groups. Vascular NCD is the second common NCD among age groups older than 65 years. However, frontotemporal NCD is the second common NCD among younger than 65 years population.

Etiology of NCD

Neurodegeneration is a normal aging process. Different levels of brain atrophies are suspected in any human being's brain after the ages of 70-80s. However, several factors, principally biological factors, hasten these senile processes.

A. Biological factors:
   1. Genetic factors: like polymorphism of apolipoprotein E4 and trisomy 21 increase the risk of Alzheimer's disease, mutations in genes encoding the microtubule associated protein tau in frontotemporal NCD. Several NCDs show family running histories. What's more, approximately 10% of frontotemporal NCD show an autosomal dominant mode of inheritance.
   2. All kinds of NCD, particularly vascular NCD, share same biological risk factors of cerebrovascular diseases like hypertension, hyperlipidemia, smoking, diabetes mellitus, etc…
   3. Substances misuse and nicotine smoking: despite hypothesized nicotine's preventive properties against Alzheimer's disease, nicotine as well as several other illicit substances and prescribing pills, particularly those with anticholinergic properties, may associated with variable degrees of cognitive deficits and consequence NCDs.

B. Psychological factors: although there are no clear evidences for underlying psychological factors, several studies revealed back earlier onsets and higher prevalence of NCDs among individuals suffered from schizophrenia or mood disorders, particularly major depressive disorder.

C. Social factors: many daily life habits may impede or delay the onset of NCDs like reading, writing, and exercises. Its thought that NCDs are more prevalent among illiterates or those who infrequently spend times in reading and writing. Lack of exercise as a risk factor of NCDs is explainable since it promotes cerebrovascular and other vascular pathologies.

Treatment of NCD

Unfortunately there is no cure, meantime, for any single brain-atrophy process. The main ambitions behind any treatment plan are to hamper the progression of brain atrophy, to reduce the core symptoms, and to provide cares:

A. Hampers the progression of brain atrophy: two groups of drugs are implemented in this process despite lack of significant efficacies:
   1. Acetylcholinesterase enzyme inhibitors like Donepezil, Rivastagmine, and Galantamine. These drugs reduce the metabolism of Acetyl Choline (Ach) in the brain.
   2. N-methyl-D-aspartate (NMDA) receptors antagonist like Memantine. Antagonism of NMDA receptors is a protective measure and thought to be neuroprotective and hence disease modifying. Memantine is licensed to be used for cases of moderate-to-severe NCDs.

B. Reduce the core symptoms: as declared above several kinds of NCDs are presented with numerous accompanied psychological signs and symptoms like psychosis, aggression, and mood changes. Therefore, relevant symptoms should be addressed properly like prescribing antidepressants for associated depression, or antipsychotics for psychotic and irritability symptoms.

   Nevertheless, its advisable to prescribe the least anticholinergic-like drugs like prescribing SSRIs instead of TCADs for depression. Also, antipsychotics should be prescribed with an extreme caution in cases of NCD with Lewy bodies, since it may aggravate the Parkinsonian features and may worsen the entire condition.

C. Non-pharmacological care-providing may be beneficial in providing daily living care and nursing, like toilet training. There are several non-pharmacological interventions like behavioral modification, music therapy, pet therapy, stimulus control, and several other measures, though with questionable effectiveness.

12 PEDIATRIC PSYCHIATRY

Children, like adults, may suffer from almost all psychiatric disorders. Indeed, most adulthood major psychiatric disorders spark during childhood and adolescence periods. For instance schizophrenia, other psychotic disorders, or personality disorders may have origins during second, or even first decades of life.
Despite the fact that neurocognitive disorders (see chapter 11) are disorders of elderly populations, DSM-5 does not identify any psychiatric disorder to be exclusive to adulthood. Likewise, there is no particular disorder limited to childhood per se.
Nevertheless, elimination disorders are, usually, turn off during adulthood. Moreover, the presentations of different psychiatric disorders among children may vary from adults' manifestations.
What is more, the neurodevelopmental disorders which characterized by childhood onsets, like intellectual disability (mental retardation), and autism spectrum disorder run their courses throughout the affected individuals' lives.
For that reason, DSM-5 has no particular chapter for 'Pediatric Psychiatry'. Instead, psychiatric disorders among children are, nearly, spread all over the chapters.
However, for the sake of readers' convenience, this book separated an individual chapter for disorders mainly identified during childhoods.
The psychiatric disorders that come to clinical attentions first during childhood include: Neurodevelopmental disorders; Disruptive, Impulse-Control, and Conduct disorders; and Elimination disorders. Additionally, this chapter will cover a newly introduced depressive disorder in DSM-5 with childhood onset called Disruptive Mood Dysregulation Disorder.

Neurodevelopmental Disorders

This category includes disorders which result from compromised development of CNS during the developmental periods. Therefore, these disorders characterized by early onsets, often before the child's school entrance.
Several disorders are listed under this group; nevertheless, in this book the focus will be only on the neurodevelopmental disorders that mainly address psychiatrists, namely: Intellectual disability, Autism

spectrum disorder, and Attention Deficit/Hyperactivity Disorder (ADHD).

Intellectual Disability

Intellectual disability (ID) is the new diagnostic term of what was known as 'Mental Retardation' in the past editions of DSM, and its equivalent to the ICD-11 (not published yet at the time of this writing) category of 'Intellectual Developmental Disorder'.

Historically, ID has been defined relying on Intelligence Quotient (IQ) scores of below 70. Likewise, the severity of ID was determined according to the same measure. However, although the name, ID, denotes a disabled intelligence, the disability is not limited to the intellectual abilities alone; further, there are adaptive functioning deficits in conceptual, social, and practical domains.

DSM-5 criteria for diagnosis

The following three criteria must be met:

A. Deficits in intellectual functions, such as reasoning, problem solving, planning, abstract thinking, judgment, academic learning, and learning from experience, confirmed by both clinical assessment and individualized, standardized intelligence testing.

B. Deficits in adaptive functioning that result in failure to meet developmental and sociocultural standards for personal independence and social responsibility. Without ongoing support, the adaptive deficits limit functioning in one or more activities of daily life, such as communication, social participation, and independent living, across multiple environments, such as home, school, and community.

C. Onset of intellectual and adaptive deficits during the developmental period.

Types of ID

A. According to severity: the severity of ID was determined in the past solely by the scores of IQ as follows:
   1. Mild: IQ = 55-70.
   2. Moderate: IQ = 40-55.
   3. Severe: IQ = 25-40.
   4. Profound: IQ < 25.

   However, in DSM-5 the same classification is still applied, though these severities are defined on the base of adaptive functioning that determines the required level of support, but not IQ scores. This new aspect in DSM-5 may be reducible to the invalidity of IQ scores at the lower ends. I.e. differentiating severe and profound IDs based on IQ scores may be, more or less, arbitrary.

B. According to underlying genetic syndromes:
   1. Syndromal ID: ID might be a feature of prenatally determined genetic syndromes, like Down (trisomy 21) syndrome, Lesch-Nyhan syndrome, Rett's syndrome, etc…
   2. Non-syndromal ID: here, the neurodevelopmental disorder is acquired due to prenatal infections or toxins, perinatal events-related encephalopathies, or early postnatal infections, exposure to toxins, trauma, hypoxia, or infantile seizure disorders.

Associated symptoms

ID frequently coupled with other behavioral and psychological symptoms like psychosis, depression, aggression, or disinhibitions. Furthermore, there may be co-morbid psychiatric disorders like schizophrenia, for instance, which is more prevalent among ID population compared to general population.

Epidemiology

General population prevalence is about 1%. Vast majority of IDs are of 'Mild' severity. The prevalence is thought to be more among male population.

Etiology of ID: see 'Types of ID' section.

Treatment

ID is a 'disability' rather than a 'disease' as the name signifies. Therefore, the main aim is how to rehabilitate and train the affected individuals to learn more adaptive behaviors and social skills. For that purpose, several training programs and behavioral psychotherapies are utilized like contingency therapy. These measures may be applied within family environment, hospitals, or even residential houses.

For possible associated psychotic symptoms, mood changes, irritabilities, or aggressions, periodical treatments by antipsychotics, anticonvulsants, or antidepressants may be beneficial.

Autism Spectrum Disorder

The Latin word *autismus* was coined by the Swiss psychiatrist Eugen Bleuler in 1910 when he was defining symptoms of schizophrenia (see chapter 2). He derived it from the Greek word *autós* (αὐτός, meaning "self") referring to autistic withdrawal of the patient to his fantasies. Bleuler viewed 'autistic withdrawal' as one of the fundamental symptoms of schizophrenia.

In its modern sense, 'autism' was adopted by an Austrian psychiatrist, Hans Asperger, at 1938 for describing an autistic condition known nowadays as 'Asperger Syndrome'.

Leo Kanner, an American immigrant from Ukraine, was the first who introduced the label 'early infantile autism' in a 1943 report of 11 children with outstanding behavioral similarities. Since then, autism has received mounting attentions of mental health professionals and media alike.

In addition to Kanner's brand of 'early infantile autism', this disorder received several others tags, like 'Kanner's autism', 'childhood autism', atypical autism', 'Asperger's disorder', 'childhood schizophrenia', and lastly, the DSM-IV' category of 'Pervasive Developmental Disorders'.

The cardinal clinical features of autism spectrum disorder are: impairment in social-life interaction and recognition, impairment in both verbal and non-verbal communications, and restricted pattern of, sometimes ritualistic, stereotyped behaviors (Fig 12-1).

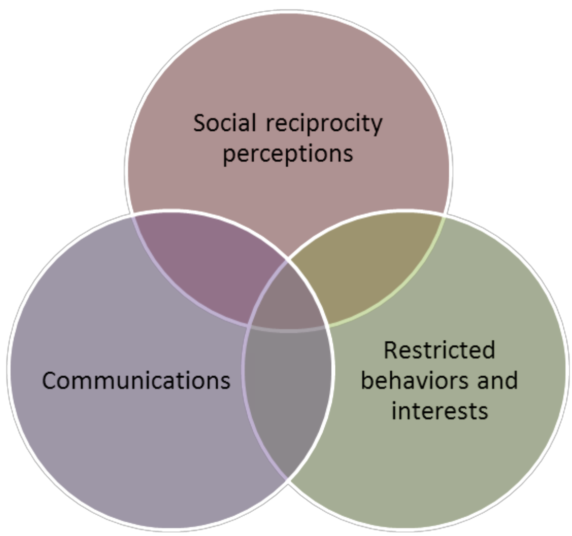

Fig 12-1: Impairment domains in autism spectrum disorder

However, since verbal and non-verbal languages are also considered parts of social life reciprocal interactions, DSM-5 addressed two main domains of impairment: Social perception, and circumscribed behaviors, attitudes, and interests (see below).

DSM-5 criteria for diagnosis

A. Persistent deficits in social communication and social interaction across multiple contexts, as manifested by the following, currently or by history:
   1. Deficits in social-emotional reciprocity, ranging, for example, from abnormal social approach and failure of normal back-and-forth conversation; to reduced sharing of interests, emotions, or affect; failure to initiate or respond to social interactions.
   2. Deficits in nonverbal communicative behaviors used for social interaction, ranging, for example, from poorly integrated verbal and nonverbal communication; to abnormalities in eye contact and body language or deficits in understanding and use of gestures; to a total lack of facial expressions and nonverbal communication.
   3. Deficits in developing, maintaining, and understanding relationships, ranging, for example from difficulties adjusting behavior to suit various social contexts; to difficulties in sharing imaginative play or in making friends; to absence of interest in peers.

B. Restricted, repetitive patterns of behavior, interests, or activities, as manifested by at least two of the following, currently or by history:
   1. Stereotyped or repetitive motor movements, use of objects, or speech (e.g., simple motor stereotypies, lining up toys or flipping objects, echolalia, idiosyncratic phrases).
   2. Insistence on sameness, inflexible adherence to routines, or ritualized patterns of verbal or nonverbal behavior (e.g., extreme distress at small changes, difficulties with transitions, rigid thinking patterns, greeting rituals, need to take same route or eat same food every day).
   3. Highly restricted, fixated interests that are abnormal in intensity or focus (e.g., strong attachment to or preoccupation with unusual objects, excessively circumscribed or perseverative interests).
   4. Hyper- or hypo-reactivity to sensory input or unusual interest in sensory aspects of the environment (e.g., apparent

indifference to pain/temperature, adverse response to specific sounds or textures, excessive smelling or touching o objects, visual fascination with lights or movement).

C. Symptoms must be present in the early developmental period (but may not become fully manifest until social demands exceed limited capacities, or may be masked by learned strategies in later life).
D. Symptoms cause clinically significant impairment in social, occupational, or other important areas of functioning.
E. These disturbances are not better explained by intellectual disability (Mental retardation) or global developmental delay. Intellectual disability and autism spectrum disorder frequently co-occur; to make co-morbid diagnoses of autism spectrum disorder and intellectual disability, social communication should be below that expected for general developmental level.

Epidemiology

Autism spectrum disorder is unique in its growing prevalence among psychiatric disorders. Although the rough estimates of its prevalence among general population is within the figure of 1-6/1000, frequencies of up to 1% have been reported recently.

Almost all studies estimated a male-to female ratio of around 4:1.

This climbing estimate of autism spectrum disorder may be due to several factors; like broadening of the diagnostic criteria to recruit sub-threshold syndromes, increased awareness and recognition of the disorder, or an increase in the predisposing factors.

Etiology

The exact etiology of autism spectrum disorder, like almost all psychiatric disorders, is still unknown. Several factors have been hypothesized to be behind this disorder; nevertheless, none of them proved to be solely responsible for the emergence of this life-long condition.

Although autism cannot, typically, be traced to a Mendelian (single-gene) mutation or to a single chromosome abnormality and none of the genetic syndromes associated with autism have been shown to selectively cause autism, studies of twins suggest that heritability may be as high as 0.9, and siblings of affected individuals are about 25 times more likely to be autistic than the general population.

Furthermore, several other environmental, though nonspecific, factors have been postulated to be responsible in the contribution of this disorder. For example, advanced mother age during pregnancy, fetal exposure to toxins and drugs, or low birth weight.

Treatment

Autism spectrum disorder is a neurodevelopmental disorder with life-long courses. The target of any therapeutic program, therefore, is not aiming any cure of the condition. Instead, the main objective is to improve the quality of lives of both the individual and parents, as well as increasing the independency of the sufferers.

While several psychosocial interventions revealed back to be ineffective, intensive early special educational programs and behavioral modifications may help children acquire some daily life self-care, social, and work skills.

The main current method is a kind of behavioral therapy named **'Applied Behavioral Analysis (ABA)'** (ABA is a kind of behavior modification relying on the principles of classical and operant conditioning that adjusts human behavior according to responses to the surrounding environmental stimuli. If it applied intensively, up to 40 hours a week for instance, the child may learn language, social, as well as academic skills). Also, numerous educational programs, speech therapies, and social-skills training programs are available options as well.

Parallel pharmacotherapies, in forms of periodical doses of antipsychotics, antidepressants, mood stabilizers, and sometime stimulants, may be beneficial in reducing associated behavioral and other psychological symptoms.

Attention-Deficit/Hyperactivity Disorder (ADHD)

ADHD was first described by an English pediatrician, Sir George Frederic Still, at 1902. Different names has been employed over time to denote this condition including: 'minimal brain dysfunction' in the DSM-I (1952), 'hyperkinetic reaction of childhood' in the DSM-II (1968), 'attention-deficit disorder (ADD) with or without hyperactivity' in the DSM-III (1980). In 1987 this was reformed to ADHD in the DSM-III-R.

DSM-5 criteria for diagnosis

A. A persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning or development, as characterized by (1) and/or (2):
   1. **Inattention:** six (or more) of the following symptoms have persisted for at least 6 months to a degree that is inconsistent

with developmental level and that negatively impacts directly on social and academic/occupational activities:

a. Often fails to give close attention to details or makes careless mistakes in schoolwork, at work, or during other activities (e.g., overlooks or misses details, work is inaccurate).
b. Often has difficulty sustaining attention in tasks or play activities (e.g., has difficulty remaining focused during lectures, conversations, or lengthy reading).
c. Often does not seem to listen when spoken to directly (e.g., mind seems elsewhere, even in the absence of any obvious distraction).
d. Often does not follow through on instructions and fails to finish schoolwork, chores, or duties in the workplace (e.g., starts tasks but quickly loses focus and easily sidetracked).
e. Often has difficulty organizing tasks and activities (e.g., difficulty managing sequential tasks; difficulty keeping materials and belongings in order; messy, disorganized work; has poor time management; fails to meet deadlines).
f. Often avoids, dislikes, or is reluctant to engage in tasks that require sustained mental efforts (e.g., schoolwork or homework; for older adolescents and adults, preparing reports, completing forms, reviewing lengthy papers).
g. Often loses things necessary for tasks or activities (e.g., school materials, pencils, books, tools, wallets, keys, paperwork, eyeglasses, mobile telephones).
h. Is often easily distracted by extraneous stimuli (for older adolescents and adults, may include unrelated thoughts).
i. Is often forgetful in daily activities (e.g., doing chores, running errands; for older adolescents and adults, returning calls, paying bills, keeping appointments).

2. **Hyperactivity and impulsivity:** six (or more) of the following symptoms have persisted for at least 6 months to a degree that is inconsistent with developmental level and that negatively impacts directly on social and academic/occupational activities:
   a. Often fidgets with or taps hands or feet or squirms in seat.
   b. Often leaves seat in situations when remaining seated is expected (e.g., leaves his/er place in the classroom, in the office or other workplace, or in other situations that require remaining in place).

c. Often runs about or climbs in situations where it is inappropriate. (Note: in adolescents and adults, may be limited to feeling restless.)
d. Often unable to play or engage in leisure activities quietly.
e. Is often "on the go," acting as if "driven by motor" (e.g., is unable to be or uncomfortable being still for extended time, as in restaurants, meetings; may be experienced by others as being restless or difficult to keep up with).
f. Often talks excessively.
g. Often blurts out an answer before a question has been completed (e.g., completes people's sentences; cannot wait for turn in conversation).
h. Often has difficulty waiting his/er turn (e.g., while waiting in line).
i. Often interrupts or intrudes on others (e.g., butts into conversations, games, or activities; may start using other people's things without asking or receiving permission; for adolescents and adults, may intrude into or take over what others are doing).

B. Several inattentive or hyperactive-impulsive symptoms were present prior to age 12 years.
C. Several inattentive or hyperactive-impulsive symptoms are present in two or more settings (e.g., at home, school, or work; with friends or relatives; in other activities).
D. There is clear evidence that the symptoms interfere with, or reduce the quality of, social, academic, or occupational functioning.
E. The symptoms do not occur exclusively during the course of schizophrenia or another psychotic disorder and are not better explained by another mental disorder (e.g., mood disorder, anxiety disorder, dissociative disorder, personality disorder, substance intoxication or withdrawal).

Epidemiology

Prevalence differs according to diagnostic criteria, and symptoms and age of onset definitions. Overall pictures in most cultures are around 5% of children and around 2.5% among adults. Males diagnosed three times more than females.

Etiology

Genetic has substantial role in ADHD. Its thought that involved genes affect dopamine neurotransmission. Also, siblings of affected

individuals have 3-4 folds increase in the risk of developing same condition. Twin studies revealed evidences of heritability of ADHD from one's parents.
Low birth weight, tobacco smoking or alcohol drinking during pregnancy, and infections have been found to be associated with ADHD.

Treatment

A. Psychological therapies and support:
   Both parents and teachers are recommended to be involved in the support of these children. Teachers and parents alike should encourage and reward desired behaviors. In this sense, they have to be oriented about principles of 'operant conditioning'.
B. Pharmacotherapy:
   Main drugs used in the treatment of ADHD are stimulants like methylphenidate and dexamphetamine, TCADs like imipramine, and most recently Atomoxetine (Strattera). These drugs thought to reduce the core hyperactivity and inattentive symptoms thru increasing dopamine and/or noradrenaline neurotransmissions.
   However, clinicians should consider possible serious side effects, especially when addressing stimulants since they associated with the risks of poor appetite, diminished growth, and poor appetite. Also, there are evidences of serious risks of suicidal behaviors associated with the uses of Atomoxetine.

Conduct Disorder

DSM-5, unlike previous editions, identifies an individual chapter, (named Disruptive, Impulse-Control, and Conduct disorders), for a pattern of maladaptive behaviors that violate the rights of others and/or bring the individual into significant conflict with norms, legislations, or authority figures.
Although any psychiatric disorder might be associated with 'rules-breaking' behaviors, but these behaviors are the outstanding presenting features in this class.
The most serious among these disorders is what is called 'conduct disorder'.
Conduct disorder is, therefore, a psychological disorder diagnosed in childhood or adolescence that presents itself through a repetitive and persistent pattern of behavior in which the basic rights of others or major age-appropriate norms are violated. These behaviors are often referred to as "antisocial behaviors."

DSM-5 criteria for diagnosis

A. A repetitive and persistent pattern of behavior in which the basic rights of others or major age-appropriate societal norms or rules are violated, as manifested by the presence of at least three of the following 15 criteria in the past 12 months from any of the categories below, with at least one criterion present in the past 6 months:

   Aggression to People and Animals

   1. Often bullies, threatens, or intimidates others.
   2. Often initiates physical fights.
   3. Has used a weapon that can cause serious physical harm to others (e.g., a bat, brick, broken bottle, knife, gun).
   4. Has been physically cruel to people.
   5. Has been physically cruel to animals.
   6. Has stolen while confronting a victim (e.g. mugging, purse snatching, armed robbery).
   7. Has forced someone into sexual activity.

   Destruction of Property

   8. Has deliberately engaged in fire setting with the intention of causing serious damage.
   9. Has deliberately destroyed others' property (other than by fire setting).

   Deceitfulness or Theft

   10. Has broken into someone else's house, building, or car.
   11. Often lies to obtain goods or favors or to avoid obligations (i.e., "cons" others).
   12. Has stolen items of nontrivial value without confronting a victim (e.g., shoplifting, but without breaking and entering; forgery).

   Serious Violations of Rules

   13. Often stays out at night despite parental prohibitions, beginning before age 13 years.
   14. Has run away from home overnight at least twice while living in the parental or parental surrogate home, or once without returning for a lengthy period.
   15. Is often truant from school, beginning before age 13 years.

B. The disturbance in behavior causes clinically significant impairment in social, academic, or occupational functioning.

C. If the individual is age 18 years or older, criteria are not met for antisocial personality disorder.

Epidemiology

The exact prevalence is unknown. The most approximate rate is within the figure of 2-10%. All studies indicated three times higher rate among male compared to female populations. Most of the cases progress to adulthood antisocial personality disorder (see chapter 8).

Etiology

A. Biological factors: conduct disorder appears to be more common among children of biological parents with alcohol or substance use disorders, personality disorders, schizophrenia, mood disorders, and conduct disorder.
B. Psychological factors: like temper tantrum, and below-average IQ children.
C. Environmental factors: like family rejection, harsh treatment, physical or sexual abuses, low level of parental supervision, parental and peer-group rewarding attitudes of the aggressive behaviors, parental criminalities, as well as several other maladaptive environmental factors may, ominously, predispose to the emergence of conduct disorder.

Treatment

There is no specific pharmacological intervention for this disorder. What can be offered, are limited to special training programs for both the child and parents, anger management, remedial teaching, or residential care for severe uncontrolled cases.

Elimination Disorders

Although incontinences of urine or feces may be symptoms of several mental and medical disorders among adults and elderly, including neurocognitive disorders (dementias); elimination disorders, by definition, are mental conditions for which the incontinence and its related personal distresses or socio-occupational impairments are the only clinical manifestations, and are not explainable within a context of another mental or medical disorders.

Elimination disorders are usually first diagnosed in childhood or adolescence. This category includes both ‘Enuresis’, which is repeated voiding of urine into inappropriate places, and ‘Encopresis’, which is repeated passage of feces into inappropriate places.

Etiology

A. Biological factors:
   1. Genetic factors: around 70% of children with enuresis have a first-degree relative who has been enuretic. Also, the concordance rates for enuresis are twice among MZ twins compared to DZ twins.
   2. Encopresis may be secondary to constipation. Therefore, careful assessment should be considered for possible causes of constipation.

B. Psychological factors: like fears, stresses, shyness, or rebellion may be the only reasons behind elimination disorders.

C. Environmental factors: like poor toilet training, or stressful family environments.

<u>Enuresis</u>

DSM-5 criteria for diagnosis

A. Repeated voiding of urine into bed, clothes, whether involuntary or intentional.

B. The behavior is clinically significant as manifested by either a frequency of at least twice a week for at least 3 consecutive months or the presence of clinically significant distress or impairment in social, academic (occupational), or other important areas of functioning.

C. Chronological age is at least 5 years (or equivalent developmental level).

D. The behavior is not attributable to the physiological effects of a substance (e.g., a diuretic, an antipsychotic medication) or another medical condition (e.g., diabetes, spina bifida, a seizure disorder).

Enuresis may be evident during daytime only, nocturnal only, or both diurnal and nocturnal. Also it may be primary (i.e., the child has never learnt urine continence), or secondary (i.e., the child has lost the learnt ability of urine continence)

Epidemiology

The prevalence is around 5-10% among 5-year-olds, 3-5% among 10-year-olds, and 1% among 15-year-olds. Nocturnal enuresis is more common among boys, whereas diurnal enuresis is less prevalent and occurs more among girls.

Treatment

A. Reassurance of the child. The parents should not blame or punish the child. Instead, rewards should be offered to successes without drawing attentions to failures.
B. Restriction of fluid intakes before bedtime.
C. Regular wakeup of the child to urinates during night times (e.g., at midnight, and 03:00 am).
D. Enuresis alarm: it consists of a detector pad attached to the night clothes, and an alarm buzzer. When the clothes get wet by urine, the detector is activated and the alarm sounds. Then the child turns off the alarm, gets up to empty the bladder, and changes the clothes. These steps may need the help of parents.
E. Medications: antidiuretic hormones like (Desmopressin) in forms of tablets or nasal sprays, and/or Imipramine (TCAD) in small doses are beneficial for those did not respond to above measures.

Encopresis

DSM-5 criteria for diagnosis

A. Repeated passage of feces into inappropriate places (e.g., clothing, floor), whether involuntary or intentional.
B. At least one such event occurs each month for at least 3 months.
C. Chronological age is at least 4 years (or equivalent developmental level).
D. The behavior is not attributable to the physiological effects of a substance (e.g., laxatives) or another medical condition except through a mechanism involving constipation.

Epidemiology

The prevalence is around 1% among 5-year-olds, and its more prevalent among boys than girls.

Treatment

There is no specific pharmacological intervention. Child reassurance, toilet training, and behavioral modifications may be satisfactory.

Depressive disorders

Any kind of mood disorder may occur in children, yet with possible different clinical manifestations. However, DSM-5 introduced a new depressive disorder, ‘Disruptive Mood Dysregulation Disorder’, with onset between 6 to18 years old.

Disruptive Mood Dysregulation Disorder

DSM-5 criteria for diagnosis

A. Severe recurrent temper outbursts manifested verbally (e.g., verbal rages) and/or behaviorally (e.g., physical aggression toward people or property) that are grossly out of proportion in intensity or duration to the situation or provocation.
B. The temper outbursts are inconsistent with developmental level.
C. The temper outbursts occur, on average, three or more times per week.
D. The mood between temper outbursts is persistently irritable or angry most of the day, nearly every day, and is observable by others (e.g., parents, teachers, peers).
E. Criteria A-D have been present for 12 or more months. Throughout that time, the individual has not had a period lasting 3 or more consecutive months without all of the symptoms in criteria A-D.
F. Criteria A and D are present in at least two of three settings (i.e., at home, at school, with peers) and are severe in at least one of these.
G. The diagnosis should not be made for the first time before age 6 years or after age 18 years.
H. By history or observation, the age of onset of criteria A-E is before 10 years.
I. There has never been a distinct period lasting more than 1 day during which the full symptom criteria, except duration, for a manic or hypomanic episode (see chapter 3) have been met.
J. The behaviors do not occur exclusively during an episode of major depressive disorder (see chapter 3) and are not better explained by another mental disorder (e.g., autism spectrum disorder, PTSD, separation anxiety disorder, persistent depressive disorder).
K. The symptoms are not attributable to the physiological effects of a substance or to another medical or neurological condition.

Since this diagnosis is newly introduced in the DSM, clear information about possible etiological factors, courses and development, and treatments are lacking yet. Nonetheless, DSM referred to the possibilities of progression of children with these symptoms pattern to adulthood anxiety or depressive disorder.

13 SEXUAL DISORDERS

Sexual disorders are a group of mental disorders, in which the main complaints of the patients are of their own sexual lives without any underlying medical conditions or substances induced.

In its latest edition, DSM provided three independent chapters for sexual disorders. These are: Sexual Dysfunctions, Gender Dysphoria, and Paraphilic Disorders.

Sexual Dysfunctions comprise disorders related to the sexual cycle, Gender Dysphoria includes one principal condition related to one's incongruence between the assigned and experienced gender, while Paraphilic Disorders embrace disorders of the sexual object 'preferences'.

Before tackling each disorder separately, its essential for the reader to differentiate between the concept of 'Sex' and 'Gender'. In English language, the term "sex" is a source of confusion for several medical and social sciences professionals since it connotes both male/female and sexuality.

APA, in its DSM-5, refers to terms "sex" and "sexual" from biological point of understanding both male and female. For instance, sex chromosomes, gonads, sex hormones, and non-ambiguous internal and external genitalia. Disorders of sex development denote conditions of inborn somatic deviations of the reproductive tract from the norm and/or discrepancies among the biological indicators of male and female which are outside the coverage of this textbook.

Likewise, 'Sexual Dysfunctions' described below, signify 'improper' functioning of the 'sexual cycle' during the sexual practice, whether a masturbation or an intercourse. Therefore, in order to understand these 'dysfunctions', the reader is called to get sense of the normal sexual cycle (See both Figure 13-1 and table 13-1).

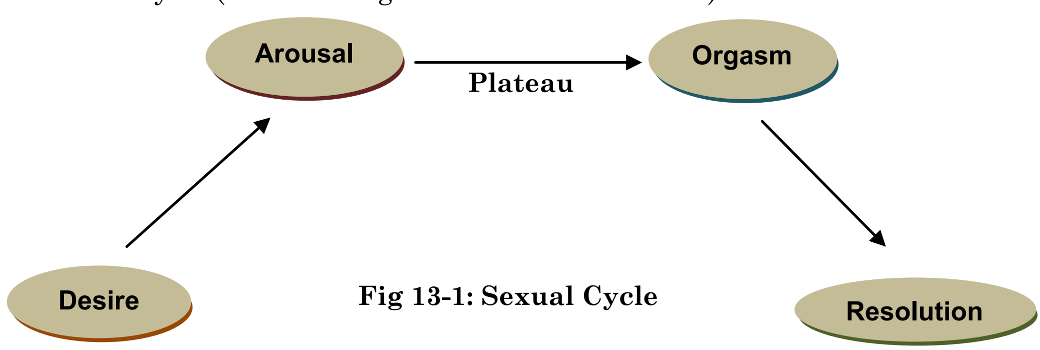

Fig 13-1: Sexual Cycle

Table 13-1: Sexual cycle response		
Phase	Woman	Man
Arousal	Clitoral enlargement Ballooning of vaginal vault Nipple erection	Penile erection Testes elevate Nipple erection
Plateau	Clitoris retracts Tenting of vagina Bartholin's secretion Sex flush	Increase in penile circumference Testes increase in size Cowper's secretion Sex flush
Orgasm	Pelvic contractions Increase in pulse and blood pressure Ejaculation	Contractions Increase in pulse and blood pressure Ejaculation
Resolution	Clitoris return to normal size Loss of vaginal congestion Descent of uterus and vagina	Penile retraction Testes descend

Consequently, 'Sexual Dysfunctions' comprise the following disorders:

A. Desire disorders: male hypoactive sexual desire disorder, and female sexual interest/arousal disorder.
B. Arousal disorders: erectile disorder, and female sexual interest/arousal disorder.
C. Orgasm disorders: premature (early) ejaculation, delayed ejaculation, and female orgasmic disorder.
D. Pain/penetration disorder: genito-pelvic pain/penetration disorder.

On the other hand, the term "Gender" is used to denote the public (and usually legally recognized) lived role as boy or girl, and man or woman. Gender is usually assigned at birth commonly as boy or girl. Sometime this assignment named 'natal gender'. Accordingly, 'Gender Dysphoria' basically refers to an individual's affective/cognitive dissatisfaction with the assigned gender.

Paraphilia, as term derived from "Para" which means "around", "beyond", and "Philia" which stands for "desire" "interest" "fondness". Hence, the term 'Paraphilia' applied to improper 'sexual target' requirement for arousal. I.e., the individual looks after 'odd' sexual objects. Usually, the sexual target of an individual is another individual of the opposite gender 'heterosexual', the same gender 'homosexual', or both genders 'bisexual'. What happened in 'Paraphilias' are intense sexually arousing fantasies, sexual urges or behaviors generally targeting: non-human objects, the suffering or humiliating of oneself or one's partner, or children or other non-consenting individuals.

Taking all together, it's vital to get thru the following concepts before considering each disorder distinctly:

A. **Genetic Sex:** refers to male karyotype (XY) and female karyotype (XX).
B. **Sexual identity:** refers to biologic sexual characteristics, such as genitalia, hormonal composition, and secondary sexual characteristics. Basically, there are two sexual phenotypes: Male and Female.
C. **Gender identity:** is the person's sense (both affects 'emotions' and cognitions) of being male or female, rather than the biologic state of being male or female. It develops in most persons by the age 2 or 3 years and fixed almost before the age of 5. Usually, gender identity corresponds to one's sexual identity. However, it is highly determined by the environmental factors through the learning processes (how the parents and others assign the baby). When this sense of identity is at variance with the anatomical sex (sexual identity), the person is said to have 'Gender Dysphoria'.
D. **Gender role:** is the social behavior that allows others to categorize the person as male or female. Although boy vs. girl or man vs woman roles are more or less universal, these roles might vary across different cultures as well as over different periods of time and eras.
E. **Sexual orientation:** is the erotic attraction felt toward persons of the opposite gender (heterosexual), the same gender (homosexual), or the both (bisexual).

Sexual Dysfunctions

Two main points could be detected in most recent, DSM-5, diagnoses of these disorders: first, all of these dysfunctions require a minimum duration of 6 months; second, its not mandatory for dysfunctions, unlike majority of psychiatric disorders, to be complicated by social or occupational dysfunctioning, although personal distress kept to be a necessity for each diagnosis.

One can uniform the DSM-5 criteria for the entire sexual dysfunctions as the following:

DSM-5 Criteria for the diagnosis of all Sexual Dysfunctions

A. Symptoms must be experienced on almost all or all sexual activity occasions.
B. Symptoms in A have persisted for the minimum of 6 months.

C. Symptoms in A cause clinically significant distress in the individual.
D. The symptoms in A are not better explained by nonsexual mental disorders, or as a consequence of severe relationship distress or other significant stressors and are not attributable to the effects of a substance/medication or another medical condition.

Classifications of Sexual Dysfunctions

A. Specify type:
1. Lifelong
2. Acquired

B. Specify type:
1. Generalized
2. Situational

C. Specify severity:
1. Mild
2. Moderate
3. Severe

Male Hypoactive Sexual Desire Disorder

It is defined as a persistent or recurrent deficient (or absence) of sexual/erotic thoughts or fantasies and desires for sexual activities.

Etiology

A. Biological factors: like hormonal disturbances; for instance, low level of testosterone or higher levels of prolactin or estrogen.
B. Psychological factors: like fear from sex or chronic stress, anxiety, or depression.
C. Social factors: like abstinence from sex for a prolonged period, lack of active willing partner, marital conflicts, or history of been a victim of rape, or sexual abuse.

Differential Diagnoses

A. Medical conditions: e.g., Parkinson disease, stroke, hypothyroidism, hypo-gonadism, etc…
B. Substances induced: e.g., Alcohol, psychotropics, antihypertensives, etc…
C. Psychiatric disorders: e.g., Depressive disorders, Anxiety Disorders, Schizophrenia, etc…

Prevalence

Its around 2% of men worldwide. It increases with age.

Treatment

A. Search for the possible differential diagnoses and treat them, if any.
B. Cognitive-behavioral therapy, including stress management, social and partnership skills training.
C. Uses of fertilized chicken eggs might be beneficial for hypoactive disorder.
D. Bupropion, which is dopamine reuptake inhibitor antidepressant, may have rule in raising the libido.
E. Other drugs like bromocriptine, and sildenafil were beneficial in some cases.

Female Sexual Interest/Arousal Disorder

Its defined by the presence of three or more of the following:

1. Absent/reduced interest in sexual activity.
2. Absent/reduced sexual/erotic thoughts or fantasies.
3. No/reduced initiations of sexual activity, and typically unreceptive to partner's attempts to initiate.
4. Absent/reduced sexual excitement/pleasure during sexual activity.
5. Absent/reduced sexual interest/arousal in response to any internal or external sexual/erotic cues (e.g., written, verbal, visual).
6. Absent/reduced genital or non-genital sensations during sexual activity.

Etiology

A. Biological factors: there appears to be a strong influence of genetic factors on vulnerability to sexual problems in female.
B. Psychological factors: similar to male hypoactive sexual desire disorder.
C. Social factors: similar to male hypoactive sexual desire disorder.

Prevalence

Low sexual desire is common among women in general. However, there is no personal distress among most of them, which is mandatory for the diagnosis to be made. Till now, the exact prevalence of this disorder is unknown, though its estimated to be rare.

Treatment

A. Portable vacuum clitoral therapy device.
B. Systemic Drugs: Like sildenafil, alprostadil (prostaglandin E), phentolamine (alpha receptor blocker), yohimbine, or hormonal replacement for post-menopausal women.
C. Psychotherapy: a form of cognitive-behavioral and educational therapy called (Sex Therapy) is beneficial in majority of conditions, and should be considered as the first line treatment before progressing to other more invasive means of therapy.

Erectile disorder (impotence)

It's defined by the presence of at least one of the following:

1. Marked difficulty in obtaining an erection during sexual activity.
2. Marked difficulty in maintaining an erection until the completion of sexual activity.
3. Marked decrease in erectile rigidity.

Etiology

A. Biological factors: like hormonal disturbances.
B. Psychological factors: like neurotic personality traits, fear and anxiety from sex.
C. Social factors: like marital conflicts.

Prevalence

It accounts for around 2% of male population. The prevalence increases with aging.

Differential diagnoses

Problems of sexual arousal might be the presentation of one or more of the following conditions; therefore, it is mandatory for the psychiatrist to exclude these conditions before diagnosing sexual arousal disorders:

A. Medical conditions: Like endocrinal dysfunctions (hypo- hyper-thyroids, cushingoid syndrome, hypogonadism, etc…), stroke, heart diseases, organs failure, spinal cord pathologies, etc…
B. Substance induced: like alpha and beta receptors blockers, anticholinergics, psychotropics, steroids, dopamine antagonists, etc…
C. Other psychiatric disorders: like schizophrenia, depressive disorders, anxiety disorders, etc…

Treatment

A. Local injection of vasodialtors, local creams containing vasodilators, portable penal vacuum, and constrictor ring fixed at the base of the penis.
B. Systemic Drugs: Like sildenafil, alprostadil (prostaglandin E), phentolamine (alpha receptor blocker), yohimbine, or hormonal replacement for post-menopausal women.
C. Psychotherapy: a form of cognitive-behavioral and educational therapy called (Sex Therapy) is beneficial in majority of conditions, and should be considered as the first line treatment before progressing to other more invasive means of therapy.

Premature (early) ejaculation

Its defined as a persistent or recurrent pattern of ejaculation occurring during partnered sexual activity within approximately 1 minute following penetration and before the individual wishes it.

Etiology

A. Biological factors: like dopamine or serotonin transporters gene polymorphism.
B. Psychological factors: like fear and anxiety from sex.
C. Social factors: like sexually inexperienced young men.

Prevalence

It is regarded to be the commonest sexual problem among men, it accounts for about 25-40% of all sexual dysfunctions in men, and prevalent in more than 20% of men worldwide.

Treatment

A. Sex therapy.
B. Anti-depressants, particularly SSRIs or Clomipramine. (getting benefits from their side effects)
C. Alpha-receptor blockers. (getting benefits from their side effects)
D. Local anesthetic ointments.

Delayed Ejaculation and Female Orgasmic Disorder

Delayed ejaculation

It is defined by the presence of one of the following without the individual's desire:

1. Marked delay in ejaculation.

2. Marked infrequency or absence of ejaculation.

Female orgasmic disorder

It is defined by the presence of one of the following without the individual's desire:

1. Marked delay in, marked infrequency of, or absence of orgasm.
2. Markedly reduced intensity of orgasmic sensations.

Etiology

A. Biological factors: like genetic predisposition, aging process, or physical tiredness.
B. Psychological factors: like fear and anxiety from sex, lack of emotions for the partner, or psychological exhaustion.
C. Social factors: like poor sexual technique by the partner, or cultural and religious expectations in case of female orgasmic disorder.

Prevalence

The condition occurs in around 15% of pre-menopausal and 35% of post-menopausal women. The condition is much rarer among men.

Treatment

Consists mainly of the combination of Cognitive-behavioral as well as educational psychotherapies (sex therapy).

Genito-Pelvic Pain/Penetration Disorder

It is defined by persistent or recurrent difficulties with one (or more) of the following:

1. Vaginal penetration during intercourse.
2. Marked vulvovaginal or pelvic pain during vaginal intercourse or penetration attempts.
3. Marked fear or anxiety about vulvovaginal or pelvic pain in anticipation of, during, or as a result of vaginal penetration.
4. Marked tensing or tightening of the pelvic floor muscles during attempted vaginal penetration.

Etiology

A. Biological factors: like post-infection without residual physical findings, or pain or inability to insert tampon prior to intercourse are important risk factors. Also, penile malformation or impaired lubrication are among risk factors.
B. Psychological factors: like fears from sex, or negative attitudes toward sex.

C. Social factors: like marital discomfort, or prior sexual abuse.

Differential diagnosis
The condition may be secondary to hymenal scarring, endometriosis, or vulvular vestibulitis.

Prevalence
It occurs in 14% of women and 0.2% of men. Prevalence increased with age.

Treatment
Gradual dilatation procedure might be beneficial in few cases. However, the main therapy came from curing underlying conditions.

Gender Dysphoria
This condition was named ‘Gender Identity Disorder’ in the previous version of DSM, DSM-IV-TR. Though, since the main suffering of the affected individual is not related to the ‘Gender Identity’ per se. Rather, its his/er unhappiness and dissatisfaction with one’s gender behind the clinical presentation. For that reason, in DSM-5, the name has been modified to denote the cognitive and emotional components (Dysphoria) of the condition instead of a pathology of the identity.
DSM-5 designed two different sets of criteria for the diagnosis of the condition, one for children and another for adolescents and adults. However, since the sufferers draw clinical attentions at during or after adolescence, and to avoid possible confusion to undergraduate medical students, the diagnostic criteria for adolescents and adults only will be presented here.

DSM-5 criteria for diagnosis
A. A marked incongruence between one’s experienced/expressed gender and assigned gender, of at least 6 months’ duration, as manifested by at least two of the following:
   1. A marked incongruence between one’s experienced/expressed gender and primary and/or secondary sex characteristics (or in young adolescents, the anticipated secondary sex characteristics).
   2. A strong desire to be rid of one’s primary and/or secondary sex characteristics because of a marked incongruence with one’s experienced/expressed gender (or in young adolescents, a

desire to prevent the development of the anticipated secondary sex characteristics).

3. A strong desire for the primary and/or secondary sex characteristics of the other gender.
4. A strong desire to be of the other gender (or some alternative gender different from one's assigned gender).
5. A strong desire to be treated as the other gender (or some alternative gender different from one's assigned gender).
6. A strong conviction that one has the typical feelings and reactions of the other gender (or some alternative gender different from one's assigned gender).

B. The condition associated with clinically significant distress or impairment in social, occupational, or other important areas of functioning.

Etiology

The exact cause is unknown, however its due to postnatal life events. The formation of gender identity is influenced by the interaction of children with parents. Normally, a boy dealt with, assigned, and valued by parents as a male gender, through offering him boy dresses, boys games, and encouraging him to imitate the father. Likewise, the reverse is true for a girl.

Sometime, in few families, the reverse might occur, through dealing with the boy as a girl and dealing with the girl as a boy. For example, in our society, boy children may be more valued within some families. Such an attitude by these families may lead to over spoiling of the male baby and dealing with him as if the baby is a girl, through dressing him like girls, doing makeup, and involving him with girls toys and plays (dolls and dollhouse). Moreover, when the family has only one boy, and because he became the center of attention, the family possibly prevents their boy to join other boys in their plays. Instead, they keep him with his sisters to be involved in their effeminate plays and attitudes. The same is true for the girl babies, when the family may deal with her more toughly, assigned for her the role of her brothers, and day by day they became more tomboyish.

Prevalence

Exactly unknown; however, male affected four times that of female.

Treatment:

A. Psychotherapy to decrease the distresses, and not to convince the reverse for the patient.
B. Real life test: the patient should keep his/her livings in the opposite gender's role for at least 1-2 years to evaluate the adjustment of the person in the new gender role.
C. If the person passed stage 2 out of problem, then hormonal therapy will be given: estrogen for male and androgens for female.
D. Surgical reassignment.
E. Helps for the patient and the family to adjust to the new situation.

Paraphilic Disorders

As mentioned at the beginning of the current chapter, the term 'Paraphilia' applied to improper 'sexual target' requirement for arousal. It does not require clinical attention and it's not regarded as a disorder per se. However, whenever the condition is coupled by personal distress or impairment in functioning, or when the paraphilic activity entails harm or risk of harm to oneself or others, then the paraphilic activity called 'Paraphilic Disorder'

DSM-5 criteria for diagnosis

A group of disorders of at least 6 months duration, which cause significant distress or impair social, or occupational life, or carry the risk of harm to oneself or others and which are characterized by recurrent, intense sexually arousing fantasies, sexual urges or behaviors generally involving: a- non-human objects, b- the suffering or humiliating of oneself or a non-consenting partner, or c- sexual activities targeting children or other non-consenting individuals.

Clinical types: see table 13-2

Etiology

The exact cause is unknown. It might be due to the absence of a usual, classical, willing, active sexual partner. The psychopathologies are those of Obsession and compulsion, feeling of sexual inadequacy, as well as psychopathologies underlie dependence (addiction) (how a maladaptive behavior became repetitive (conditioned) due to enforcement by the rewarding system in the limbic system through possible excessive dopamine neurotransmission).

However, the condition might occur secondary to:

A. Anti-social personality disorder.
B. Temporal lobe (complex-partial) epilepsy.
C. Post-encephalitic neuropsychiatric syndrome.
D. Frontal lobe tumors.
E. Bilateral temporal lobe lesions.
F. Multiple sclerosis.

Table 13-2: Paraphilic Disorders

Paraphilias	**Sexual Object of Preference**
Exhibitionism	Repeated exposure of genitals to unprepared non-consenting strangers
Voyeurism	Observing an unsuspecting, non-consenting naked, disrobing, or engaging in sexual activity
Frotteuristic	Touching or rubbing against non-consenting person
Fetishism	Uses of the opposite gender's objects, like shoes, panties, etc…
Transvestic Fetishism	Getting sexual pleasure by wearing clothes of the opposite gender
Pedophilia	Involving children (younger than 16) in getting sexual pleasure
Necrophilia	Involving cadavers in getting sexual pleasure
Zoophilia (Bestiality)	Involving animals in getting sexual pleasure
Sadism	Inducing pain, torture, humiliation to the partner to get sexual pleasure
Masochism	Inducing pain, torture, humiliation to oneself to get sexual pleasure
In addition to many other types…Klismaphilia, Urophilia, Coprophilia, telephone scatologia, etc…	

Prevalence

Exactly unknown, but probably paraphilias are disorders of men. Only few cases of women have been recorded till now.

Treatment

A. Treat underlying conditions, if any.
B. Behavioral therapy, called aversive therapy (deconditioning).
C. Social skill trainings.
D. Drugs, like: antipsychotics, SSRIs, clomipramine, lithium, anti-androgens like medroxyprogesterone, or cyproterone.

14 SLEEP-WAKE DISORDERS

Sleep disturbances may be symptoms of almost all psychiatric disorders as well as several medical conditions. However, sleep specialists, psychiatrists, and other clinicians commonly come across a group of patients who are complaining primarily, and may be solely, from a cluster of sleep-related signs and symptoms.

Sleep-wake disorders can be classified into four main groups:

A. Quantitative changes in sleep durations in which there are reduction, increase, or disruption of sleep times:
   1. Insomnia disorder
   2. Hypersomnolence disorder
   3. Narcolepsy
   4. Circadian rhythm sleep-wake disorders

B. Qualitative changes in sleep pattern in which there are parallel psychological or behavioral signs and symptoms with sleep (parasomnias).
   1. Non-REM sleep arousal disorders
   2. Nightmare disorder
   3. REM sleep behavior disorder

C. Breathing-related sleep disorders:
   1. Obstructive sleep apnea hypopnea
   2. Central sleep apnea
   3. Sleep-related hypoventilation

D. Restless legs syndrome

Breathing-related sleep disorders' patients mainly address specialists in respiratory system, internists, or sleep specialists; and Restless legs syndrome patients mainly consult neurologists. Therefore, this chapter will focus only on the first two groups of disorders which are focuses of psychiatrists' attentions.

What is biological clock?

It is a tendency of an organism to express regular periodic changes or rhythms in physiological and behavioral functions at approximately the same time each day, each month, or once a year. Examples of biological clock include: circadian rhythm, menstrual cycle, and any seasonal patterns of physiological and behavioral activities.

Biological clock determined by chemical agents (neurotransmitters, hormones, etc…), process of learning (evolution) and environmental stimuli.

What is circadian rhythm?
'Circadian' is a Latin phrase derived from 'circa = about' and 'dies = day'. Hence, the term 'circadian' stands for "about a day". Circadian rhythm, therefore, is a biological activity that follows cycles that repeat over 24 hours. Examples include: normal sleep-wake cycle, volume of urine excreted/24 hours, body temperature changes/24 hours, food and drink intakes/24 hours, Electro-Encephalography (EEG) activities' changes/24 hours, etc…

What is sleep (Somnus) (*Hypnos*-God of Sleep)?
It is an altered state of consciousness in which the brain, for a time, gives up some of its functions.

Why do we sleep? What are the functions of sleep?
A. Sleep is a time of rest to both the body and the mind.
B. Sleep comes about in evolution to preserve energy and protect individual during the 24 hours.
C. Bulk of growth hormone secretions occur during sleep, sleep deprivation leads to impaired growth.
D. During sleep, the mind consolidates newly learned data (consolidation of memory).

What are the stages of sleep?
Sleep has two main stages:
A. Non-Rapid Eye Movement (REM) sleep:
   1. Stage 1
   2. Stage 2
   3. Stage 3
   4. Stage 4

   Note: both stage 3 and 4 also named: (Slow Wave Sleep), (Deep Sleep), (Orthodox Sleep), (Non Dream Sleep), (Synchronized Sleep), or (Passive Sleep).
B. Rapid Eye Movement (REM) sleep: also named: (Paradox Sleep), (Dream Sleep), (De-Synchronized Sleep), or (Active Sleep). (See tables 14-1 and 14-2).

How long does average sleep last?
Infants spend more than 15 hours in sleep. Half of this time is REM sleep. Total sleep time, then, decreased to around 6-9 hours for an average normal adult. REM sleep is reduced as well to constitute around 25% of total sleep time. Furthermore, total sleep time may be as low as 4 hours/24-hour among older age groups.
Sleep stages pass through a cyclical pattern over around 90 minutes and as follows: [Stage 1- Stage 2- Stage 3- Stage 4- REM – REM – Stage 4- Stage 3- Stage 2- REM]
As its noticed, REM sleep replaces Stage 1 sleep from the second cycle. An average normal adult sleep witnesses 4-5 cycles. REM sleep tends to predominate the second half of sleep, and near early morning, the entire sleep may be REM sleep. This is why on awakening in morning, dreams are more recalled.

Insomnia Disorder

DSM-5 criteria for diagnosis

A. A predominant complaint of dissatisfaction with sleep quantity or quality, associated with one (or more) of the following symptoms:
   1. Difficulty initiating sleep. (In children, this may manifest as difficulty initiating sleep without caregiver intervention).
   2. Difficulty maintaining sleep, characterized by frequent awakenings or problems returning to sleep after awakenings. (In children, this may manifest as difficulty returning to sleep without caregiver intervention).
   3. Early-morning awakening with inability to return to sleep.

B. The sleep disturbance causes clinically significant distress or impairment in social, occupational, educational, academic, behavioral, or other important areas of functioning.
C. The sleep difficulty occurs at least 3 nights per week.
D. The sleep difficulty is present for at least 3 months.
E. The sleep difficulty occurs despite adequate opportunity for sleep.
F. The insomnia is not better explained by and does not occur exclusively during the course of another sleep-wake disorder (e.g., narcolepsy, a breathing-related sleep disorder, a circadian rhythm sleep-wake disorder, a parasomnia).
G. The insomnia is not attributable to the physiological effects of a substance (e.g., a drug of abuse like a stimulant, a medication).
H. Coexisting mental disorders and medical conditions do not adequately explain the predominant complaint of insomnia.

Table 14-1: stages of sleep		
Sleep Non REM	% of total normal adult sleep	EEG Pattern
Stage1	5%	Theta Waves,4-7c/s (small, regular, slow)
Stage 2	45%	12-14c/s (Sleep Spindle and K-Complex)
Stage 3	12%	Delta Waves, up to 4c/s (large, slow, irregular)
Stage 4	13%	Delta Waves, up to 4c/s (large, slow, irregular)
REM	25%	Beta Waves,14-25c/s (small, rapid, irregular)

Table 14-2: Comparison between Non REM sleep and REM sleep:		
	Non-REM	REM
Eye movement	Absent	Very rapid 10-20/s
Vital signs	Deactivated	Activated
Muscles	Increased relaxation	Almost completely paralyzed (except heart, diaphragm, smooth muscles and eyes muscles)
Metabolic rate of brain	Decreased to 25-30% of wakefulness	Comparable with wakefulness
Dreams	Only 25% of times record dreams and are not vivid, more like normal thinking and more related to what is happened in the awakened day	Almost always recorded if awakened at it, and are vivid, surrealist
Night mares	No	Yes
Night terror, enuresis, walking	Yes	No
Penile/Clitoral erection	No	Yes

Epidemiology

Insomnia disorder is the commonest sleep disorder. The prevalence rate may reach up to 20% of population. The prevalence is more among female population.

Etiology

Although there are evidences of higher rates of this disorder among MZ twins compared to DZ twins, it's not obvious this difference is due to genetic predispositions or shared environmental factors yet. Most noticeable etiological factors may be anxious and worry temperaments, and uncomfortable sleep venues like noises, light, or high or low temperature.

Treatment

A. Sleep hygiene: it is a kind of educational therapy aiming in modifying and improving behavioral habits related to sleep, as follows:
   1. The sleep environment (bedroom) should be familiar, relaxed, dark, and quiet.
   2. The sleeping room (bedroom) should be used for sleep and sex only. I.e., avoid eating, watching TV, or other social activities in the bedroom.
   3. The individual should be encouraged to stick to bedtimes. I.e., going to bed and wakes up at the same times each day.
   4. Address the bed only when tired.
   5. The following behaviors may improve both the quality and quantity of sleep:
      a. Regular mild daily exercise, like walking for instance.
      b. Taking a warm shower immediately before going to bed.
      c. Drinking a glass of warm milk before bed time.
   6. Avoiding the following behaviors may improve both the quality and quantity of sleep:
      a. Caffeine-beverages drinking late in evening and night.
      b. Excessive alcohol drinking.
      c. Excessive smoking.
      d. Excessive daytime sleep (noon naps).
      e. Large dinner late at night.
      f. Staying in bed for long time awake.

Although sleep hygiene seems to be easily applicable, though it's hard to find individuals with sleep disorders to stick-well to these instructions. Instead, they prefer easier aids of sleep pills.

B. Sleep pills: despite the availability of several sleep pills, clinicians and psychiatrists are, generally, directed to keep these pills as a last option since there is a great risk of tolerance, withdrawal features, and sedatives-related disorder. Sleep pills include mainly Benzodiazepines, and Non-Benzodiazepine GABA agonists (Z-drugs). However, several other psychotropic drugs with sedative properties may be used cautiously in smaller doses like antipsychotics, antidepressants, antihistamines, etc… (See chapter 10 and 16).

Hypersomnolence Disorder

DSM-5 criteria for diagnosis

A. Self-reported excessive sleepiness (hypersomnolence) despite a main sleep period lasting at least 7 hours, with at least one of the following symptoms:
   1. Recurrent periods of sleep or lapses into sleep within the same day.
   2. A prolonged main sleep episode of more than 9 hours per day that is non-restorative (i.e., unrefreshing).
   3. Difficulty being fully awake after abrupt awakening.

B. The hypersomnolence occurs at least 3 times per week, for at least 3 months.

C. The hypersomnolence is accompanied by significant distress or impairment in cognitive, social, occupational, or other important areas of functioning.

D. The hypersomnolence is not better explained by and does not occur exclusively during the course of another sleep-wake disorder (e.g., narcolepsy, a breathing-related sleep disorder, a circadian rhythm sleep-wake disorder, or a parasomnia).

E. The hypersomnolence is not attributable to the physiological effects of a substance (e.g., a drug of abuse like a sedative, a medication).

F. Coexisting mental disorders and medical conditions do not adequately explain the predominant complaint of hypersomnolence.

Epidemiology

Although the prevalence is thought to be high, the exact rate, worldwide, is exactly unknown yet. This is may be due to the unwillingness of sufferers to consult healthcare providers.

Etiology

This condition may run in family through an autosomal dominant mode of inheritance. Although hypersomnolence may increase during stresses, little is known about psychological or social factors of this condition.

Treatment

A. Sleep hygiene: see above.
B. Stimulants but with cautions (see chapter 10).

Narcolepsy

DSM-5 criteria for diagnosis

A. Recurrent periods of an irrepressible need to sleep, lapsing into sleep, or napping occurring within the same day. These must have been occurring at least 3 times per week over the past 3 months.
B. The presence of at least one of the following:
   1. Episodes of cataplexy, defined as either (a) or (b), occurring at least a few times per month:
      a. In individuals with long-standing disease, brief (seconds to minutes) episodes of sudden bilateral loss of muscle tone with maintained consciousness that are precipitated by laughter or joking.
      b. In children or in individuals within 6 months of onset, spontaneous grimaces or jaw-opening episodes with tongue thrusting or a global hypotonia, without any obvious emotional triggers.
   2. Hypocretin deficiency, as measured using cerebrospinal fluid (CSF) hypocretin-1 immunoreactivity values (less than or equal to one-third of values obtained in healthy subjects tested using the same assay, or less than or equal to 110 pg/mL). Low CSF levels of hypocretin-1 must not be observed in the context of acute brain injury, inflammation, or infection.
   3. Nocturnal sleep polysomnography showing rapid eye movement (REM) sleep latency less than or equal to 15 minutes, or multiple sleep latency test showing a mean sleep latency less than or equal to 8 minutes and two or more sleep-onset REM periods.

Epidemiology

It is relatively rare condition with an overall prevalence of around 0.03% of general population.

Etiology

Narcolepsy is thought to has strong genetic bases since the prevalence among first-degree relatives raised, exceptionally, up to 2% and the concordance rate among MZ twins is around 25-32%.

Infections, and head trauma may trigger the onset of Narcolepsy also. Furthermore many researchers suggest similarities between Narcolepsy and Epilepsy including epileptic EEG discharges; hence, they may share similar etiological factors as well.

Treatment

Same as hypersomnolence. Trials of anticonvulsants may be beneficial in some cases.

Circadian Rhythm Sleep-Wake Disorders

DSM-5 criteria for diagnosis

A. A persistent or recurrent pattern of sleep disruption that is primarily due to an alteration of the circadian system or to a misalignment between the endogenous circadian rhythm and sleep-wake schedule required by an individual's physical environment or social or professional schedule.
B. The sleep disruption leads to excessive sleepiness or insomnia, or both.
C. The sleep disturbance causes clinically significant distress or impairment in social, occupational, and other important areas of functioning.

Types

A. **Delayed sleep phase type:** a pattern of delayed sleep onset and awakening times, with an inability to fall asleep and awaken at a desired or conventionally acceptable earlier time.
B. **Advanced sleep phase type:** a pattern of advanced sleep onset and awakening times, with an inability to remain awake or asleep until the desired or conventionally acceptable later sleep or wake times.
C. **Irregular sleep-wake type:** a temporally disorganized sleep-wake pattern, such that the timing of sleep and wake periods is variable throughout the 24-hour period.
D. **Non-24-hour sleep-wake type:** a pattern of sleep-wake cycles that is not synchronized to the 24-hour environment, with a consistent daily drift (usually to later and later times) of sleep onset and wake times.
E. **Shift work time:** insomnia during the major sleep period and/or excessive sleepiness (including inadvertent sleep) during the major awake period associated with a shift work schedule (i.e., requiring unconventional work hours).

Epidemiology
The exact prevalence is unknown. Advanced-type thought to be the commonest type, but Delayed-type appears to be as common as 7% among adolescents.

Etiology
Although advance-type showed an autosomal dominant mode of inheritance and non-24-hour found to be more common among blind individuals, most of the etiological factors believed to be due to environmental-triggers as well as habitual learning of particular maladaptive phase of sleeping.

Treatment
Setting back the environmental factors with sleep hygiene might be satisfactory.

Parasomnias
Here, there are qualitative changes in the sleep pattern rather than quantitative decrease, increase, or disruption in the sleep duration.

Non-Rapid Eye Movement Sleep Arousal Disorder
DSM-5 criteria for diagnosis
A. Recurrent episodes of incomplete awakening from sleep, usually occurring during the first third of the major sleep episode, accompanied by one of the following:
   1. **Sleepwalking (Somnambulism):** repeated episodes of rising from bed during sleep and walking about. While sleepwalking, the individual has blank, staring face; is relatively unresponsive to the efforts of others to communicate with him/er; and can be awakened only with great difficulty.
   2. **Sleep terrors:** recurrent episodes of abrupt terror arousals from sleep, usually beginning with a panicky scream. There is intense fear and signs of autonomic arousal, such as mydriasis, tachycardia, rapid breathing, and sweating, during each episode. There is relative unresponsiveness to efforts of others to comfort the individual during the episodes.

B. No or little (e.g., only a single visual scene) dream imagery is recalled.

C. Amnesia for the episodes is present.

D. The episodes cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
E. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of misuse, a medication).
F. Coexisting mental and medical disorders do not explain the episodes of sleepwalking or sleep terrors.

Epidemiology
Both sleepwalking and sleep terrors are common as isolated episodes. However, the prevalence of sleepwalking as a disorder is probably within the range of 1%-5%. The prevalence is higher among younger age groups. Sleep terrors, also, is common among children. The picture may be more than one-third of 18 months old children population. The prevalence decreases by aging, to the lowest, around 2.2% among adults.

Nightmare Disorder
DSM-5 criteria for diagnosis
A. Repeated occurrence of extended, extremely dysphoric, and well-remembered dreams that usually involve efforts to avoid threats to survival, security, or physical integrity and that generally occur during the second half of the major sleep episode.
B. On awakening from the dysphoric dreams, the individual rapidly become oriented and alert.
C. The sleep disturbance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
D. The nightmare symptoms are not attributable to the physiological effects of a substance (e.g., a drug of misuse, a medication).
E. Coexisting mental and medical disorders do not explain the predominant complaint of dysphoric dreams.

Epidemiology
Its more common among adolescents than other age groups. The rate may reach as high as 4%.

Rapid Eye Movement Sleep Behavior Disorder
DSM-5 criteria for diagnosis
A. Repeated episodes of arousal during sleep associate with vocalization and/or complex motor behavior.

B. These behaviors arise during REM sleep and therefore usually occur more than 90 minutes after sleep onset, are more frequent during the later portions of the sleep period, and uncommonly occur during daytime naps.
C. Upon awakening from these episodes, the individual is completely awake, alert, and not confused or disoriented.
D. Either of the following:
   1. REM sleep without atonia on polysomnographic recording.
   2. A history suggestive of REM sleep behavior disorder and an established synucleinopathy diagnosis (e.g., Parkinson's disease, multiple system atrophy).
E. The behaviors cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (which may include injury to self or the bed partner).
F. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of misuse, a medication).
G. Coexisting mental and medical disorders do not explain the episodes.

Epidemiology

It accounts for around 0.5% of general population. The prevalence may be higher among patients with psychiatric disorders.

Etiology of Parasomnias

A. Biological factors: Parasomnias are more common among family members with similar conditions. Fever and psychotropic drugs uses may account for episodes of non-REM sleep arousal and REM sleep behavior disorders respectively.
B. Psychological factors: like emotional stresses, traumas, and personality disturbances.
C. Environmental factors: like sleep deprivation, or irregular sleep schedules.

Treatments of Parasomnias

A. Sleep hygiene, as well as reassuring both the child and parents, and advises about how keeping the individual safe during the episodes.
B. Sleepwalking, sleep terrors, and nightmare disorders may respond to lower doses of TCADs.
C. All Parasomnias may get benefits from minor doses of Benzodiazepines like Clonazepam.

15 FEEDING AND EATING DISORDERS

Feeding and Eating disorders are characterized by a persistent disturbance of eating or eating-related behavior that results in an altered consumption or absorption of food that significantly impairs physical health or psychosocial functioning.

Until late 1970s, eating disorders were believed to be uncommon. Following the description of bulimia nervosa, they have increasingly been seen as conspicuous and disabling. Although eating disorders sound predominantly western-culture bound disorders, recently, nevertheless, with the spread of western media and fashions, the condition used to affect individuals from other cultures as well including ours, especially among young ladies.

APA in it's DSM-5 enumerated five main classes of feeding and eating disorders including: Pica, Rumination Disorder, Avoidant/Restrictive Food Intake Disorder, Anorexia Nervosa, Bulimia Nervosa, Binge-Eating Disorder, beside two additional Other Specified and Unspecified Feeding or Eating Disorders.

In this chapter, only the two well-known disorders to common people as well as professionals, Anorexia Nervosa and Bulimia Nervosa, will be tackled.

Etiology

A. Genetics: concordance rates for Anorexia Nervosa (AN) in Mono Zygotic twins are significantly higher than those for Di Zygotic twins. Furthermore, there is an increased risk of both AN and Bulimia Nervosa (BN) among first degree relatives of same disorders. Also, rates of mood disorders and OCD are increased among relatives of AN.
B. Psychological factors: low self-esteem, beside obsessional traits, in addition to displacement defense mechanism are among psychological influences for eating disorders.
C. Environmental factors: historical and cross-cultural variability in the prevalence of AN supports its association with cultures and settings in which 'thinness' is valued. Occupations and avocations that encourage thinness, such as modeling and elite

athletics, are also associated with increased risk. Furthermore, early childhood physical as well as sexual abuses may incline victims towards future eating disorders.

ANOREXIA NERVOSA (AN)

Although the name suggests 'loss of appetite' to be key feature of the victims, no apparent patients complain, however, from anorexia. Instead, the name 'Anorexia Nervosa' was erroneously coined by the English physician William Gull in 1868 to merely underscore the psychological origin of the condition.

DSM-5 criteria for diagnosis

A. Restriction of energy (food and drink) intake relative to requirements, leading to a significantly low body weight according to age, sex, development, and physical health.
   The severity of low weight is defined according to Body Mass Index (BMI) which is calculated by dividing weight in kgs by the square of height in meter. For instance if an individual's weight is 80 kgs and height is 1.8 m, then the BMI will be = $80/(1.8)^2 = 24.7$:
   Normal BMI = 18.5 – 24.9
   Overweight =25 – 29.9
   Obese = 30 – 34.9
   Severe obesity = 35 – 39.9
   Morbid obesity ≥ 40

   Mild AN = 17 – 18.5
   Moderate AN = 16 – 16.9
   Severe AN = 15 – 15.9
   Extreme AN < 15
B. Intense FEAR of gaining weight or of becoming fat, or persistent behavior that interferes with weight gain like excessive exercise, or purging behaviors (self-induced vomiting, or misuse of laxatives, diuretics, or enemas).
C. Disturbance in the way in which one's body weight or shape is experienced, unjustified influence of body weight or shape on self-evaluation and elf-esteem, or persistent lack of recognition of the seriousness of the current low body weight.

Clinical presentation and associated features

AN commonly presents during adolescence or young adulthood. It rarely begins before puberty or after age 40. The onset of the disorder is often associated with a stressful life event, such as leaving home for college.

Associated clinically presentations are the outcomes of the starvation and purging behaviors. These might include:

A. Amenorrhea in females
B. Anemia
C. Loss of bone mineral density
D. Immune system suppression
E. Hypotension, bradycardia, and/or arrhythmias
F. Lanugo hair development
G. Emaciation
H. Constipation
I. Disturbance of electrolytes and hormones
J. Depression
K. Disturbance of libido
L. Symptoms of OCD
M. Increase in suicide’s risk to around 12/100000/year.

Prevalence

Male-to-female ratio is around 1:10. 12-month prevalence among females is around 0.4%. The condition is more prevalent in mid-to-high class societies, yet less common in Middle East. Mortality rate is about 10% due to starvation or suicide.

Treatment

Initial treatment should be stressed on correction of the significant consequences of starvation (reversing the nutritional abnormalities) with hospitalization if necessary.

After weight stabilization, calorie intake is gradually increased to an amount necessary to gain 2 to 5 lb/week. (*Note*: 1 Ib = 0.45 kg) (Monitor the individual carefully, because sometimes they may consume large amounts of fluid before being weighed or hide heavy articles under their clothing).

Cognitive–behavioral therapy through modification of irrational overemphasis on weight. Rewards and punishments based on absolute weight, not on eating behaviors.

Family therapy designed to reduce conflicts about control by parents.

Antidepressants or Antipsychotics might help in reducing the obsessional 'Fat Phobia' and in regaining weight.

BULIMIA NERVOSA (BN)

The term *bulimia* comes from Greek *βουλιμία* (insatiable hunger), a compound of *βους* (bous), ox + *λιμός* (līmos), hunger. Literally, bulimia nervosa means disease of hunger affecting the nervous system. Bulimia nervosa was coined by the British psychiatrist Gerald Russell in 1979.

DSM-5 criteria for diagnosis

A. Recurrent episodes of binge eating. An episode of binge eating is characterized by:
   1. Eating an amount of food that is definitely larger than what most individuals would eat in a similar period of time under similar circumstances.
   2. A feeling that one cannot stop eating or control what or how much one is eating.

B. Recurrent inappropriate compensatory behaviors in order to prevent weight gain, such as fasting, excessive exercise, or purging behaviors (self-induced vomiting, or misuse of laxatives, diuretics, or enemas).

C. Both A and B occur, on average, at least once a week for three months.

D. Self-evaluation is unduly affected by body shape and weight.

E. The disturbance doesn't occur exclusively during episodes of AN.

Clinical presentation and associated features

The age of onset is similar to that of AN. Typically, individuals with BN are within the normal weight or over weight (see above BMIs). The binge eating frequently begins during or after an episode of dieting to lose weight. Experiencing stressful life events or frequent down-moods also precipitate the binge eating. For instance a student might indulge in the binge eating after receiving low grades in school examination.

Associated features are similar to that of AN, but the frequent and more severe purging behavior here might lead to more severe GIT complications like esophageal tear as well as dental erosions.

Prevalence

It's similar to AN. However, the 12-month prevalence is around 1% to 1.5%.

Treatment

Cognitive behavioral therapy is the key treatment. However, antidepressant medications particularly SSRIs are usually employed (They respond even if they are not depressed).

PART III

Therapies

In

PSYCHIATRY

16 PHARMACOTHERAPY

This chapter concerned with only drug-therapy used in psychiatry. Other forms of therapy like physical therapies and psychotherapies are dealt with, separately, in the next two chapters.

The main drug-groups used in psychiatry are: Antipsychotics, Antidepressants, Mood Stabilizers, and Anxiolytics. Nevertheless, several other kinds of drugs may be employed to treat side effects of above drugs or associated medical conditions.

Antipsychotics

Antipsychotics, as the name denotes, are group of drugs that therapeutically used to reduce the core psychotic symptoms (hallucinations, delusions, formal thought disorders, psychomotor excitement, and agitation). There are several other terms used interchangeably with antipsychotics, like: anti-dopamine, major tranquilizers, and neuroleptics.

Mechanisms of actions

A. Dopamine receptors antagonism: which is the main therapeutic mechanism of action and the only mode of action among traditional 'typical' antipsychotics. However, most recent antipsychotics, like Aripiprazole, may have dual effects on dopamine receptors.

B. Serotonin receptor antagonism: this is a property of the new 'atypical' antipsychotics. The benefit of anti-serotonin property is in reducing the 'anti-dopamine' side-effects and the possibilities of reducing negative symptoms of schizophrenia (see chapter 2).

Depot preparations differ from regular preparations in their pharmacokinetic properties in which longer times, usually weeks, are necessary for the plasma steady-state levels to be achieved. Therefore, depot antipsychotics are useful alternatives for patients who are non-adherent to therapy.

Types of antipsychotics: see table 16-1

Table 16-1: Antipsychotics

	Mechanism of action	Generic Name	Trade Name	Doses
Typical (first generation antipsychotics)	Blockade of dopamine receptors (mainly D2 receptor)	Chlorpromazine	Largactil	300-1000mg/ Day
		Haloperidol	Serenase	2-30mg/ Day
		Trifluperazine	Stelazine	10-30mg/ Day
Atypical (second generation antipsychotics)	Blockade of both dopamine and serotonin receptors	Olanzapine	Zyprexa, Olexar, Olan	5-20mg
		Risperidone	Risperdal, Respirox	2-16mg/ Day
		Quetiapine	Seroquel, Seroquen	400-800mg/ Day
		Ziprasidone	Geodon	80-160mg/ Day
		Aripiprazole	Abilify	10-30mg/ Day
		Molindone	Moban	15-225mg/ Day
		Paliperidone	Invega	3-15mg/ Day
		Iloperidone	Fanapt	8-32mg/ Day
		Asenapine	Saphris	10-40mg/ Day
		Clozapine	Leponix, Clozaril	350-900mg/ Day
Depot Antipsychotics		Fluphenazine decanoate	Modecate	6.25-50mg/ Week
		Flupenthixol decanoate	Depixol	12.5-400mg/ Week
		Haloperidol decanoate	Haldol	12.5-75mg/ Week
		Olanzapine pamoate		405mg/ Month
		Risperidone suspension	Risperdal Consta	25-50mg/ 2 weeks
		Paliperidone suspention	Xeplion	25-150mg/ Month

Indications of antipsychotics

A. Treatment of psychotic disorders, like schizophrenia.
B. An alternative or adjuvant therapy to mood stabilizers for bipolar disorders.
C. An alternative or adjuvant therapy to antidepressants for depressive disorders.
D. An alternative or adjuvant therapy to anxiolytics for anxiety disorders.
E. For treatment-resistant OCD.

F. For symptomatic treatment of personality disorders.
G. For behavioral control of several neurodevelopmental disorders like autism-spectrum disorder and intellectual disability, and neurocognitive disorders.
H. For behavioral control in the emergency settings. For instance, in the management of delirium or other causes of agitation.
I. For treatment of resistant cases of feeding disorders like anorexia nervosa.
J. For treatment of neuromedical conditions like Huntington's disease and Tourette's syndrome.
K. Symptomatic treatment of nausea and vomiting, vertigo, and hiccup.

Side effects of antipsychotics

Antipsychotics have broad-spectrum effects on central neurochemical receptors. Besides antagonizing both dopamine and serotonin receptors, antipsychotics block several other neurotransmitters' receptors like adrenergic, histamine, muscarinic and nicotinic receptors. Most of their side effects may be explainable within this wide range of 'blocking' effects.

A. Due to anti-dopamine property

1. Extra-pyramidal: (common with typical drugs) Parkinsonism, Akathisia (objective and subjective sense of restlessness), Tardive Dyskinesia (any abnormal hyperactive movement except tremor can be regarded as tardive dyskinesia), Dystonia, and Neuroleptic Malignant Syndrome (NMS) (see below).
2. Hyperprolactinemia and subsequent galactorrhoea, amenorrhoea, gynaecomastia, hypogonadism, and sexual dysfunction (Aripiprazole is least causing hyper prolactinemia due to its dual effect on dopamine).
3. Depression.
4. Sexual dysfunction (decreased sexual desires).

Extra pyramidal Side Effects (EPS)

EPS are the side effects that occur with the use of dopamine antagonist agents like:

A. Antipsychotics (more with typical agents).
B. Antiemetic like metoclopramide.
C. Agents used for vertigo like Prochlorperazine (Stemitel).

Neuroleptic Malignant Syndrome

Is a medical emergency, idiosyncratic reaction, may occurs even after single small sized dose, characterized by elevated temperature, impaired consciousness (semi delirious), body rigidity, profound sweating, and many other features including autonomic liability like increased or decreased heart rate or blood pressure.
On laboratory examination, there might be an elevated muscle enzymes (creatine phosphokinase) and leukocytosis.

Prevalence

It is more in male and young patients, and early in the course of treatment. It may occur in response to rapid increase in the dosages of Antipsychotics. This condition occurs in around 0.02-2.4% of those who receive anti dopamines. The mortality rate is within the frame of 10-20%.

Possible Causes of Death in NMS

A. Excessive sweating resulted in hypovolumic shock.
B. Excessive sweating resulted in hypovolumic acute renal failure.
C. Myoglubinurea (due to muscle damage resulted from extreme rigidity) leading to acute renal failure.
D. Electrolyte disturbances.
E. Aspiration as a result of extreme muscle contractions and difficulty in swallowing.
F. Respiratory failure.

Differential Diagnosis

A. Meningitis.
B. Encephalitis.
C. Catatonia.

Management

A. Stop anti-dopamine medications.
B. Admit the patient to intensive care unit.
C. Rehydrate the patient with parentral fluids.
D. Pack for the elevated temperature.
E. Correct electrolyte disturbances (Na, K, Ca).
F. Check vital signs half an hourly in addition to regular follow up.
G. Benzodiazepine may be beneficial.
H. Bromocriptine (dopamine agonist) may be valuable, start with small doses and increase gradually to maximum of 60 mgs /day.

I. Dantroline (muscle relaxant) can be used up to 10 mgs/day, but in intensive care unit.

Acute Dystonia

Dystonia can be defined as uncontrolled spasm and posturing of a group of muscles.

Common clinical types

A. Oculogyric crisis
B. Blepharospasm
C. Torticollis
D. Opisthotonus
E. Pleurothotonus

Prevalence

Dystonia occurs more in young, male patients and early in treatment (more than 90% occur within the first 5 days of starting medications), it's more with typical Antipsychotics, especially with Haloperidol, and Trifluperazine.

Management

Muscle spasms respond well to Anticholinergics (Benztropine 1-4mg/day, Procyclidine 5-15mgs/day, Benzexhol 5-15mgs.day) or Antihistamine (Diphenihydramine 25-200mgs/day).
Consider reduction in the dosage or changing the Antipsychotic.

B. Due to anti-serotonin property

1. Sexual dysfunctions.
2. Depression.

C. Due to anti-histamine property

1. Sedation. (Aripiprazole and Pimozide are less sedating)
2. Increased appetite, obesity, and type 2 Diabetes Mellitus might be a complication, especially with Olanzapine and Clozapine.

D. Due to anti-adrenergic property (blockade of alpha and beta receptors)

1. Hypotension.
2. Postural hypotension.
3. Sexual dysfunctions (impaired ejaculation, and/or retrograde ejaculation).
4. Depression.

E.Due to anti-cholinergic property

1. Dry mouth.
2. Blurred vision, and closed angle glaucoma.
3. Urine retention.
4. Constipation and paralytic ilieus.
5. Sexual dysfunction (impaired erection).
6. Cardiac arrhythmias.

F.Other side effects

1. Hypersensitivity reaction.
2. Skin and retina pigmentation, especially with first generation antipsychotics.
3. Increased risk of seizure, especially with Clozapine.
4. Anemia.
5. Agranulocytosis, especially with Clozapine.

Antidepressants

Antidepressant drugs classified to the following groups according to their mechanisms of action:

A. **Tricyclic Antidepressants (TCADs):** (table 16-2)

TCADs are the oldest antidepressants. The first TCAD discovered during 1950s was imipramine.

Mechanism of action

The main mechanism of action of TCADs is the inhibitory effects on both serotonin and noradrenaline reuptakes and hence increasing the availabilities of these two neurotransmitters in the synaptic clefts. Therefore, TCADs are serotonin-noradrenaline reuptake inhibitors (SNRIs). Additionally, TCADs exhibit antagonist properties over a number of serotonergic and adrenergic receptors.

Table 16-2: TCADs		
Generic name	Trade name	Daily Doses
Imipramine	Tofranil	10-250 mgs
Amitriptyline	Tryptizol	10-250 mgs
Clomipramine	Anafranil	10-250 mgs
Nortriptyline	Pamelor	10-150 mgs
Trimipramine	Surmontil	30-300 mgs
Lofepramine	Lomont	140-210 mgs
Doxepin	Adapin	10-300 mgs
Dosulepin	Prothiaden	75-225 mgs

Side effects

The main side effects of TCADs are related to their anticholinergic properties (see antipsychotic side effects). Other side effects may include drowsiness, sweating, impaired attention, concentration, and memory, and change in appetite.

B. **Selective Serotonin Reuptake Inhibitors (SSRIs):**

(table 16-3)

Since the first introduction of Fluoxetine early in 1980s, SSRIs received an outstanding clinical attentions although several clinical trials showed no superiority over the older TCADs. Nevertheless, the advantage of SSRIs may be in their more tolerable side effects and higher toxicity dosages.

SSRIs mode of action is through increasing the availability of serotonin in the synaptic clefts for post-synaptic actions.

Table 16-3: SSRIs

Generic name	Trade name	Daily Doses
Fluoxetine	Prozac	20-60 mgs
Sertraline	Zoloft	50-200 mgs
Paroxetine	Paxil	20-50 mgs
Citalopram	Celexa	20-60 mgs
Escitalopram	Lexapro	5-20 mgs
Fluvoxamine	Luvox	100-300 mgs

Side effects

Although SSRIs, except paroxetine, lack the anticholinergic side effects of TCADs, distressing GIT side effects may be experienced by users like: dyspepsia, nausea, vomiting, diarrhea, or anorexia. Other possible side effects include sexual dysfunctions, bleeding tendencies, insomnia, and irritabilities.

Paroxetine, unlike other SSRIs, has anti-muscarinic properties, and may cause sedation.

C. **Monoamine Oxidase Enzyme Inhibitors (MAOIs):**

(table 16-4)

MAOI was intended basically for the treatment of tuberculosis. However, researchers noticed that patients received MAOI were exceptionally ‘happy’. The first implication of MAOIs as antidepressants was in early 1950s.

Mechanism of action

As the name MAOI indicates, the mechanism of action of these drugs is thru preventing the metabolism of monoamines like serotonin, noradrenaline, adrenaline, and dopamine.

However, MAOIs carry the risk of serious interactions with drugs or foods containing amines and tyramine (like cheese and wine), the condition which may lead to an emergency 'hypertensive crisis' and death.

Therefore, most of MAOIs have been withdrawn from markets since 1970s. Only few of them are still licensed for resistant and atypical cases of depression.

Table 16-4: MAOIs		
Generic name	Trade name	Daily Doses
Isocarboxazid	Marplan	10-60 mgs
Phenelzine	Nardil	15-90 mgs
Tranylcypromine	Parnate	10-30 mgs
Moclobemide	Aurorix	150-900 mgs

Side effects

The most serious side effect of MAOIs is the hypertensive crisis-reaction when co-administered with amine-containing drugs or tyramine-containing foods. Other side effects include sleep and appetite disturbances, nervousness, postural hypotension, gastric upset, and hepatotoxicity.

D. **Atypical antidepressants:** (table 16-5)

Table 16-5: Atypical Antidepressants			
Mechanism of action	**Generic name**	**Trade name**	**Daily doses**
Serotonin-Noradrenaline Reuptake Inhibitors (SNRIs)	Duloxetine	Cymbalta	60-120 mgs
	Venlafaxine	Effexor	75-375 mgs
Serotonin Antagonists and Reuptake Inhibitors (SARIs)	Nefazodone	Serzone	100-600 mgs
	Trazodone	Desyrel	150-600 mgs
Noradrenaline Reuptake Inhibitors (NRIs)	Reboxetine	Edronax	4-6 mgs
	Atomoxetine	Strattera	20-100 mgs
Noradrenaline-Dopamine Reuptake Inhibitor (NDRI)	Bupropion	Wellbutrin	150-450 mgs
Noradrenergic and Specific Serotonergic Antidepressants (NaSSAs)	Mirtazapine	Remeron	15-45 mgs
	Mianserin	Lumin	30-90 mgs

Side effects: (table 16-6)

Table 16-6: side effects of atypical antidepressants	
Atypical antidepressant	Side effects
Duloxetine	GIT disturbance, sleep disturbance, anorexia, headache, dry mouth
Venlafaxine	GIT disturbance, sleep disturbance, anorexia, headache, dry mouth, sexual dysfunctions
Nefazodone	GIT disturbance, sedation, tremor, headache, tachycardia, postural hypotension, priapism
Trazodone	GIT disturbance, sedation, tremor, headache, tachycardia, postural hypotension, priapism
Reboxetine	GIT disturbance, insomnia, headache, dry mouth, tachycardia, urinary retention, erectile dysfunction
Atomoxetine	GIT disturbance, insomnia, anorexia, headache, impulsivity, suicidal behaviors
Bupropion	Irritability, anxiousness, anorexia, insomnia, headache, , alopecia, pruritus, seizure
Mirtazapine	Increased appetite, weight gain, sedation, edema, headache
Mianserin	Sedation, rash, jaundice, joints pain.

Indications of antidepressants

Antidepressants may be indicated for the treatment of the following psychiatric and medical conditions:

A. Depressive and bipolar disorders.
B. Anxiety disorders.
C. OCD, but in larger doses.
D. Schizoaffective disorder.
E. Associated depressive symptoms of other psychiatric disorders.
F. Associated impulsivities of other psychiatric disorders.
G. Trauma and stress related disorders.
H. An alternative option for the treatment of enuresis (TCADs).
I. An alternative option for the treatment of ADHD.
J. Medical conditions, like migraine, irritable bowel syndrome, and fibromyalgia.

Mood Stabilizers

Mood stabilizers are group of drugs mainly used in psychiatry for controlling the manic and hypomanic mood episodes. However, this group has its place in controlling impulsive and maladaptive behaviors associated with several psychiatric disorders in addition to evidences of its efficacies in controlling depressive disorders.

Mood stabilizers include, basically, Lithium and a group of antiepileptic drugs. Nevertheless, antipsychotics and benzodiazepines are also considered mood stabilizers to some extends as well, and can be considered, particularly antipsychotics, as an alternative or augmentative therapies to lithium or antiepileptic drugs.

Lithium

Although lithium is a naturally occurring element, it was only at 1817 that Johan August Arfwedson, noticed the presence of a new element while analyzing petalite ore. This element formed compounds similar to those of sodium and potassium.
Lithium was used for the treatment of gout in the 19th century since it dissolves uric acid before being withdrawn due to its toxicity.
An Australian psychiatrist, John Cade, was the first who discovered the mood stabilizing effects of lithium at 1949. Since then, several lithium preparations are available for prescription, including: Lithium Carbonate, Lithium Citrate, Lithium Orotate, Lithium Bromide, and Lithium Chloride.
Lithium Carbonate is the commonest preparation in use, whereas Bromide and Chloride preparations fell out of use in 1940s because of their toxicities.

Mechanism of action

The exact mood stabilizing mechanism is unknown yet. However, lithium inhibits the formation of cyclic adenosine monophosphate (cAMP) and weakens the formation of different inositol mediators. These properties may have neuro-stability effects. Lithium thought to upgrade levels of group of neurotransmitters, mainly glutamate and serotonin. Additionally, lithium found to promote neurons' plasticity, a finding which promises preventive effects of lithium on the development of neurocognitive disorders.

Dosage prescription, serum level, and monitoring

Lithium characterized by narrow therapeutic window. I.e., the plasma level necessary for the mood stabilizing effects is closer to the toxic plasma level.
Lithium prescribed under several trade names like Priadel and Camcolit, Eskalith, Lithobid, etc…
Most of the formulations contain lithium carbonate or lithium citrate in the doses or 200, 300, 400, or 450 mgs.

Before starting lithium therapy, baseline screening, including renal function test, thyroid function test, ECG, and weight, should be measured. Scheduling the dosage is plasma level-dependent. Most studies indicated a therapeutic plasma level of 0.4-0.8 mmol/L. However, levels of up to 1-1.2 mmol/L may be tolerable.

On the other hand, side effects may appear after the level of 1 mmol/L and early features of toxicity may start after 1.2mmol/L plasma level. However, serious toxic effects strike after 2 mmol/L concentrations.

Therefore, frequent and close observation of lithium plasma level is mandatory, particularly in the first 6 months of therapy. For instance, its recommended to measure lithium concentration 4 days after first administration, then once weekly for the subsequent three weeks, and 7 days after every dose change until the desired plasma level is reached. Afterward, every 3 months lithium level, and every 6 months renal function test, thyroid function test, and BMI should be monitored.

Lithium monitoring should be performed more frequently if other drugs were co-administered with lithium, as well as if it is taken during hot climates when there is risk of excessive sweating, because lithium elimination is, more or less, sodium dependent.

Side effects

Most of the side effects are plasma level dependents. These may include:

A. GIT disturbance like nausea, vomiting, dyspepsia.
B. Tremor, which is fine at the beginning, but may end up with coarse tremor in the toxic levels.
C. Polyuria, polydipsia, and there is risk of nephrogenic diabetes insipidus. In higher doses there are risk of interstitial nephritis and renal failure.
D. There may be around 20% chance of hypothyroidism.
E. Exacerbation of skin conditions like acne and psoriasis.
F. Ankle edema.
G. Weight gain.

Lithium toxicity

Lithium toxicity is a top medical emergency, if its not managed promptly, possibly ends up with death. Although early features of toxicity may appear after 1.2mmol/L, clear features draw attentions after 1.5 mmol/L, and levels above 3mmol/L may be fatal.

Risk factors

The crucial risk factor is the alterations in sodium level which may be due to the following factors:

A. Medical conditions: like Addison's disease, medical conditions associated with altered renal efficiency (***Note***, lithium is the only psychotropic agent which metabolized solely by kidney), or any medical condition associated with high fever and excessive sweating.
B. Dehydration.
C. Old-age groups.
D. Hot climate and weather conditions.
E. Low-salt diets.
F. Drugs: any drug alters kidney's ability of handling sodium may change the metabolism of lithium and hence, reduces kidney's ability of lithium elimination. These drugs include: ACE-inhibitors (e.g., enalapril), angiotensin II receptor antagonists (e.g., candesartan), Thiazide-related and loop diuretics, Non-steroidal anti-inflammatory Drugs (NSAIDs) (e.g., diclofenac), Carbamazepine, and SSRIs.

Clinical Features

Starting features include increase anorexia, diarrhea, nausea, and vomiting. These features are followed by CNS effects like coarse tremor, muscle weakness, muscle twitching, ataxia, and drowsiness. After 2mmol/L lithium levels, attacks of more severe symptoms may be obvious like renal failure, delirium, slurred speech, nystagmus, seizure, coma, and death.

Treatment

On receiving a suspected lithium-toxicity patient in the emergency room, the following measures should be applied:

A. IV cannula access and immediate blood drop for measuring lithium serum level, renal function test, thyroid function test and full blood count.
B. Life-saving measures should be done accordingly without waiting the results of (A). For instance, if the patient was delirious or comatose, the medical staff should deal with these conditions according to guidelines.
C. Provide IV fluids immediately.
D. Nausea, vomiting, or tremor can be treated symptomatically.

E. The life-saving measure would be osmotic or forced alkaline diuresis. However, thiazide-related and loop diuretics should never be used.
F. If lithium' level over crossed 3 mmol/L, then peritoneal or hemodialysis are often necessary.

Anticonvulsants

Although the exact mood stabilizing mechanism of action is not clear yet, several anticonvulsants are utilized as mood stabilizers; like, Valproate, Carbamazepine, Phenytoin, Lamotrigine, Topiramate, etc…

In this section, the focus would be only on the main two anticonvulsants employed for the treatment of bipolar disorders, Valproate and Carbamazepine.

Valproate

Several formulations of valproate are available like: Valproate, Valproic Acid, Sodium Valproate, and Semi-Sodium Valproate. Sodium Valproate is commonly used in UK, whereas in USA, Valproic acid is widely used.

Although valproic acid was first synthesized in 1882, it was only in 1967 when first employed as an antiepileptic drug.

The exact mechanism of action is not clear. However, valproate thought to stabilizes neurotransmissions through synapses and neuro-membranes via depletion of inositol and inhibition of voltage-gated sodium channels. Additionally, valproate inhibits the catabolism of GABA inhibitory neurotransmitter, which in turn may help in calming down the over-excited brain.

Carbamazepine

Carbamazepine was discovered in Switzerland in 1953 and marketed for the first time to treat 'trigeminal neuralgia' in 1962. It was only in 1962 when carbamazepine first used as an antiepileptic. However, around 10 years later, two Japanese physicians, Takezaki and Hanaoka, discovered its anti-manic effects when they treated manic patients resistant to antipsychotics. Similar to valproate, exact mood stabilizing effect is not understood yet. But, it thought to share similar actions to that of Valproate on the nervous system. (See table 16-7)

Table 16-7: Valproate and Carbamazepine Mood Stabilizers			
Mood Stabilizer	Trade name	Doses	Side effects
Valproate	Depakine, Depakote	Up to 25-40 mgs/kg/day	Teratogenic effects, nausea, gastric upset, cognitive impairment, sedation, weight gain, alopecia, tremor, low platelet count
Carbamazepine	Tegretol, Equetro	Up to 20-25 mgs/kg/day	Teratogenic effects, agranulocytosis, aplastic anemia, hepatitis, Steven-Johnson Syndrome, cognitive impairment, sedation, weight gain, ataxia.

Mood stabilizers and pregnancy

Mood stabilizers showed to carry teratogenic effects. For instance, lithium has a well-known association with cardiac malformation like Ebstein's anomaly and septal defect, and carbamazepine and to lesser extends valproate are associated with neural-tube defects. Therefore, most states-guidelines prefer avoidance of mood stabilizers during pregnancy and adopting antipsychotics as alternative mood stabilizers at the meantime.

If the continuous uses of mood stabilizers are mandatory, then an extreme cautions and close monitoring should be in place. Its, also, preferable to pair anticonvulsants with folic acid supplements a couple of months prior to pregnancy.

Anxiolytics

Anxiolytic agents include three main kinds of drugs; Benzodiazepines, Non-Benzodiazepines, and Buspirone. Apart from Buspirone, the other two groups exert their anxiolytic effects through enhancing the inhibitory GABA neurotransmission. Additionally, several other agents are also regarded to be anxiolytics, like barbiturates and barbiturate-like substances (not used now), antidepressants, antipsychotics, and mood stabilizers.

Benzodiazepines (BZD)

The first BZD, chlorodiazepoxide, discovered by a Croatian chemist, Leo Sternbach, in 1955 and delivered to markets in 1960.

BZDs have agonistic properties over both GABA-A and –B receptors.

Types

BZDs can be categorized to three main groups according to their half-lives (see table 16-8):

Table 16-8: Types of BZDs

BZDs	Half-life	Examples	Trade names	Doses
Short-acting BZDs (hypnotic BZDs)	1-12 hours	Midazolam	Dormicum	7.5 mgs
		Brotizolam	Lendormin	0.25 mgs
		Triazolam	Halcion	0.25-0.5 mgs
Intermediate-acting BZDs	12-40 hours	Lorazepam	Ativan	0.5-8 mgs
		Alprazolam	Xanax	0.5-2 mgs
		Clonazepam	Rivotril	0.5-2 mgs
Long-acting BZDs (Anxiolytic BZDs)	40-250 hours	Chlorodiazepoxide	Librium	5-200 mgs
		Diazepam	Valium	5-200 mgs
		Flurazepam	Dalmadorm	15-30 mgs

Indications

A. Anxiolytics
B. Hypnotics
C. Muscle relaxants
D. Anticonvulsants
E. Amnestic-dissociative effects, hence may be useful in anesthesia processes in blocking distressing per-theater experiences.
F. Useful in controlling acute psychosis and agitation. It is thought that BZDs in higher doses may block dopamine receptors as well.
G. May reduce core delirious symptoms.
H. May be used to control manic symptoms temporarily at the beginning of therapy.
I. Used in the treatments of alcohol and substance-related disorders.

Side effects

BZDs are regarded to be the safest psychotropic drugs. The only serious side effects are the risk of respiratory depression when higher doses administered, particularly thru parenteral routes, as well as their risks of recreational uses and dependence (see chapter 10):

A. **Sedation** is the most common and universal side effect. Patients should be cautioned about driving after taking benzodiazepines. Tolerance to sedative effects often occurs during the first few weeks of treatment.

B. **Cognitive Dysfunction.** Anterograde amnesia is common after benzodiazepine use, especially with high potency agents (e.g., alprazolam) or short acting agents (e.g., triazolam).

C. **Miscellaneous Side Effects**

1. Benzodiazepines may produce ataxia, slurred speech, and dizziness.
2. Respiratory depression can occur at high doses, especially in combination with alcohol, or in the presence of respiratory disorders such as chronic obstructive pulmonary disease.
3. Behavioural disinhibition.

Non-Benzodiazepine GABA Receptors Agonists (Z-drugs)

Z-drugs (table 16-9) are a group of psychoactive drugs although structurally differ from BZDs, but their pharmacodynamics properties are almost similar to BZD. Z-drugs bind to GABA-A receptors and activate BZD sites on the receptor complex.

Since this group of drugs do not act on GABA-B receptor, they almost lack other properties of BZDs like muscle relaxation, anticonvulsant property, and antipsychotic properties. Instead, the main indications of Z-drugs are as sedative/hypnotics and, to a lesser extends, as anxiolytics.

Z-drugs characterized by very short onsets of action, usually within 7-30 minutes. Therefore, they are recommended to be taken at bedtime.

Side-effects profile is similar to that of BZDs with the exception of lower risks of dependence than BZDs.

Table16-9: Z-drugs		
Drug	**Trade name**	**Dose**
Zolpidim	Ambien	5-10 mgs
Zopiclone	Imovane	3.75 – 7.5 mgs
Zaleplon	Sonata	10 mgs

Buspirone (BuSpar)

Buspirone is a non-benzodiazepine anxiolytic agent of the azaperone class. Unlike BZDs and non-BZDs, Buspirone does not act on GABA receptors. Instead it has a partial agonist activity on serotonin ($5HT_{1A}$) receptor. Therefore, Buspirone is not addicting and has no withdrawal syndrome or tolerance, and it does not produce sedation or potentiate the effects of alcohol.

Additionally, Buspirone antagonizes several presynaptic dopamine receptors and has partial agonist property on alpha-1 receptors. All these mechanisms of actions add an antidepressant property to Buspirone as well.

Indications

A. Buspirone (BuSpar) is indicated for anxiety disorders such as generalized anxiety disorder; however it is not effective for panic disorder. Instead, it may exacerbate panic symptoms.
B. It may also be an effective adjunctive agent in the treatment resistant depression. It may be added in a dosage of 15-60 mg/day if a patient has had a suboptimal response to a 3-6 week trial of an antidepressant.
C. Buspirone can be used to counter sexual side effects of SSRIs.

Dosage

A. The starting dose is 5 mg two to three times a day. Gradually increased to a maximum dosage of 60 mg per day over several weeks.
B. Many patients respond to a total dose of 30 - 40 mg per day in two to three divided doses.
C. At least two weeks are required before clinical improvement occurs. It is common to see the first signs of improvement after 3-6 weeks.

Side Effects

Buspirone is generally well tolerated; the most common side effects are nausea, headaches, dizziness, and insomnia.

17 PHYSICAL THERAPIES

There are several non-pharmacological physical attempts to treat psychiatric disorders which are, directly or indirectly, stimulation therapies like Electro-Convulsive Therapy (ECT), Trans-cranial Magnetic Stimulation (TMS), Vagal-Nerve Stimulation (VNS), and Bright-Light Therapy (Phototherapy).
These "stimulation" therapies, in mental health, are principally in practices for the treatment of depressive disorders. However, except for ECT, there are little evidences to support the efficacies of these "stimulation" therapies.
Therefore, the focus of the chapter will be on ECT only. For other "stimulation" therapies, readers are advised to address related resources.

Electro-Convulsive Therapy (ECT)

ECT is a kind of physical psychiatric therapy in which seizures are electrically induced in patients with major psychiatric disorders to get relieves from their symptoms.
Modern ECT is dated back to 1934 when a Hungarian neuropsychiatrist, Ladislas J. Meduna, inaccurately believed that epilepsy and schizophrenia symptoms cannot co-occurs at the same time. He noticed that a group of epileptic patients with co-morbid schizophrenia got relieves from psychotic symptoms after bouts of seizure. Meduna relied on camphor and metrazol in inducing seizure.
Three years later, two Italian neuropsychiatrists, Ugo Cerletti and Lucio Bini, experimented electrical shocks instead of metrazol in seizure induction on human.
Through 1940s and 1950s, ECT became popular worldwide. However, the therapies were "unmodified", i.e., the seizures were induced without muscle relaxants, conditions which led to full-scale convulsions and serious side effects like fractures, and joint dislocations.
Due to the steady development of antidepressants during 1950s and onward, alongside with the negative representations in media, ECT received noticeable criticisms both among publics and professionals.
Nowadays, ECT is almost the last line of the life-threatening psychiatric conditions like suicide, and resistant-to-therapies' major psychiatric disorders.

Mechanism of action

Despite decades of researches, nobody knows the exact therapeutic mechanism of action. It is thought seizures may redistribute or upgrade the levels, though temporarily, of several neurotransmitters besides pruning of normally dense synaptic connections in the hippocampus deep in the temporal lobe.

Types of ECT procedure

The procedure, induction of seizure electrically, can be done with anesthesia **(Modified)** or without anesthesia **(Unmodified)**. Also the procedure can be done either by placing both electrodes on one side of head only **(Unilateral)** or on both sides **(Bilateral)**. Unmodified ECT carries more risk of fracture and dislocation of joints, whereas modified ECT is paired with the side effects of anesthesia as well. Bilateral ECT is more therapeutically effective than unilateral ECT, though it has more serious side effects like more severe cognitive impairments. The commonest mode of ECT procedure nowadays is the Modified Bilateral ECT.

Indications of ECT

National Institute for Clinical Excellence (NICE) (2010) recommended sparing ECT for achieving rapid and short-term improvement of severe symptoms when other treatment options has proven unsuccessful or when conditions considered to be life-threatening like:

A. Severe depressive disorders.
B. Postpartum psychosis.
C. Catatonia.
D. A prolonged or severe manic episode.

Additionally, although its not recommended by any guideline, researchers found beneficial effects of ECT for treatment-resistant conditions of OCD and non-catatonic schizophrenia. However, these findings are in need of replications.

Dosages of ECT

The dose of ECT depends on the seizure-threshold which is widely variable among individuals. Therefore, during the first therapy, the electricity voltage should be titrated until the seizure is apparently induced then subsequently the same dosage is applied. However, several studies found a dosage of 150 millicoulombs (mC) is satisfactory depending on the fact that two-thirds of populations' seizure-threshold is between 100 and 200 mC.

The therapist should take in account that seizure-threshold is higher among males and old-age groups. Also, higher doses are necessary for those who are epileptics, received ECT in the past, and those receiving anticonvulsant therapies.
The frequency of ECT sessions depends on the clinical experiences. Usually a frequency of 6-12 sessions are satisfactory. Fewer than 6 or more than 12 may be practiced as well. However, most of patients who did not respond to 6-8 sessions, do not respond to more sessions. Hence, ECT is preferable to be terminated if there is no improvement after the eighth session.
ECT sessions are usually distributed over two or three days per a week. There is little evidence supporting the superiority of three sessions/week over two. Instead, closer sessions carry more risk of cognitive impairments.
Overall, typical ECT encompasses two ECT sessions weekly for three subsequent weeks.

Patient's preparation for ECT
Before starting ECT, the following preparatory steps should be taken in count:

A. Its advisable for the patient to be drug-off for around 10-14 days prior to first therapy. However, since its not clinically applicable to stop all psychotropic agents and as ECT may be an emergency life-saving measure with no prior time to take-off drugs, keeping the patient drug-free only the day before therapy may be satisfactory. For mood stabilizers like anticonvulsants, 48 hours drug-off may be necessary. Nevertheless, clinicians can continue prescribing while patients undergoing ECT taking in account careful attentions to anticonvulsants (because they raise the seizure thresholds), SSRIs (because they may prolong the seizure), and TCADs and antipsychotics (because they reduce seizure thresholds).
B. Both the clinician and medical staff should clear up the ECT's technique and side effects, reassure the patient, and obtain therapy-consent either directly from the patient or patient's family members depending on the country's ECT regulations.
C. The patient should take nothing by mouth for at least 5 hours prior to the therapy as a preparation for anesthesia and to minimize aspirations.
D. Before the therapy the ECTist should check patient's dentures removal, any loose or broken teeth, and empty bladder.

E. Previous ECT records should be investigated for possible side effects, complications, ECT dosages, etc…
F. Finally, ECT theatre should be well-equipped with all necessary kits like oxygen, vacuum, etc...

Side effects

In addition to the side effects of anesthesia, the seizure itself has several adverse consequences:

A. Semi-delirious states immediately after awakening.
B. Cognitive impairments, especially memory impairments in forms of both retrograde- and anterograde amnesias.
C. Multiple over times ECTs may fasten or predispose to neurocognitive disorders.
D. Tongue or lips bite.
E. Bone and teeth fractures
F. Pains, especially headache and muscle pain.
G. Inadequate placements of electrodes might result in local hair and skin burns.
H. Very rare complications include arrhythmias, pulmonary embolism, aspiration pneumonia, stroke, prolonged apnea, and status epilepticus.
I. Mortality rate is relatively low, around 3-4 per 100 000 treatments, which is analogous to anesthesia's mortality.

Contraindications

There is no absolute contraindication for ECT per se. However, presence of co-morbid serious medical conditions that increase the risk of anesthesia are regarded as relative contraindications for ECT, these include: respiratory infections, serious heart diseases, cerebral or aortic aneurysm and raised intracranial pressure.

18 PSYCHOTHERAPIES

Although the term psychotherapy initially signified 'hypnosis', Frederik van Eeden (1892) was the first to give the appropriate definition of psychotherapy as "the treatment of disorders of the mind or personality by psychological or psychophysiological methods".

In other terms, psychotherapy is a kind of therapy agreed upon between two persons. However, in this sense, psychotherapy is indistinguishable from counseling, caregiving, or friendship.

To keep it within the medical concept, psychotherapy can be defined as a form of non-pharmacological and non-physical therapy, depending on verbal or non-verbal means to relief or treat psychological distresses or disorders, for which it should be delivered by qualified and well-trained professionals to one, two, or a group of individuals.

Psychiatrists are not the only professionals authorized to provide psychotherapies; instead, clinical psychologists, social workers, licensed counselor, or any psychotherapy-trained practitioner can, however, afford psychotherapy.

Classification of psychotherapy

There are more than 450 kinds of psychotherapy. However, one can group them according to their techniques, number of clients involved in the therapy, or the level of complexity:

A. Techniques: examples include
   1. Psychodynamic
   2. Cognitive
   3. Behavioral
   4. Interpersonal
   5. Counselling
   6. Crisis intervention; and several other techniques

B. Number of participants:
   1. Individual: this therapy is indicated when the therapy needs to be tailored to the specific problems of that individual, or when group therapy is distressful because of the sensitive nature of the problem or client's social anxiety.
   2. Couple/marital: it is chosen when the psychological distresses raised by marital conflicts.

3. Family: this kind is usually adopted for behavioral problems among children and adolescents thought to be due to the adverse family environment.
4. Group: whether small or large number groups selected for those who share similar problems like phobias, or when other clients may contribute positively in the therapies; for instance Alcohol-use disordered patients may offer significant personal experiences in dealing with withdrawal or craving symptoms to other patients.

C. Level of complexity:
1. Part of all healthcare systems: example includes counselling to deals with emotional problems or to help clients in health-related decisions like therapeutic-abortion.
2. Part of mental-health systems: examples include modest forms of cognitive-behavioral or brief psychodynamic psychotherapies for the treatment of milder forms of depressive or anxiety disorders.
3. Provided by only specially well-trained psychotherapists: examples include particular kinds of behavioral therapy like (ABA for the treatment of Autism), more complex cognitive-behavioral therapies for major depressive or anxiety disorders, or long-psychodynamic psychotherapies for more complex anxiety or other psychiatric disorders.

Note: a psychotherapeutic session may combine more than one kind of therapy. Well known, and commonly, practiced combined psychotherapy is Cognitive-Behavioral Therapy (CBT).

Shared features of psychotherapies

Although psychotherapies differ from each other in their techniques, indications, and durations; researchers found that there are common therapeutic features shared by all kinds of therapies, these include:

A. The therapeutic relationship: is the most important among shared factors. However, this relationship may result in problems when become deeper.

In any kind of relationship, there is two main kinds of reactions, transference and countertransference. **Transference** is the redirection of feelings and desires of those unconsciously retained from childhood toward a new object. For instance, its not uncommon for people to transfer feelings from parents to their partners or children. A clear example may be the mistrust somebody feels toward an individual looking like his/er ex, unfaithful, spouse. Within psychotherapeutic concept, however, the client/patient transfer "unconsciously determined" feelings and desires from persons in his/er life experience toward the therapist. The quality of the transferred emotions depends on the cultural backgrounds, kind of therapy, kinds of problems tackled in the therapy, as well as the quality of the therapeutic relationship. Transference often demonstrated in forms of erotic attractions to the therapist. Nevertheless, other kinds of emotions may be in place like rage, parent-figure, mistrust, or on the other hand, an extreme dependence, or placing the therapist in a guru, prophet-like, or god-like statuses.

Countertransference, however, is the redirection of therapist's feelings toward the client. Theoretically, it may take any form of transference's feelings. Yet, practically sexual attractions or rage are the commonest countertransference reactions.

Both transference and countertransference were initially described by Sigmund Freud. Freud stated that therapists can gain clinical benefits from both these reactions. Nevertheless, most psychotherapeutic researchers pointed out that any countertransference sign may indicate the inadequacy of therapy-training, and that the therapist requires more personal analytic training to upgrades him/er self to be more impartial and professional.

B. Sympathetic and active listening: helping and showing the patient that his/er feelings are understood are essential in establishing the therapeutic relationship which paves the road for future compliances with the therapeutic programs.
C. Catharsis: it is vital to let the patient discharge his/er emotional sufferings and conflicts.
D. Rebuilding of self-confidence: there are patients who no longer believe on, trust, or help themselves. Therefore, reestablishing self-confidence and morale is a critical step-upward towards improvement.
E. Offering information, advices, guidance, and suggestions.

Psychoanalysis and psychodynamic psychotherapies

Psychoanalysis was the first attempt named 'psychotherapy'. This kind of therapy relies on Freud's notion that one's behavior is determined by forces derived from unconscious mind. Thereby,

stresses are derived from unconscious conflicts which may basically be of sexual or aggressive natures.
The principal approach of psychoanalysis is to gradually reveal 'repressed' experiences in the unconscious mind and integrate them into the patient's personality, conscious mind, or Ego. Therefore, the therapist in this form of psychotherapy acts as an interpreter of what the client said.

Techniques

There are three elementary techniques of psychoanalysis:

A. Free association: here, the client lies on a couch staring away from the therapist and says whatever comes to mind. The therapist depends mainly on "tongue slip" to unfold the unconscious conflicts.
B. Interpretation of dreams: since dream is the "Royal road to the unconsciousness" as Freud claimed.
C. Interpretation of transference (see above).

Indications

Clients who are appropriate for therapies related to psychoanalysis should be:

A. Not suffering from psychotic disorders.
B. Younger than 60s-year old.
C. Intelligent.
D. Have good interpersonal relationships.
E. Have appropriate social communication skills.
F. Can afford both time and money for therapy.

Cognitive Therapy (CT)

Unlike psychoanalysis, cognitive therapy is not concerning about discovering the origins (etiologies) of current mental statuses. Instead, CT stresses on modifying the way we understand, approach, or recognize the environment (cognitive framework).
According to cognitive theory, the individual's perception of environment, rather than the true environment, determines our emotions, reactions, as well as behaviors. Therefore, stressful emotions like depression or anxiety, and maladaptive behaviors are stemmed from "faulty" cognitive frameworks.
Hence, the principal goals of CT are to correct the negative thoughts and perceptions in order to deliver out more proper cognitive outline. CT commonly combined with Behavioral therapy

(BT) to form a more comprehensive therapeutic approach called Cognitive-Behavioral Therapy (CBT).

Behavioral Therapy (BT)

Similar to CT, BT stresses on symptomatic modifications, yet, of behaviors. BT relies on the principles of 'conditional learning theory' like 'reinforcement' and 'punishment'. Behavioral theorists believe that or emotions, reactions, and behaviors are learnt and acquired from our environment, past experiences, or may even be evolutionally determined; for instance, we learned to become panic when faced life-threatening animals (e.g., snake).

Consequently, distressful emotions (like depression, phobias, obsessions), as well as maladaptive improper behaviors noticed in almost all psychiatric disorders (e.g., avoidance in anxiety disorders, self-neglect in depressive disorders, impulsivities in personality disorders, hyperactivities in ADHD, inability to deals with daily-living issues in intellectual disability or neurocognitive disorders) can be modified to turnout patients' emotions and behaviors less distressful and more adaptive respectively.

What is more, behavioral psychologists do believe that our behaviors influence our cognitive framework (see above) and vice versa, the reason led to the necessity of delivering both CT and BT together in what is called CBT.

Indication of CBT

A. Depressive disorders. This is the main indication. CBT's efficacy is comparable to that of pharmacotherapy, and may be superior in milder conditions.
B. Anxiety disorders: like GAD, Phobias, Panic disorder.
C. Obsessive-Compulsive and related disorders.
D. Alcohol and substance-related disorders.
E. Neurodevelopmental disorders like Intellectual Disability, Autism spectrum Disorder and ADHD.
F. Neurocognitive disorders.
G. Personality disorders.
H. Elimination disorders.
I. Sexual disorders
J. Psychotic disorders: despite lack of solid ground for its efficacy, there are little evidences support CBT in schizophrenia and other psychotic disorders.

Interpersonal Psychotherapy (IPT)

IPT was grown for patients who cannot tolerate neither psychoanalysis nor CBT. Basically, it depends on the principles of both psychoanalysis and CBT. However, the main assumption of IPT is that human being is a medium of complex interactions with other humankinds as well as oneself. Hence, distressful reactions like anxiety and depression might be due to conflicts/deficits in these current (not past) interactions. In this line, IPT stresses on four main problems behind depression or anxiety symptoms:

A. Grief and loss.
B. Role disagreements like marital disputes for instance.
C. Role transition like a single individual became married recently, or an individual became parent for example.
D. Interpersonal deficits such as loneliness.

Indications

Although theoretically can be used for almost all psychiatric, except psychotic, disorders, the main clinical utilizations of IPT are in non-psychotic depressive disorders, eating disorders, and borderline personality disorder.

Counselling

It is a broad term stands for helping an individual to overcome a stressful situations (like grief) through advices, guidance, and providing information. Nevertheless, as a psychotherapeutic term, counselling symbolizes more elaborative, systematic, and wider techniques.

Essentially, counselling relies on the shared features of psychotherapy (see above). However, different kinds of counselling do exist which may depend on psychoanalytic, cognitive, behavioral, or even interpersonal approaches. Examples of counselling include 'problem-solving counselling' to identify life problems and help to overcome them sequentially; 'debriefing' for the survivors of disasters; 'grief counselling' to overcome complicated or prolonged grief; as well as several other kinds of counselling.

Crisis Intervention (CI)

It is an emergency psychological care intended to help individuals in crisis situations to bring back psychosocial balances. Crisis can be defined as any difficult situation as perceived or experienced by an individual to exceeds the mental powers, resources, and copying mechanisms to deal with. The crisis may be on an individual level like rape for example, or on a societal level like mass trauma, for instance school mass shooting, or a terroristic attack. Therefore, crisis depends on the tolerability-threshold of an individual. What is crisis for a person may be not for one else. CI technique is comparable to that of counselling, particularly interpersonal counselling and problem-solving counselling.

APPENDIX-A PSYCHIATRIC CASE SHEET

Demographic Data	
Name	
Age	
Gender	
Education	
Employment	
Address	
Marital Status	
Number and gender of children	

Chief Complaint(s) and Duration

History of Present Illness (HOPI)

*Please attach additional sheets should above space is not enough

Past psychiatric history: note: you can add/delete 'Occasions' columns as it's pertinent.									
		Occasions							
		1	2	3	4	5	6	7	8
Approximate date									
Diagnosis (experience)									
Received Treatment?	No								
	Yes								
If yes:	Where								
	What								
	Benefits								
	Side-effects								
Suicide									

Medico-surgical History	
1-	4-
2-	5-
3-	6-

Family History: note: you can add/delete 'children' and 'siblings' columns as it's pertinent.

	Father	Mother	Children		Siblings		Spouse	Others
			#1	#2	#1	#2		
Name								
Age								
Education								
Occupation								
Address								
Marital status								
Social circumstances								
Relation with the patie								
Major health problems								
Psychiatric history								
Alcohol/drug misuse								
Notes								

Personal History	Notes
Pregnancy	
Delivery	
Early childhood	
School age	
Education	
Occupation	
Relationships	
Sexual history	
Leisure and habits	
Forensic history	
Current social venue	

Pre-morbid Personality

Mental State Examination	
Appearance	
Behavior	
Speech	
Mood	
Thought's content	
Thought's form	
Perception	
Consciousness	
Orientation	
Attention and concentratio	
Memory	
Insight	

- **Does the client need physical examination?**
- **Does the client need specific lab investigation?**
- **What are the possible differential diagnoses?**
- **What is the most probable diagnosis?**

APPENDIX-B MEDICAL PRESCRIPTION TERMS

Many terms used in prescriptions come from the Latin, such as the letter **c** = con (with) and from Greek such as **collyr** = collyrium (poultice, eye wash).

Letter	Meaning	Letter	Meaning
a.c.	Before meal	non rep.	Do not repeat
ā ā	Of each	o.h.	Every hour
ad lib.	Freely, As desired	o.m.	Every morning
anti.	Antidote	o.n.	Every night
aq. dest.	Distilled water	p.c.	After meal
bid/BID	Twice daily	p.r.n./prn	As needed, On need
C	With	q.	Every
caps.	Capsule	q.2hr.	Every 2 hours
contra.	Contraindication	q.4hr.	Every 4 hours
d.	Daily	q.d.	Once a day
dc/DC	Discontinue	q.h.	Every hour
fl.	Fluid	qid/QID	Four times daily
ft. puiv.	Make a powder	q.o.d.	Every other day
ft.	Make	q.s./qs	A sufficient quantity
gtt.	A drop	s.	Without
h.	Hour	sem.	Half
h.s.	Just before sleep	s.i.d./SID	Once a day
hor. som.	Just before sleep	sig.	Write on label
int.	Internal, interior	Ss	Half
liq.	Solution	Stat	Immediately
m.	A fluid measure	Syr	Syrup
M.	Mix	t.i.d./TID	Three times daily
noct.	At night	ut dict	As directed

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